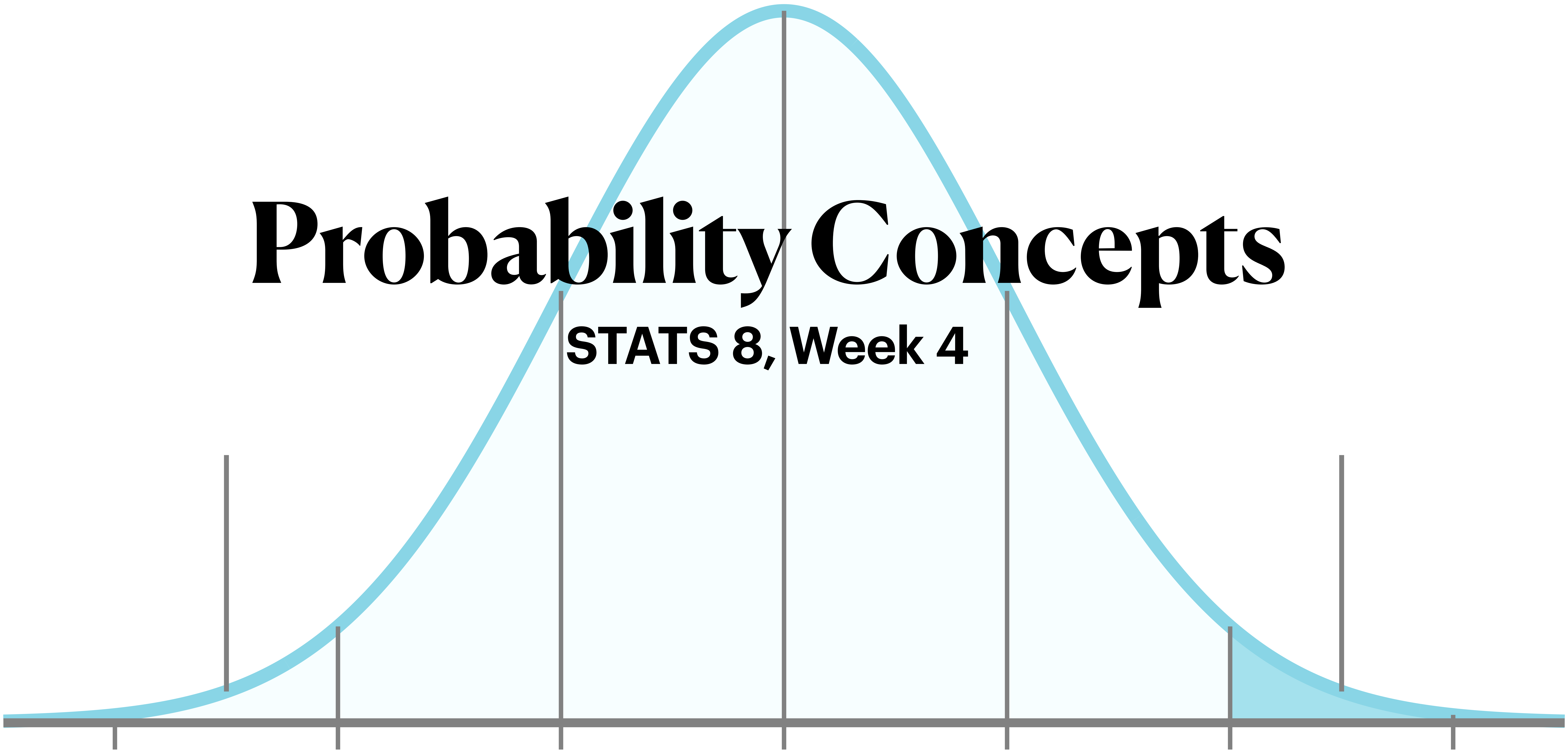
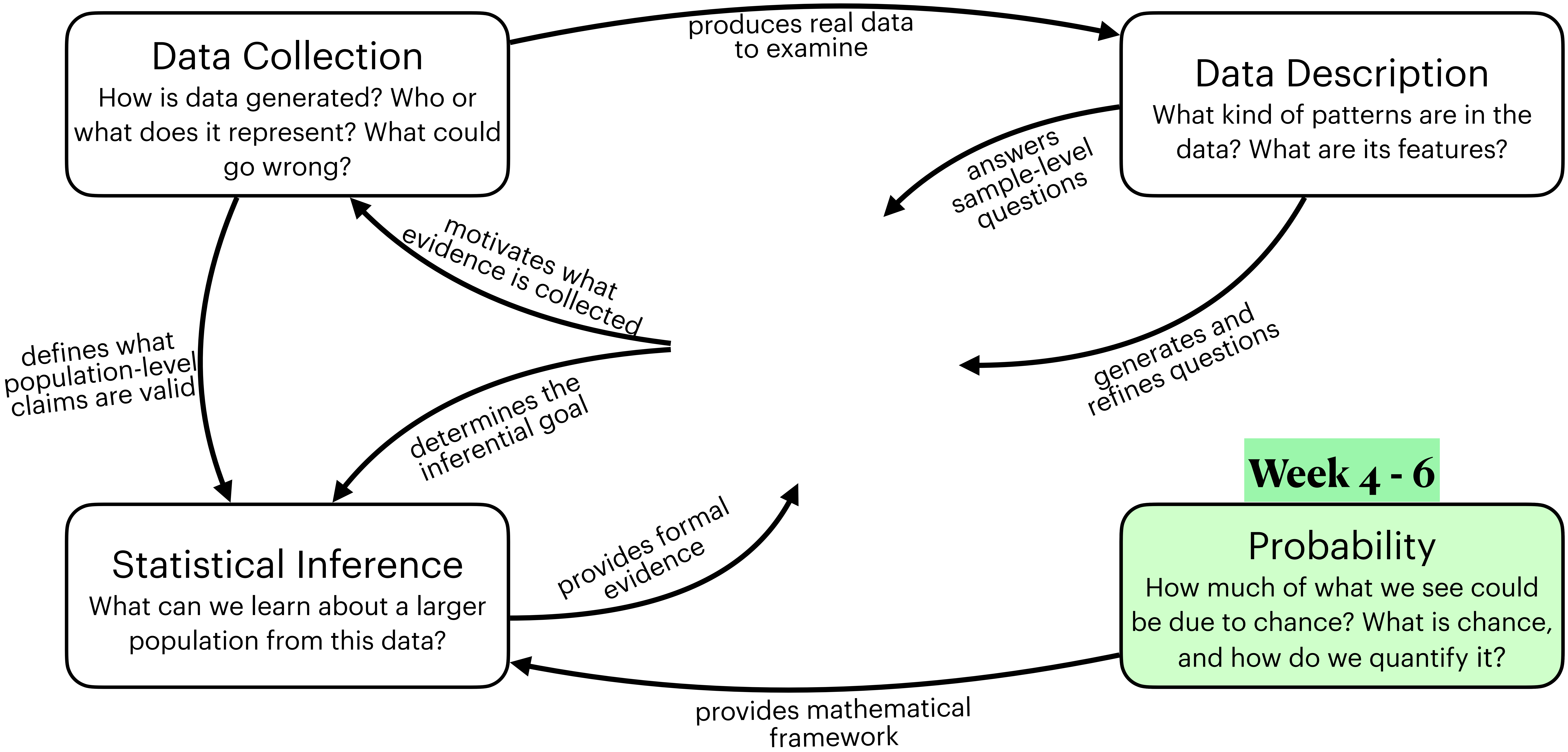
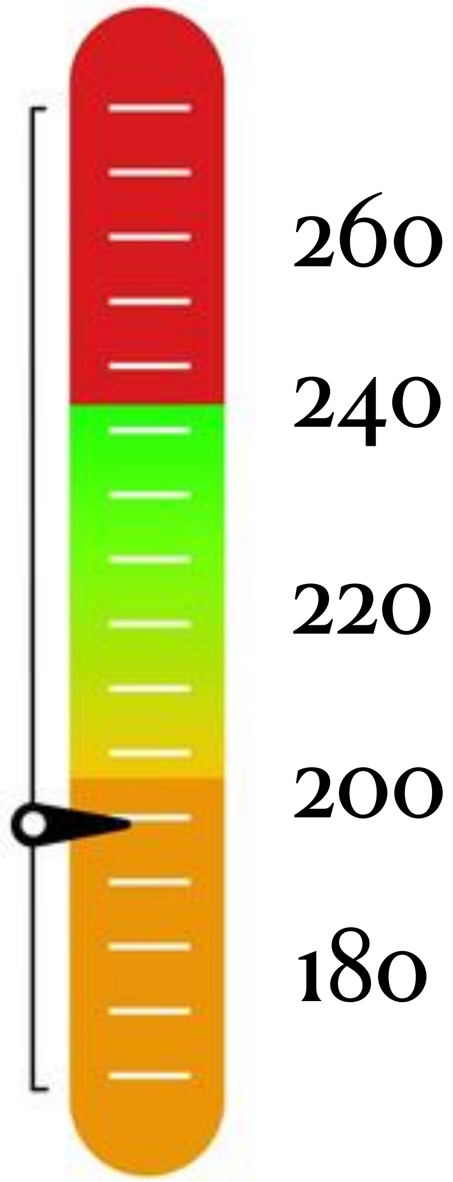
Probability Concepts

STATS 8, Week 4

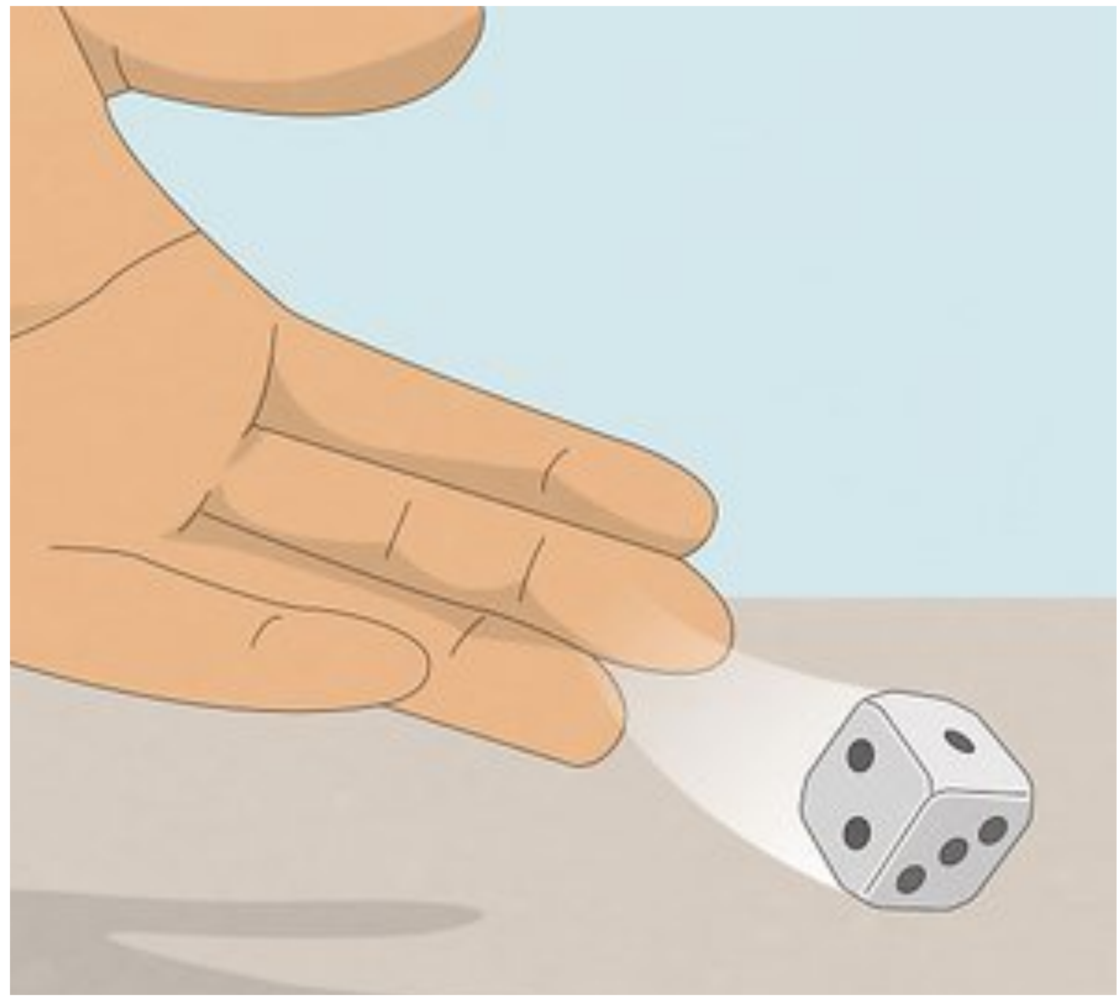


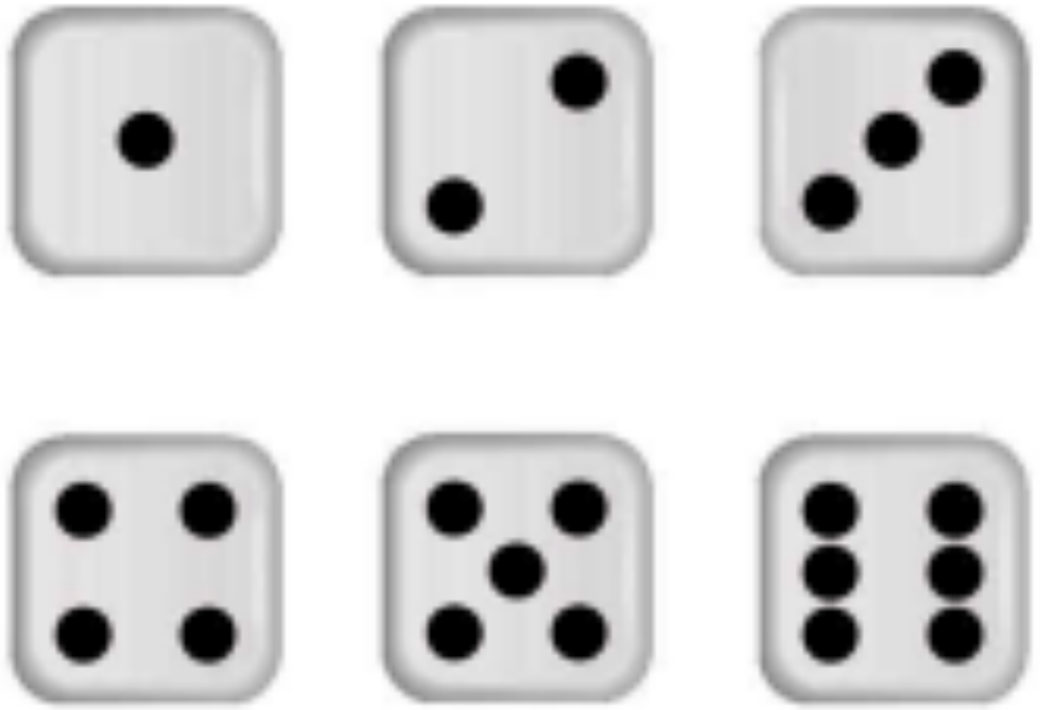

Data Collection
How is data generated? Who or what does it represent? What could go wrong?
produces real data to examine
Data Description
What kind of patterns are in the data? What are its features?
answers sample-level questions
motivates what evidence is collected
defines what population-level claims are valid
determines the inferential goal
generates and refines questions
Week 4 - 6
Statistical Inference
What can we learn about a larger population from this data?
provides formal evidence
Probability
How much of what we see could be due to chance? What is chance, and how do we quantify it?
provides mathematical framework



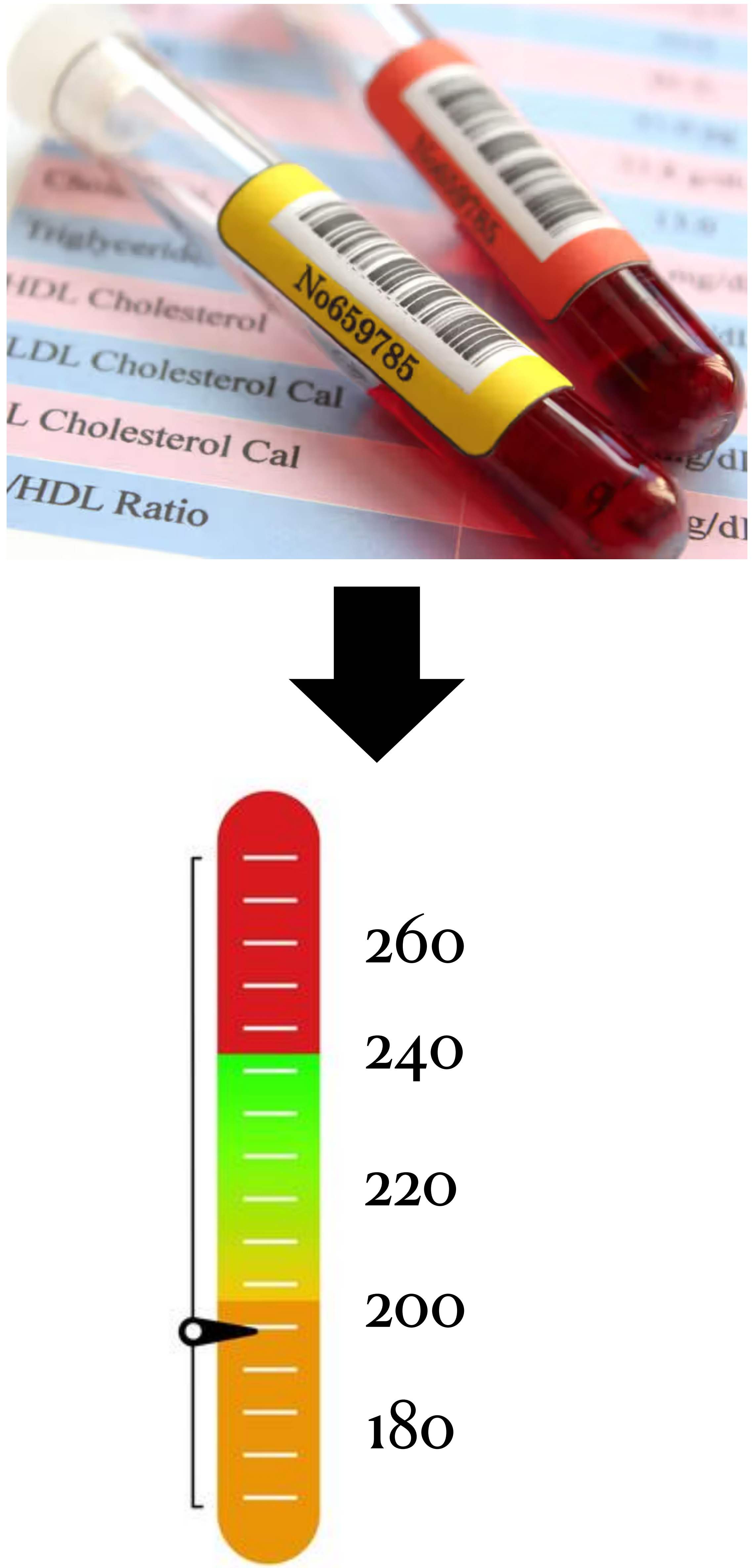
Randomness

- A **random** circumstance is a circumstance with an uncertain outcome
  - Classic toy examples include flipping a coin, rolling a die
  - Possibly more relevant: whether it will rain tomorrow, a person's cholesterol level
- The **sample space** (S) is the collection of all possible outcomes of the random circumstance

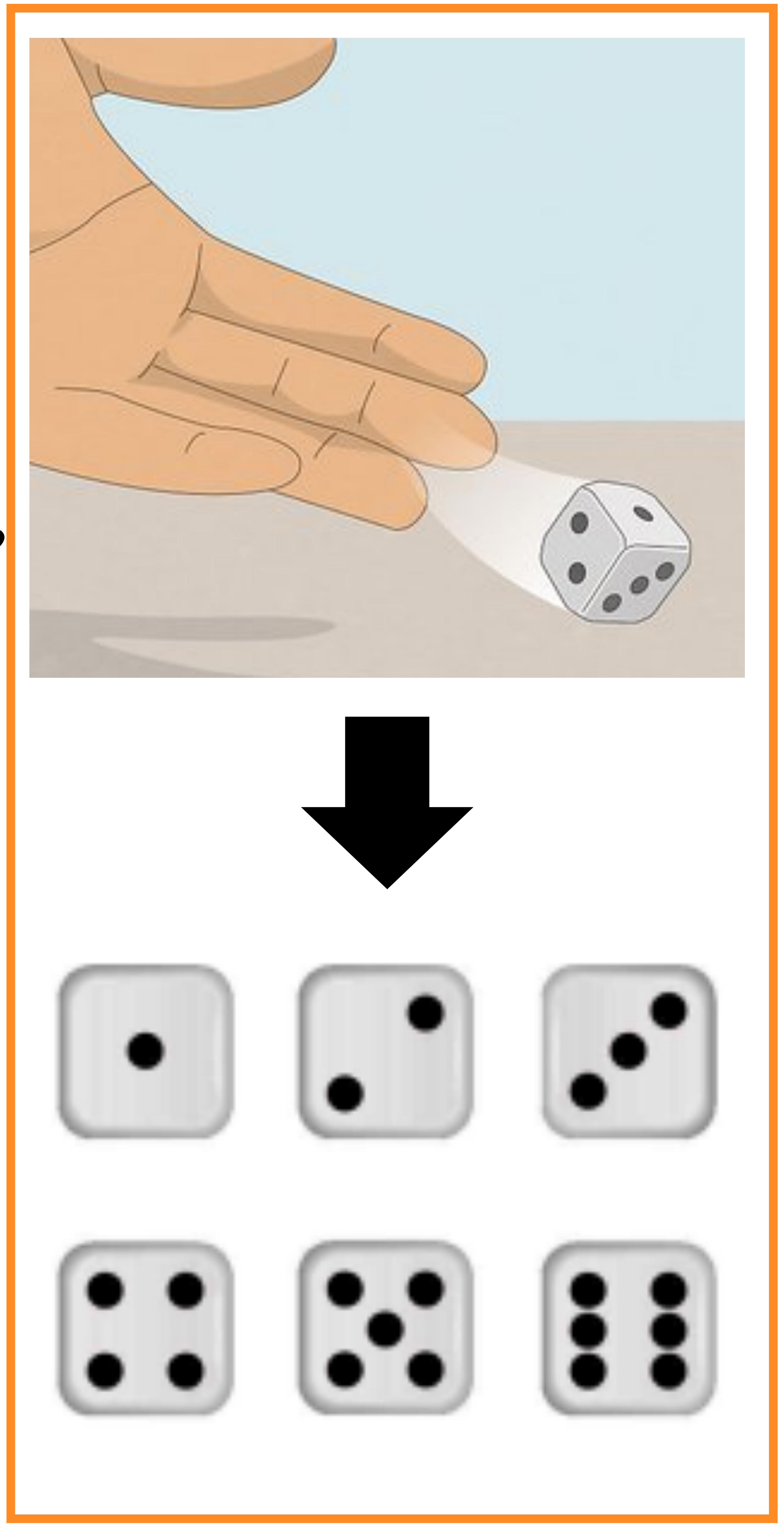


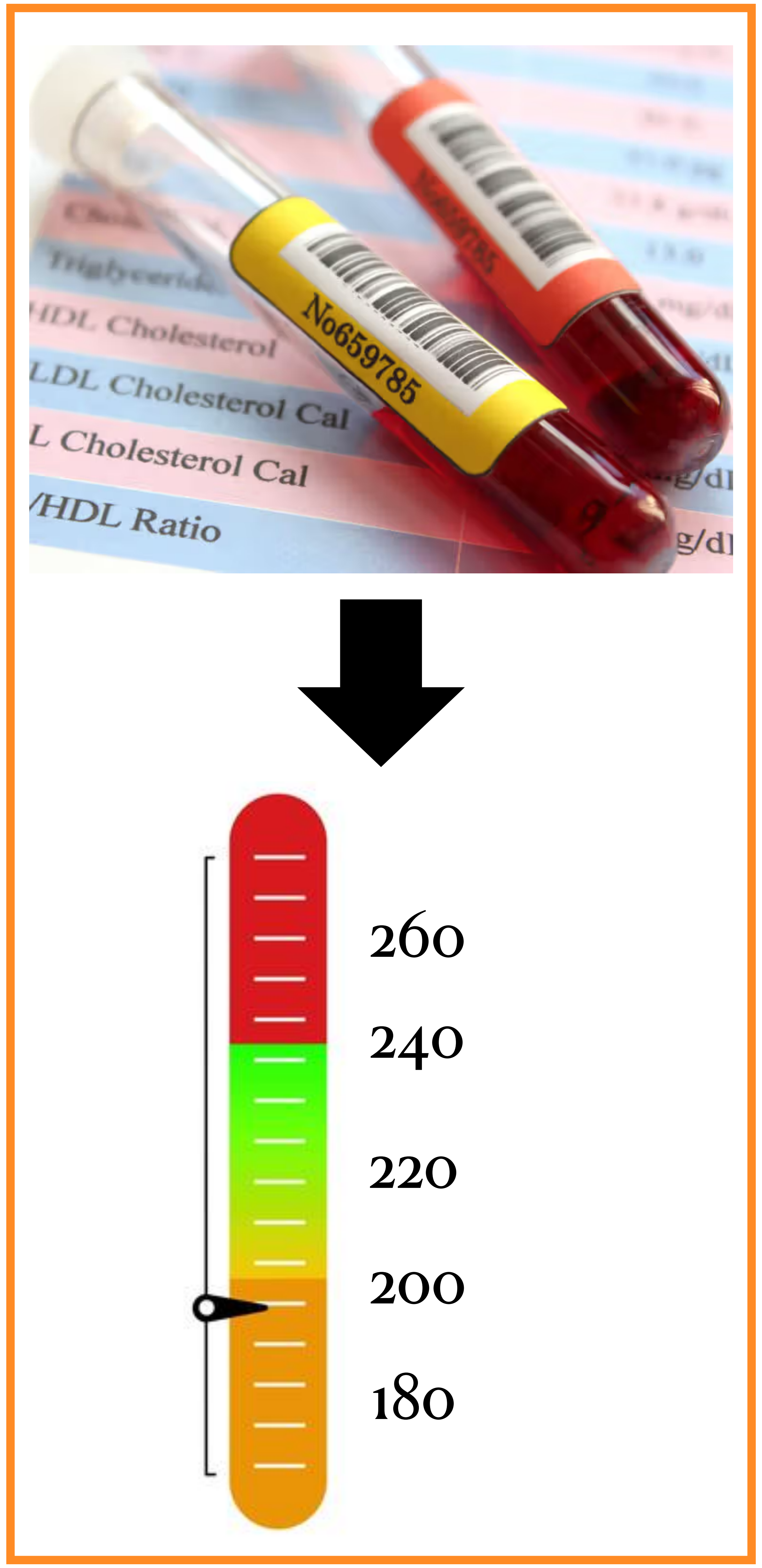


Randomness



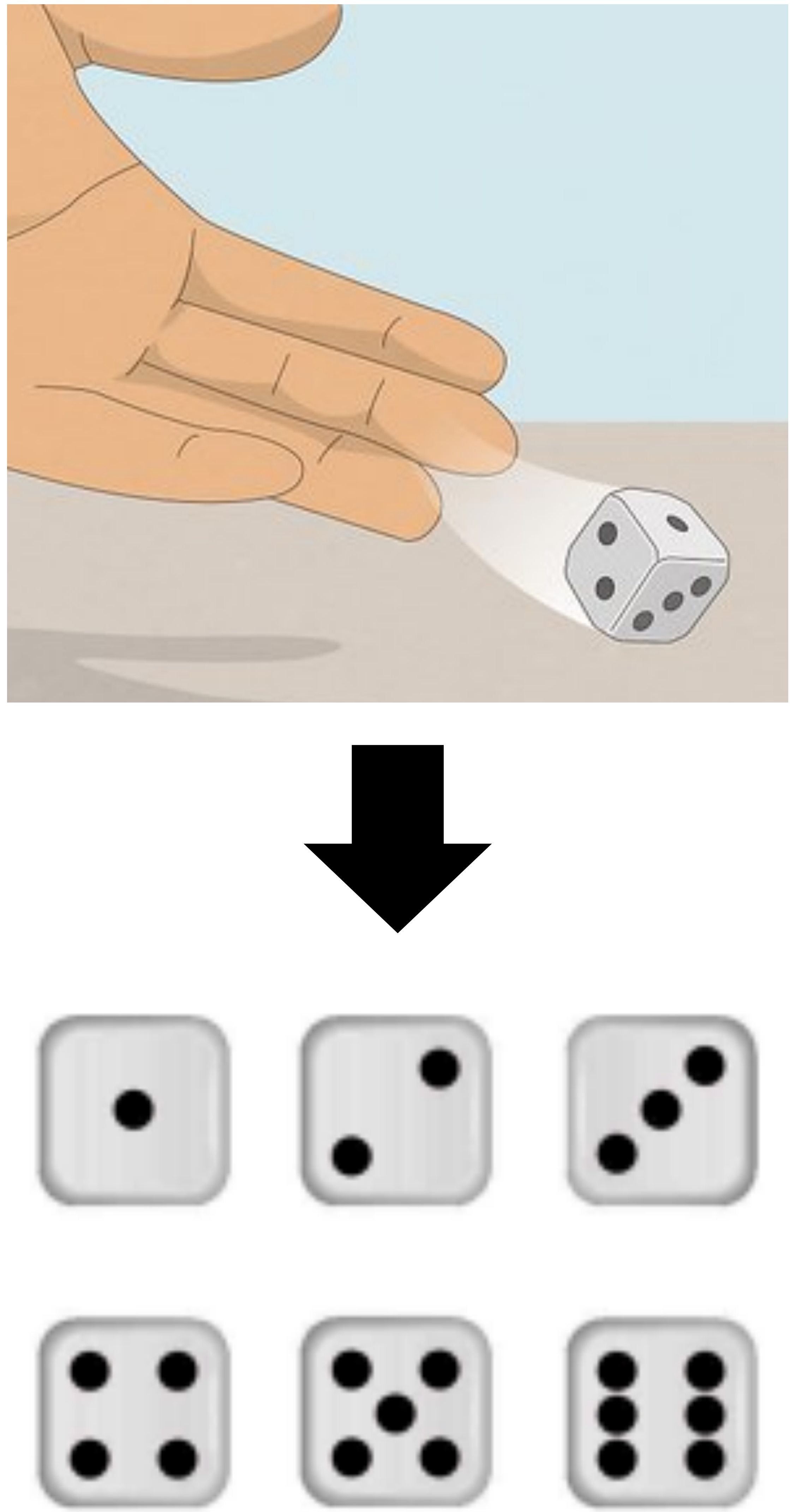
- A **random** circumstance is a circumstance with an uncertain outcome
    - Classic toy examples include flipping a coin, rolling a die
    - Possibly more relevant: whether it will rain tomorrow, a person's cholesterol level
- The **sample space** (S) is the collection of all possible outcomes of the random circumstance
    - Discrete sample space — "countable"



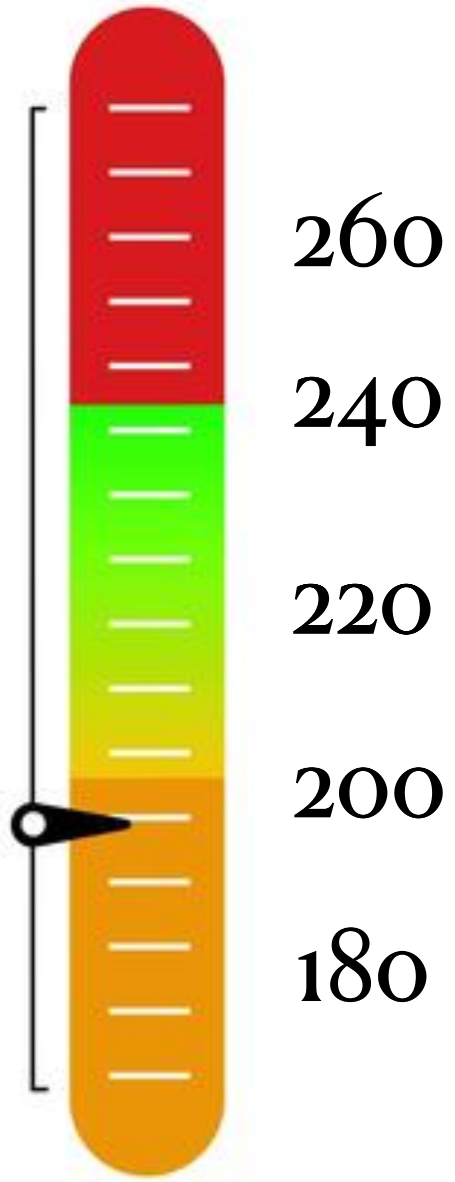


Randomness

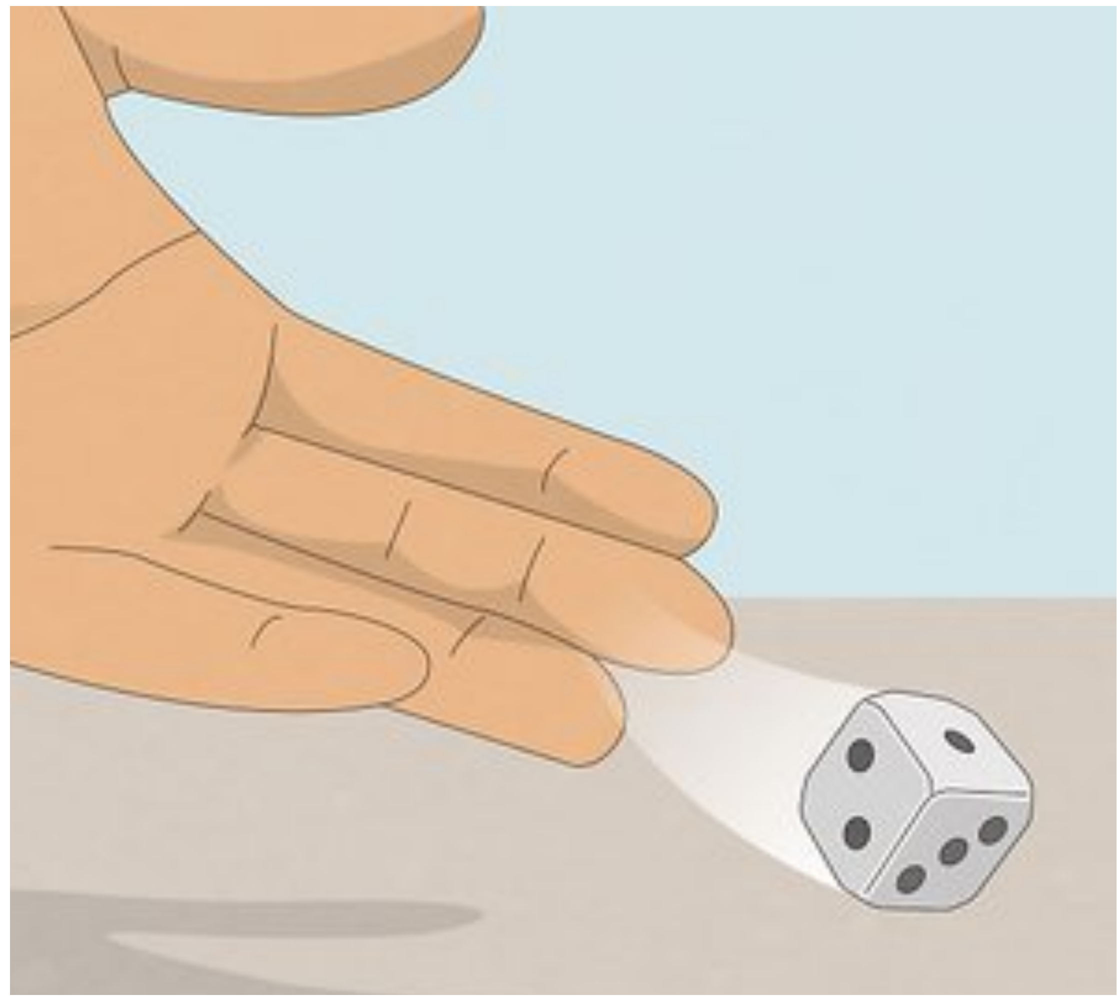
- A **random** circumstance is a circumstance with an uncertain outcome
  - Classic toy examples include flipping a coin, rolling a die
  - Possibly more relevant: whether it will rain tomorrow, a person's cholesterol level
- The **sample space** (S) is the collection of all possible outcomes of the random circumstance
  - Continuous sample space — "measurable"

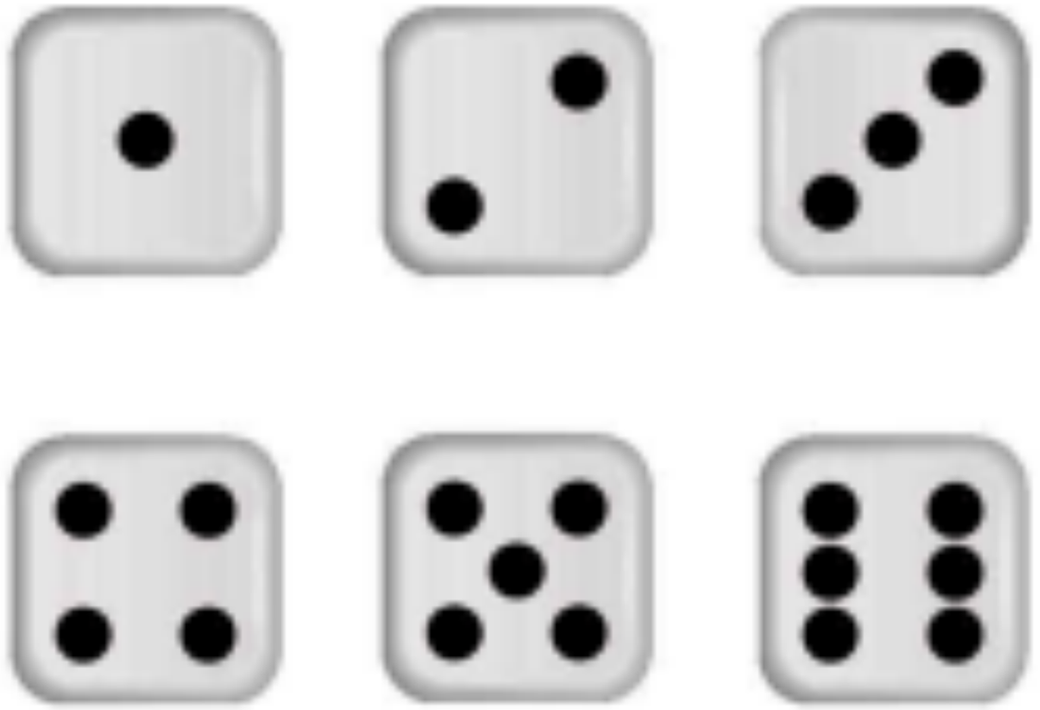


Randomness



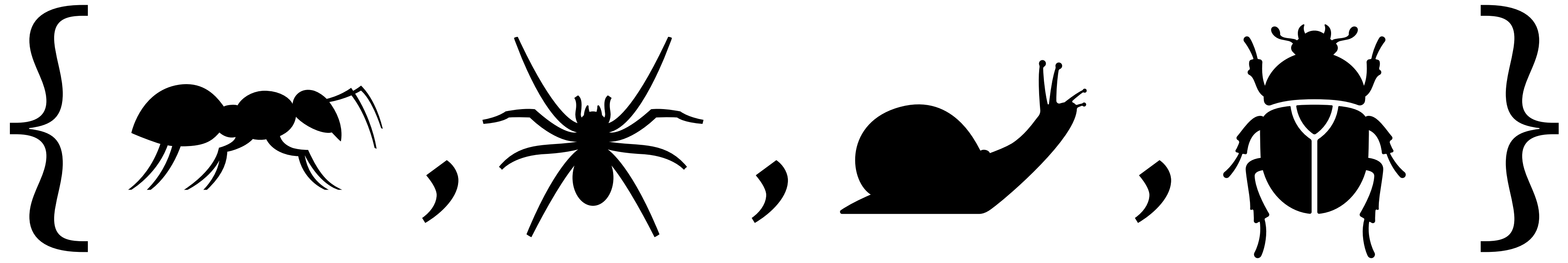
- A **random** circumstance is a circumstance with an uncertain outcome
  - Classic toy examples include flipping a coin, rolling a die
  - Possibly more relevant: whether it will rain tomorrow, a person's cholesterol level
- The **sample space** (S) is the collection of all possible outcomes of the random circumstance
  - (Discrete vs. continuous... is this ringing any bells from when we talked about quantitative variables in Topic 2?)





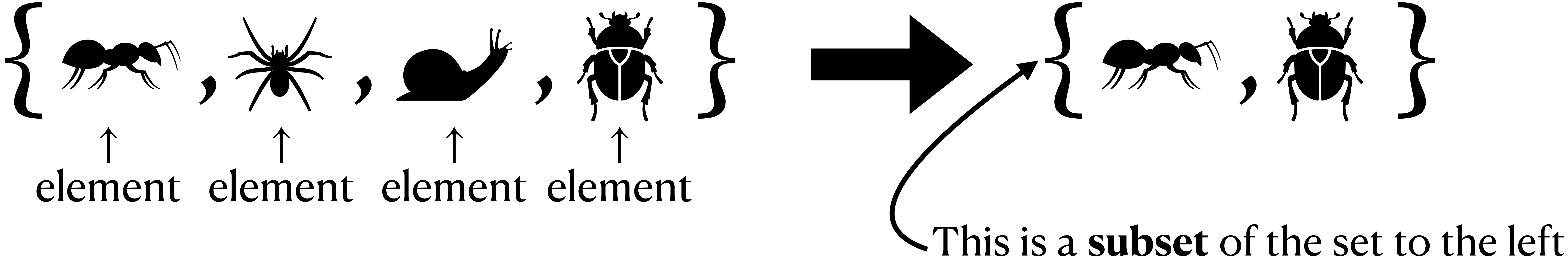
Intro Set Theory

- A **set** is a collection of things
  - A thing in the set is called an **element**
- Usually write it as a list in curly brackets: {1,2,3,4,5,6}
- Recall I defined the **sample space** as the collection of all possible outcomes of a random circumstance
  - The sample space is a **set.** Flip a coin twice? The **sample space** is S = {HH, HT, TH, TT}
- Later, we will talk about **probability** as being a measure of the size of a set



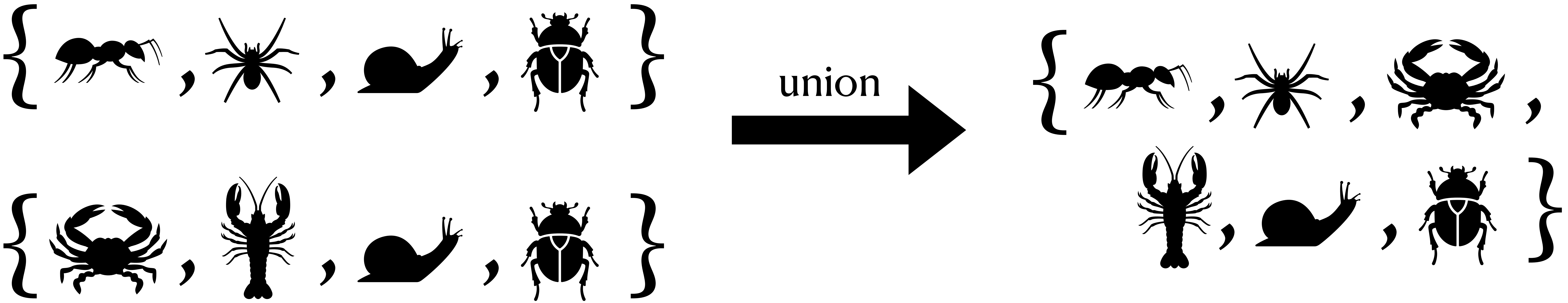
Subsets

- A **set** is a collection of things
  - A thing in the set is called an **element**
- A **subset** of another set is a set that contains only the elements also in the other set



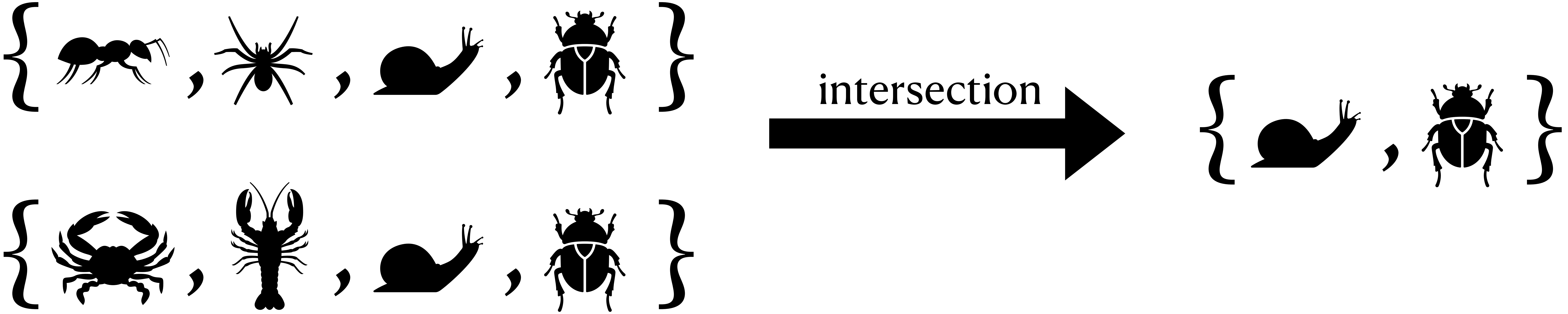
Unions of Sets

- A **set** is a collection of things
  - A thing in the set is called an **element**
- The **union** of multiple sets is the combination of all unique elements from multiple sets into a new set
  - Q: Is this element in set A **or** set B? If yes, then it is in the union.

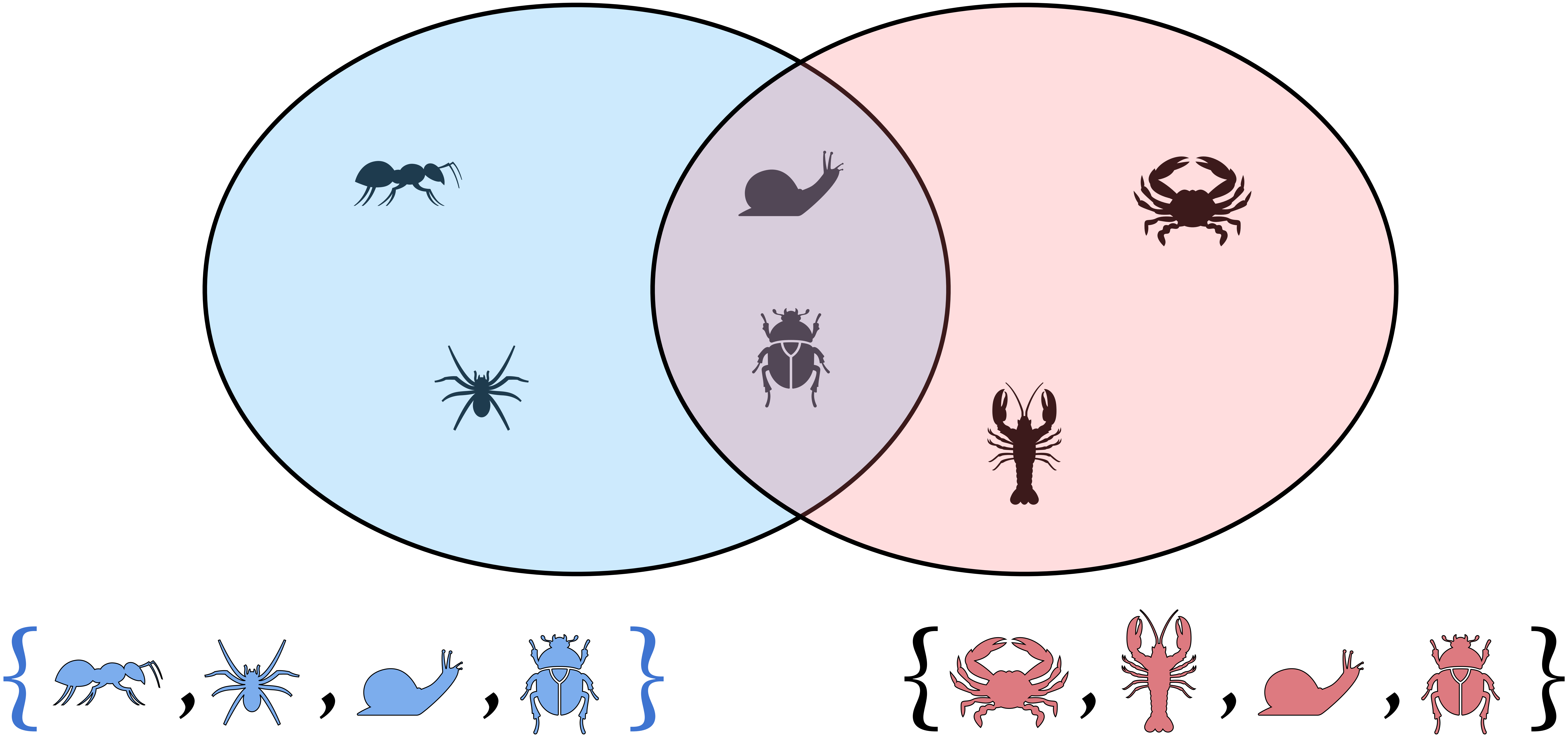


Intersections of Sets

- A **set** is a collection of things
  - A thing in the set is called an **element**
- The **intersection** of multiple sets is a new set containing only the elements common to all sets
  - Q: Is the element in set A **and** set B? If yes, then it is in the intersection.

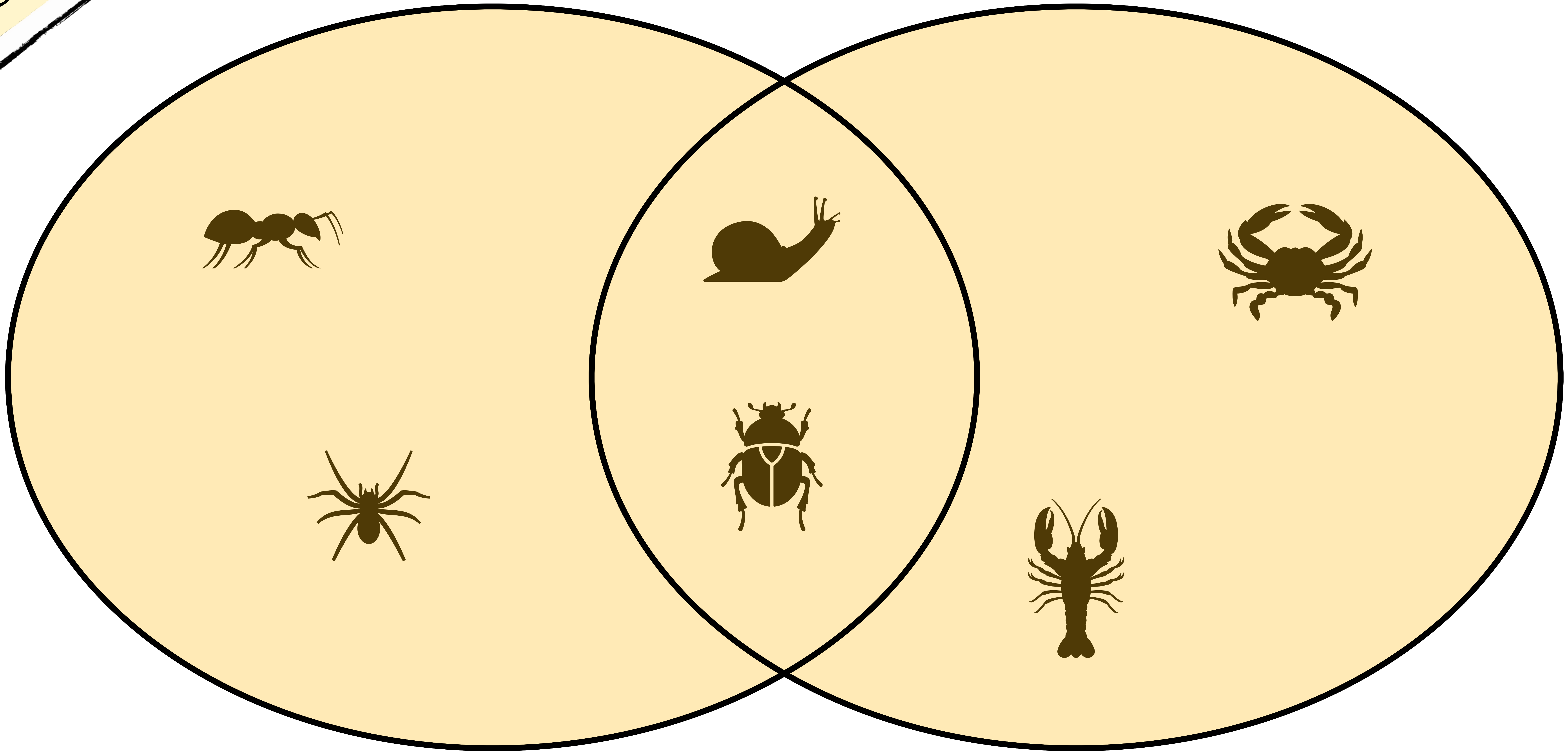


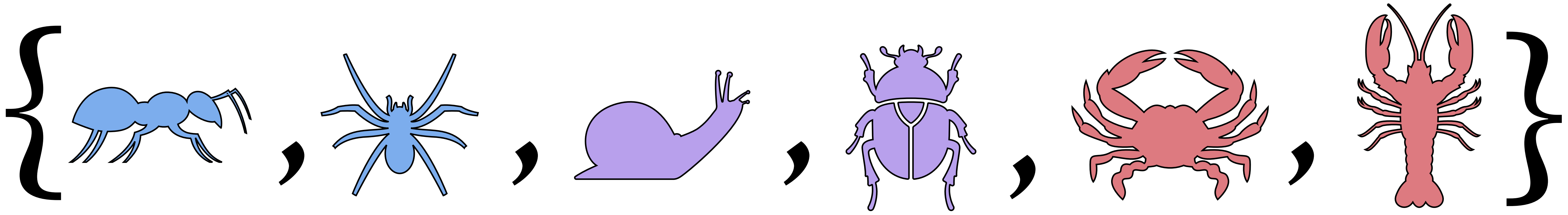
Venn diagrams for sets



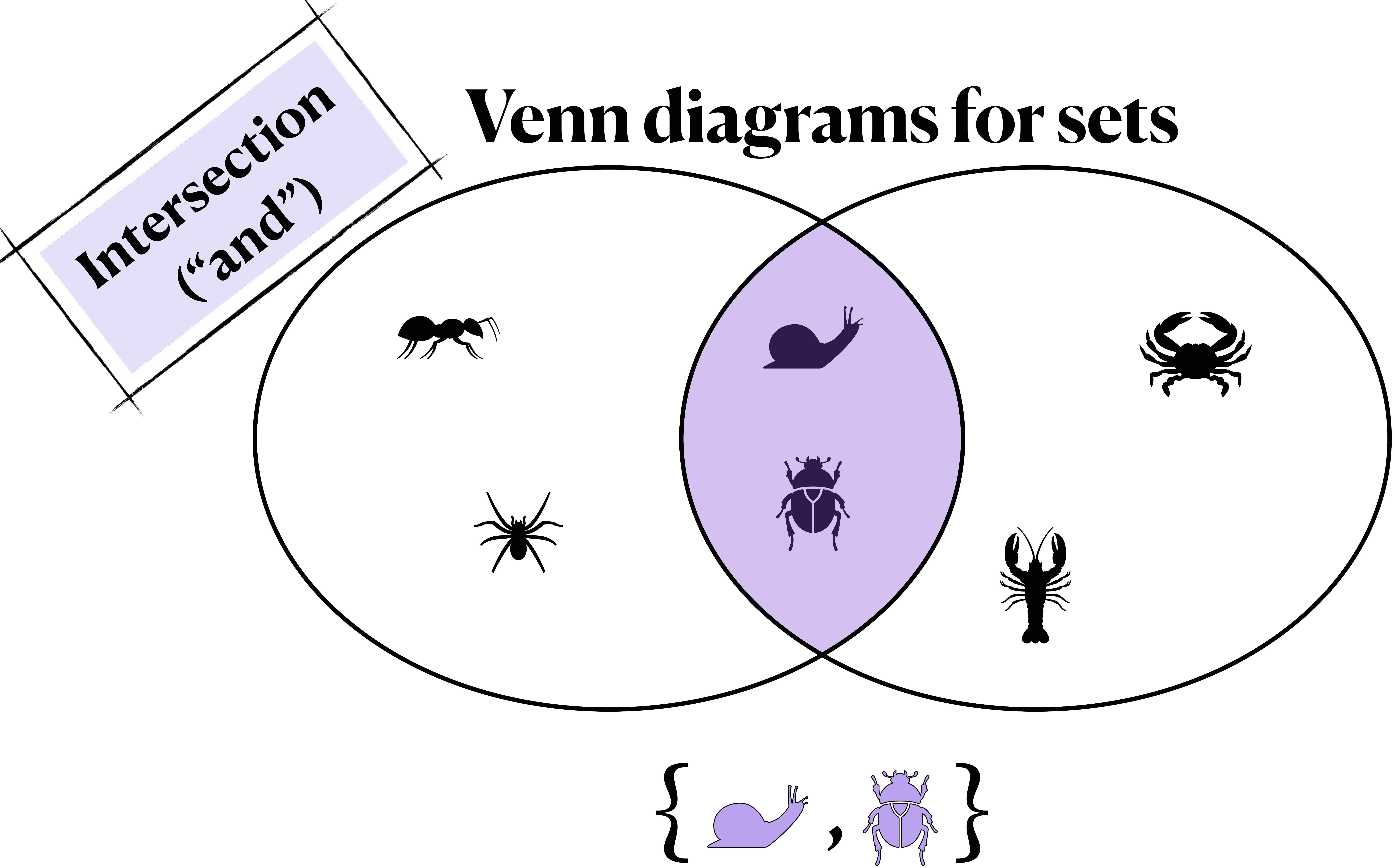
Venn diagrams for sets

Union ("or")





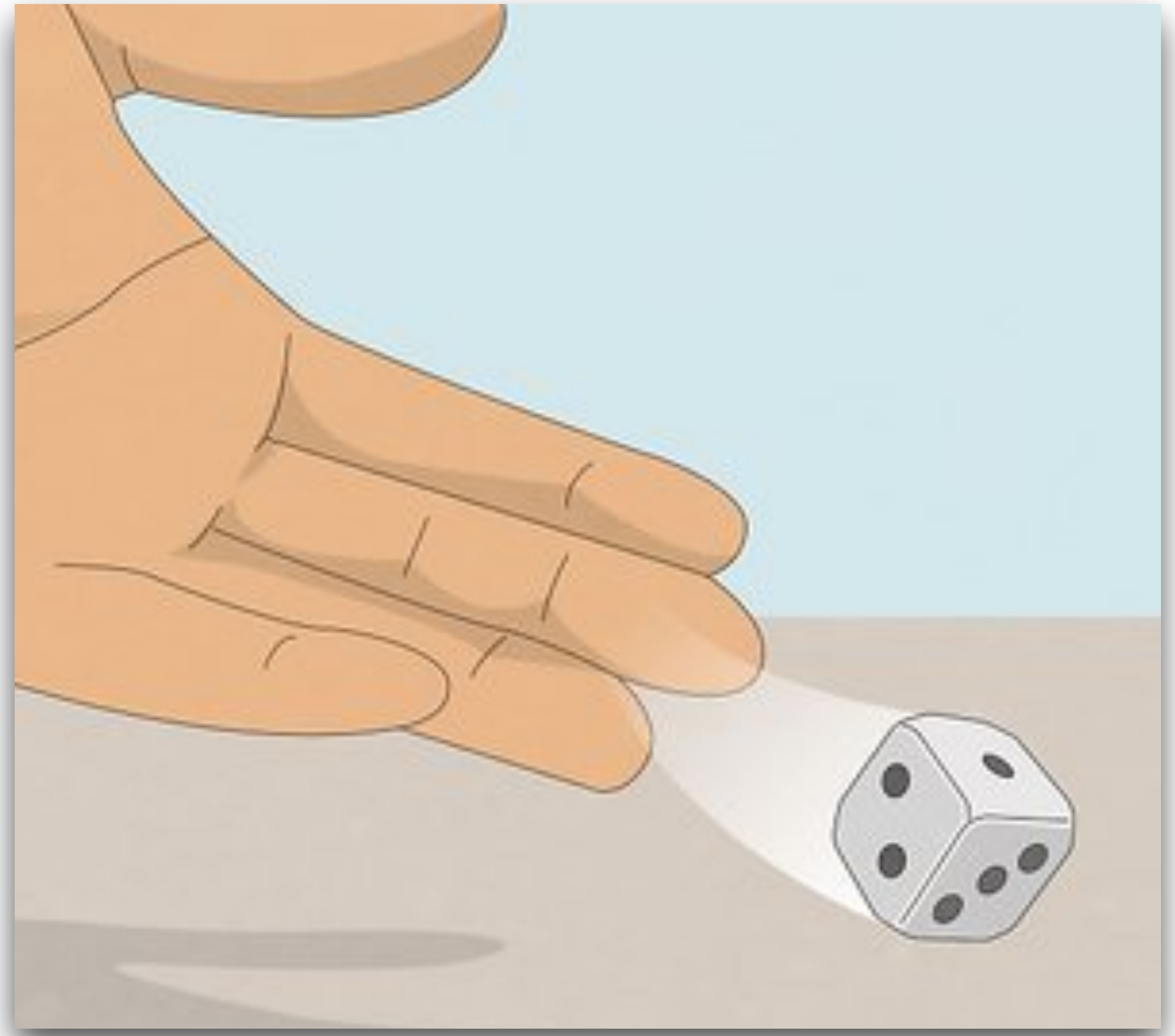
Venn diagrams for sets



Practice talking about sets…

PollEv.com/stats8

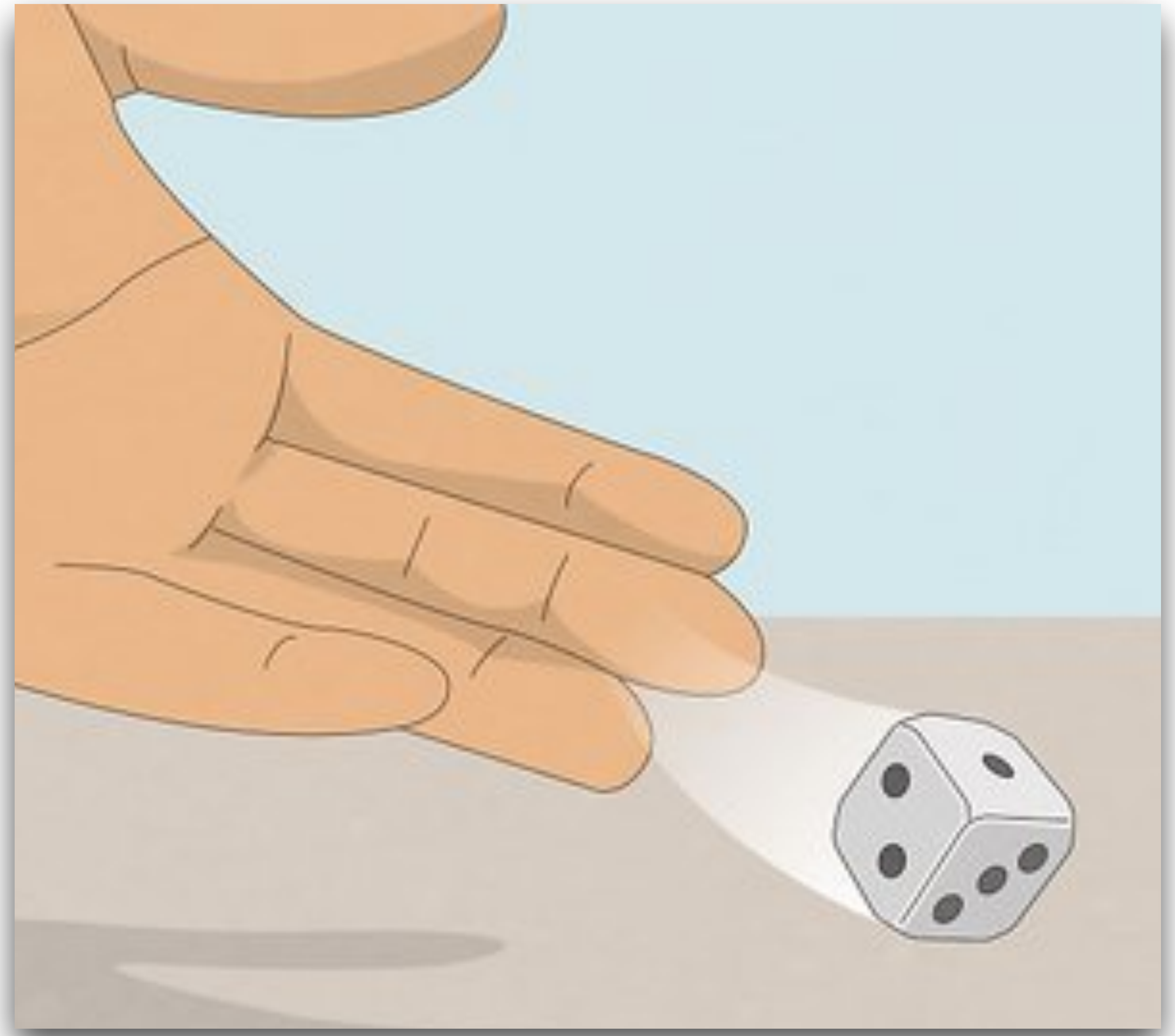
- Random circumstance: roll a fair, six-sided die.
- What is the **sample space** of this random circumstance? Write it as a set, as in the examples on the previous slides.



Practice talking about sets...

PollEv.com/stats8

- Random circumstance: roll a fair, six-sided die.
- What is the **sample space** of this random circumstance? Write it as a set, as in the examples on the previous slides.



{1, 2, 3, 4, 5, 6}

Practice talking about sets…

PollEv.com/stats8

- Random circumstance: roll a fair, six-sided die.
- Which of the following sets is a **subset** of the sample space? (Select all that apply)

  A. {1, 2, 3, 4}

  B. {10, 11}

  C. {3}

  D. {1.8, 4, 6}

Practice talking about sets…

PollEv.com/stats8

- Random circumstance: roll a fair, six-sided die.
- Which of the following sets is a **subset** of the sample space? (Select all that apply)

  A. {1, 2, 3, 4}

  B. {10, 11}

  C. {3}

  D. {1.8, 4, 6}

Practice talking about sets…

PollEv.com/stats8

- Random circumstance: roll a fair, six-sided die.
  - Let A be the set of outcomes where the number rolled is even
  - Let B be the set of outcomes where the number rolled is at least 3
- What is the **union** of the sets A and B?

Practice talking about sets…

PollEv.com/stats8

- Random circumstance: roll a fair, six-sided die.
  - Let A be the set of outcomes where the number rolled is even
  - Let B be the set of outcomes where the number rolled is at least 3
- What is the **union** of the sets A and B?

{2, 3, 4, 5, 6}

Practice talking about sets…

PollEv.com/stats8

- Random circumstance: roll a fair, six-sided die.
  - Let A be the set of outcomes where the number rolled is even
  - Let B be the set of outcomes where the number rolled is at least 3
- What is the **intersection** of the sets A and B?

Practice talking about sets...

PollEv.com/stats8

- Random circumstance: roll a fair, six-sided die.
  - Let A be the set of outcomes where the number rolled is even
  - Let B be the set of outcomes where the number rolled is at least 3
- What is the **intersection** of the sets A and B?

{4, 6}

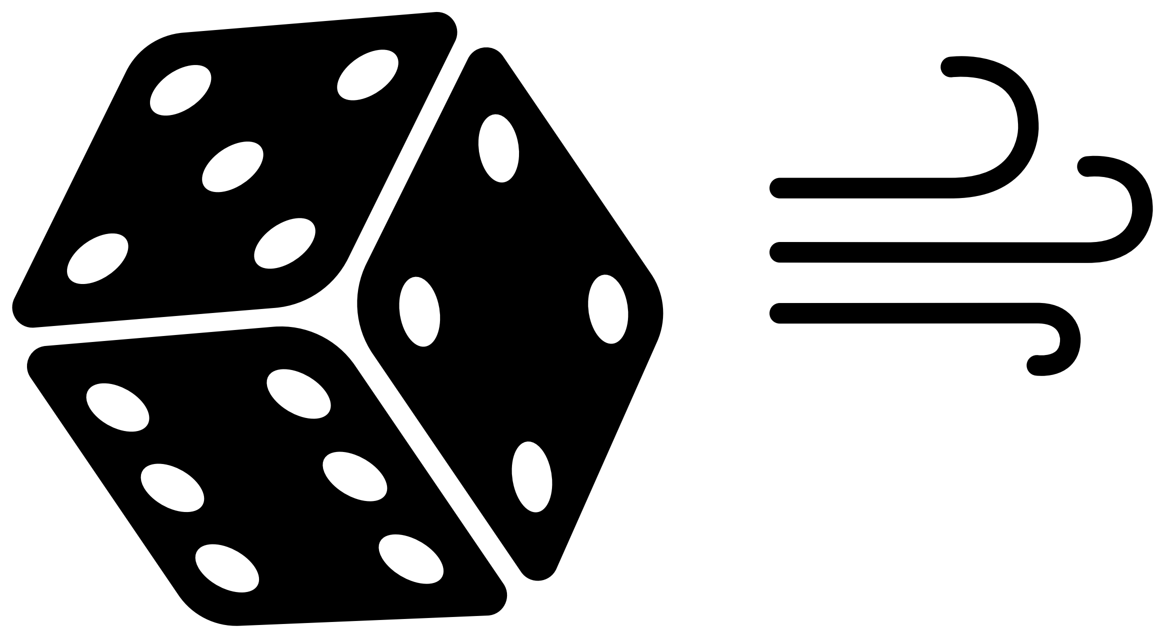
Probability

- Mathematically, **probability** is a number between 0 and 1 that is assigned to possible outcomes of a random circumstance
- In practice, **probability** should give us information on how likely it is that particular outcomes will be the result of a random circumstance
  - Wait, we were just talking about set theory, where did this come from?
    - We talk about probability as being a measure of the relative "size" of a set (compared to the size of the sample space)
    - We'll start abstract, then get more concrete

Set theory → Probability

a.k.a., why did we talk about this?

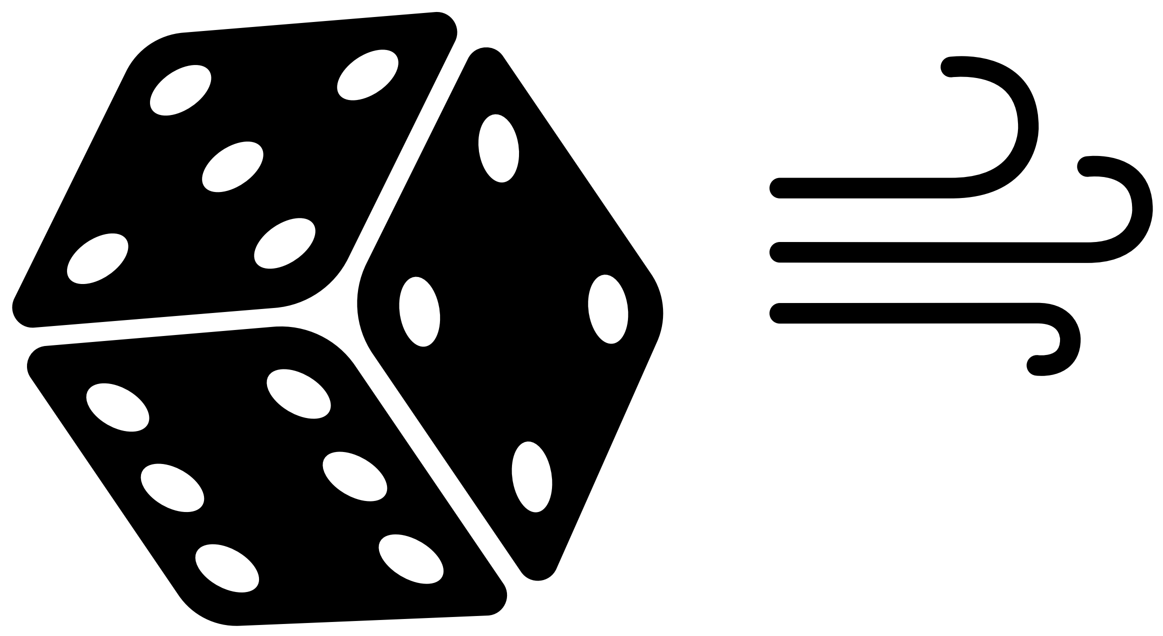
- We talk about **probability** as being a measure of the "size" of a set
- An **event** is a subset of the sample space, so to find the probability of the event we talk about the size of the subset
  - Case 1: all outcomes in the sample space have the same probability
  - Case 2: outcomes in the sample space have different probability
- Let's use the simple example of rolling a die again



Set theory → Probability

a.k.a., why did we talk about this?

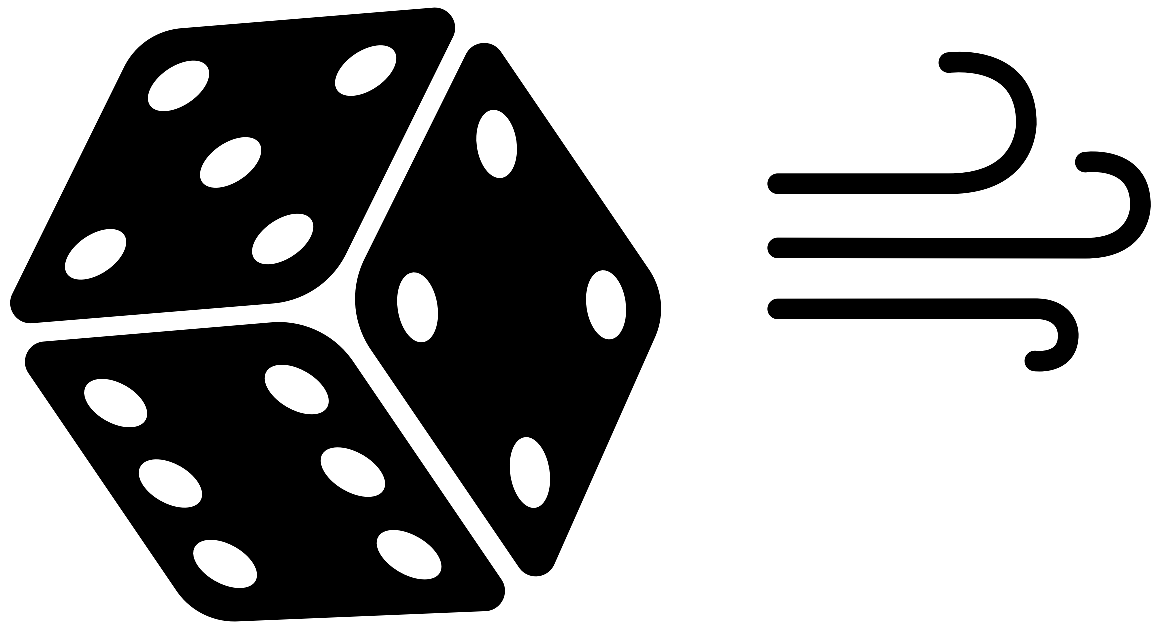
- We talk about **probability** as being a measure of the "size" of a set
- An **event** is a subset of the sample space, so to find the probability of the event we talk about the size of the subset
    - **Case 1: all outcomes in the sample space have the same probability**
    - Case 2: outcomes in the sample space have different probability
- Let's use the simple example of rolling a die again



Probability

Outcomes with equal probabilities

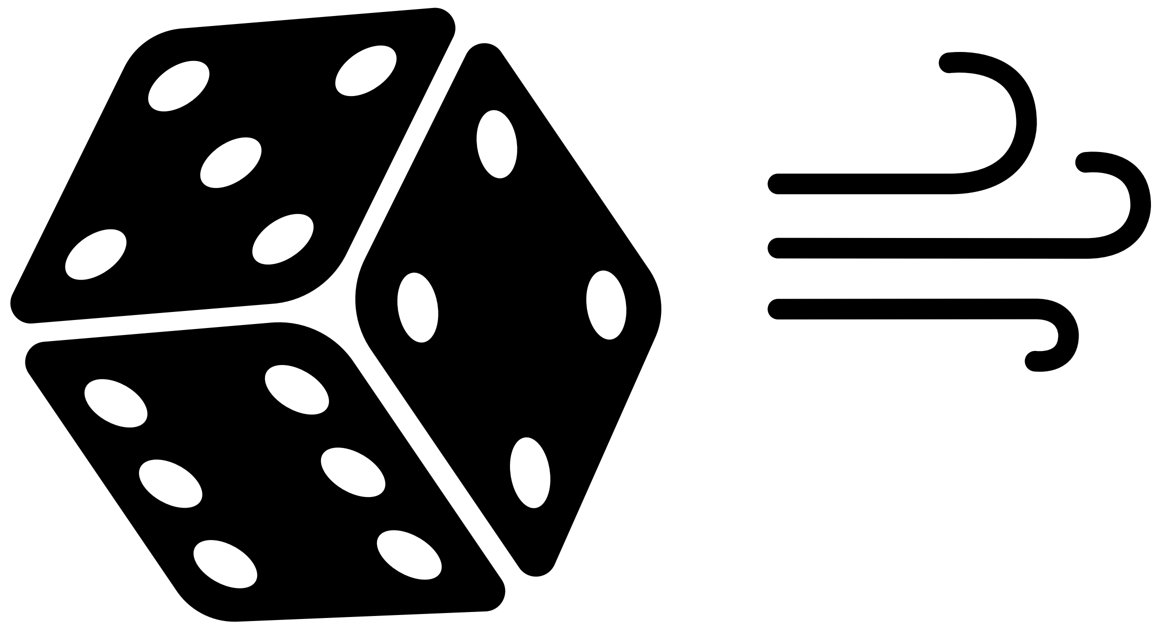
- What is the probability of getting an <u>even number</u> when you roll a fair die?
  - S = {1, 2, 3, 4, 5, 6}
  - The subset of interest, the “event” = {2, 4, 6}
  - Probability measures the “size” of the **event** relative to the size of the **sample space**
    - When all elements in the sample space have equal probability, the probability of the event is just the proportion of the sample space the event takes up
    - Because all 6 outcomes are equally likely, the event takes up 3 of the 6 equally likely “pieces” of the sample space
      - Probability = 3 / 6 = 0.5 or 50%



Probability

Outcomes with equal probabilities

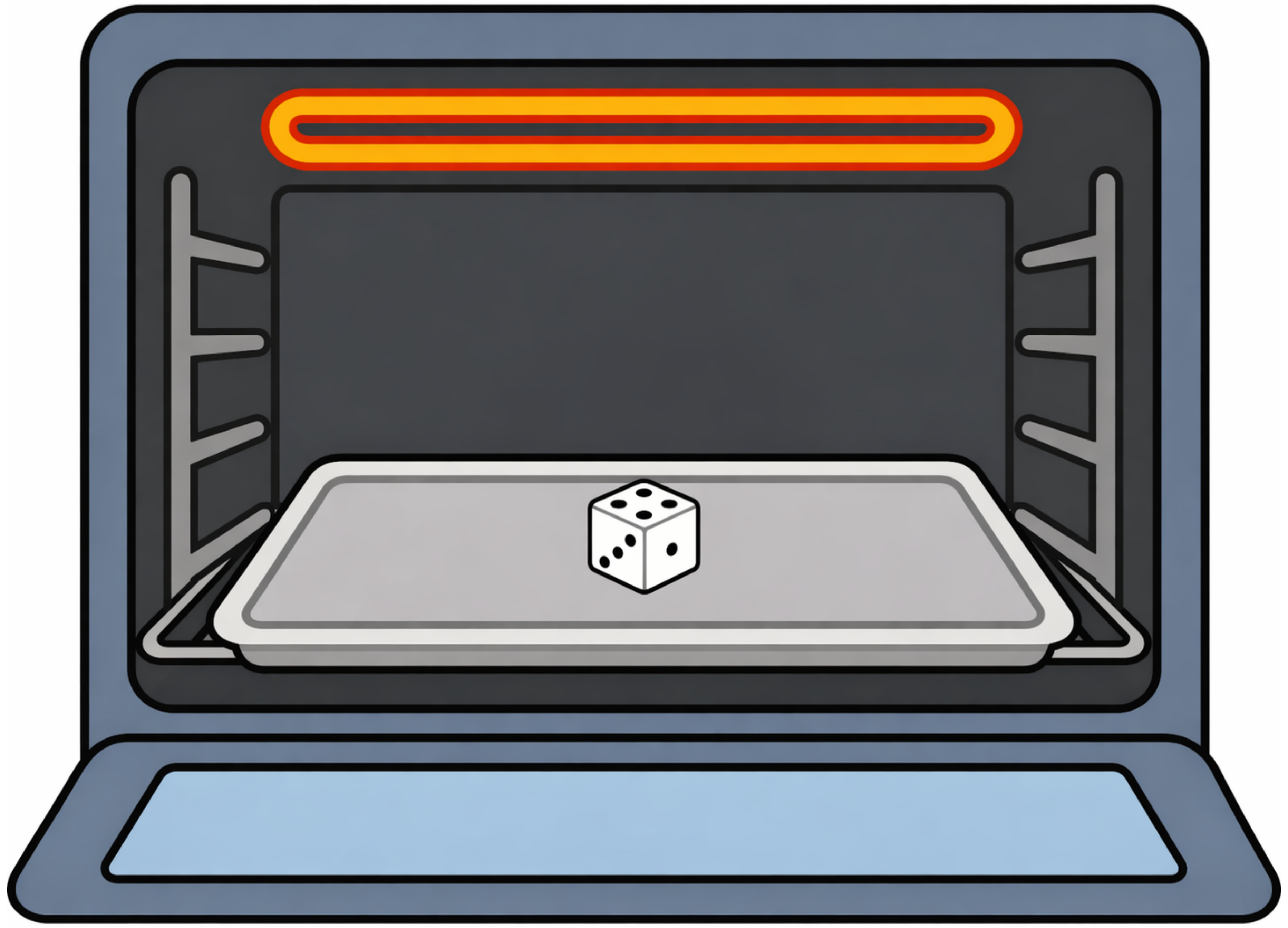
- What is the probability of getting some number, any number, when you roll a fair die?
  - S = {1, 2, 3, 4, 5, 6}
  - The subset of interest, the "event" = {1, 2, 3, 4, 5, 6}
  - Probability measures the "size" of the **event** relative to the "size" of the **sample space**
    - The event is the entire sample space, so the event takes up all of the sample space
    - The probability of the sample space is 1, or 100%



Probability

Outcomes with ~~equal~~ probabilities

- Fun fact if you didn't know, you can make an unfair die (a "loaded" die) by baking it in the oven

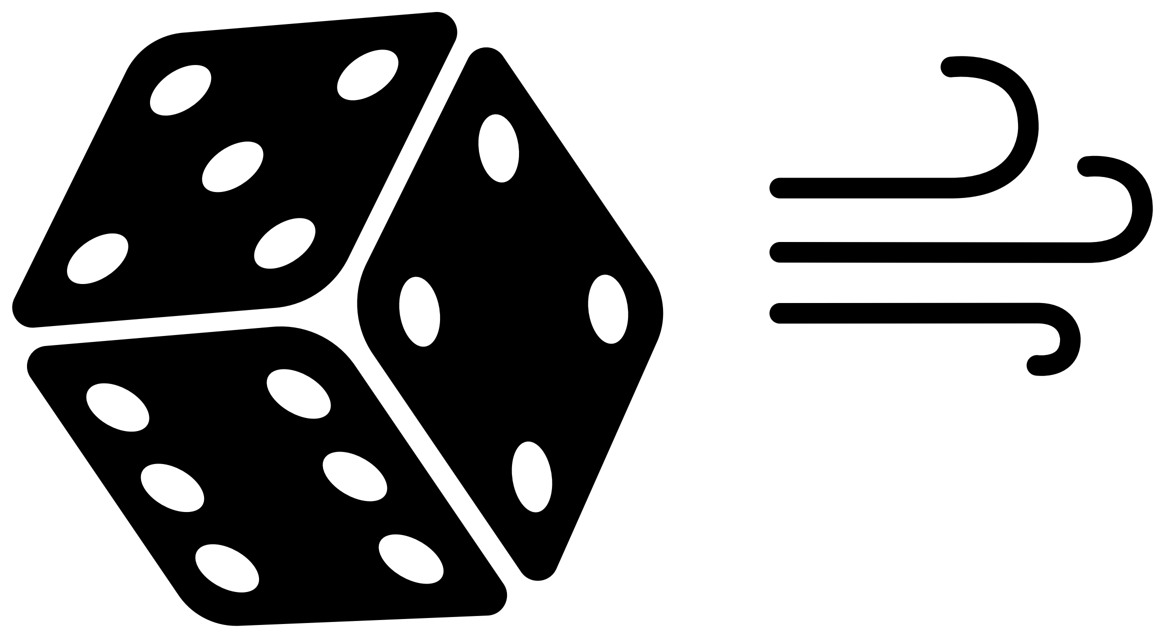


- What if I do this, and the 6 outcomes in the sample space have different probabilities?

Set theory → Probability

a.k.a., why did we talk about this?

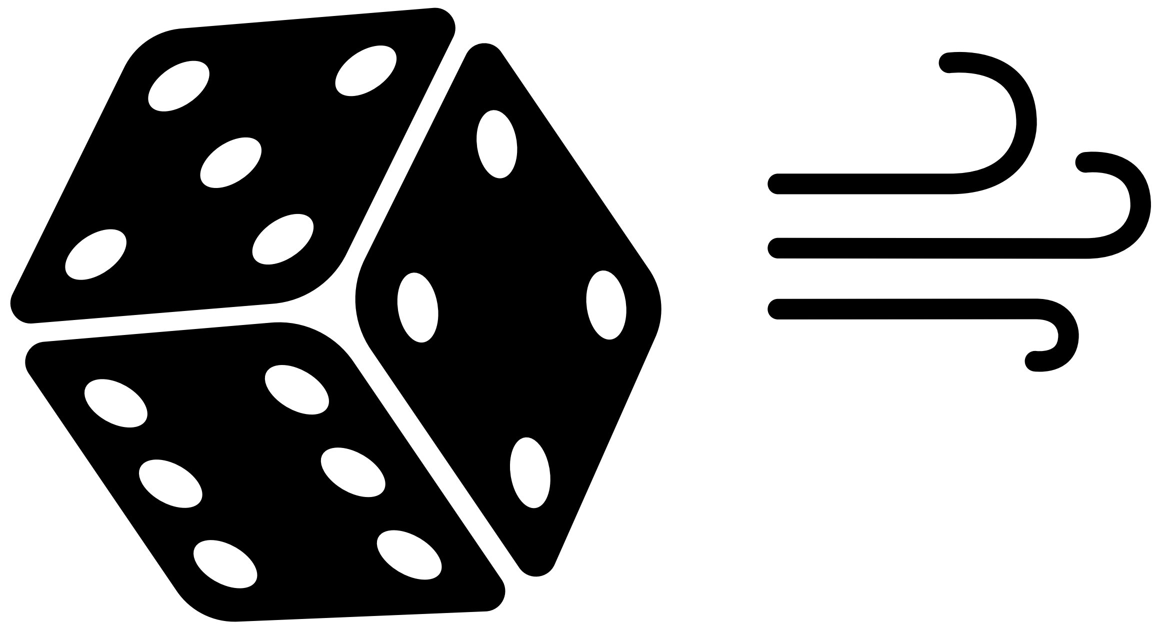
- We talk about **probability** as being a measure of the "size" of a set
- An **event** is a subset of the sample space, so to find the probability of the event we talk about the size of the subset
    - Case 1: all outcomes in the sample space have the same probability
    - **Case 2: outcomes in the sample space have different probability**
- Let's use the simple example of rolling a die again (but now we've baked it)



Probability

Outcomes with unequal probabilities

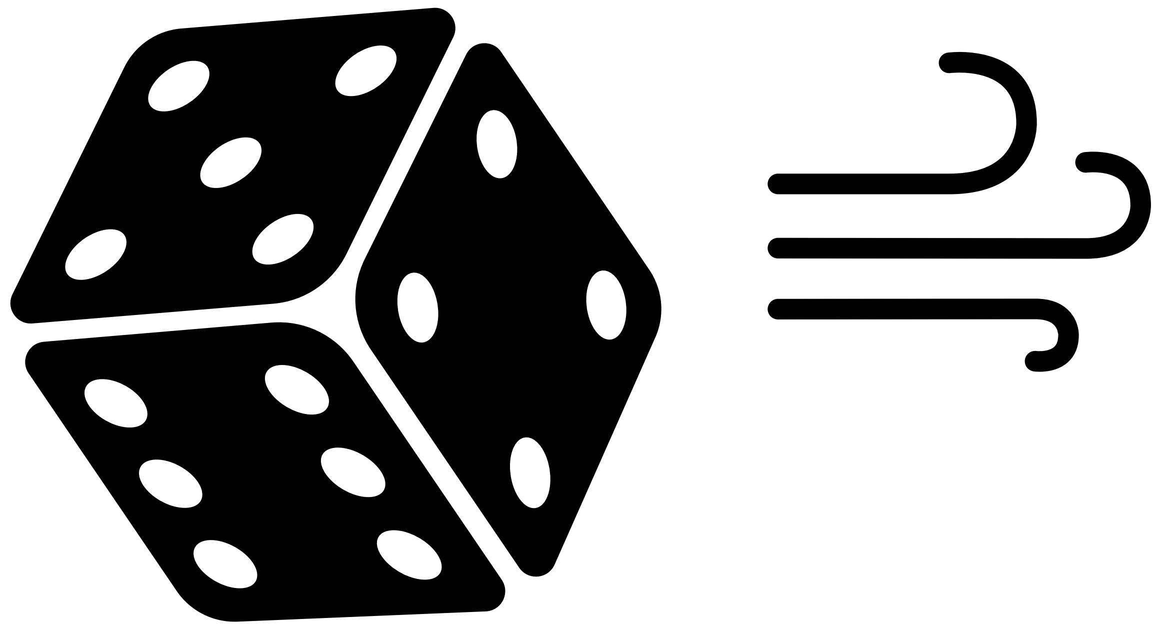
- What is the probability of getting some number, any number, when you roll an **unfair** die?
  - S = {1, 2, 3, 4, 5, 6}
  - The subset of interest, the "event" = {1, 2, 3, 4, 5, 6}
  - Probability measures the "size" of the **event** relative to the "size" of the **sample space**
    - The event is *still* the whole sample space, so it has probability 1



Probability

Outcomes with unequal probabilities

- What is the probability of getting an even number when you roll an **unfair** die?
  - S = {1, 2, 3, 4, 5, 6}; probability of S is 1 or 100%
  - The subset of interest, the "event" = {2, 4, 6}
- Before, we could measure relative size by counting outcomes, but now, we can't
  - 1, 2, 3, 4, 5, and 6 aren't equally likely to appear!
  - The **event** is still a set, and **probability** is still its size, but now the 3 outcomes ***do not contribute equally to that size***
  - How do we figure out those unequal probabilities?



Frequentist Interpretation of Probability

- In practice, **probability** should give us information on how likely it is that a particular event will be the result of a random circumstance

* Frequentist interpretation of probability — this will be the definition we take for this class, and it will shape all of the methods we learn in the future:
    - Under this interpretation, the **probability** of an event is the limit of its relative frequency in infinitely many trials
        - Let's break this down with the loaded die

$$\lim_{n \to \infty}$$

Probability

Frequentist Interpretation

- **Probability** of an event is the limit of its relative frequency in infinitely many trials
- What is the probability of getting a 3 when you roll an *unfair* die, where you don't know how much each side is weighted?
  - Roll the die repeatedly, and record:
    - Number of 3s
    - Number of rolls
  - The more you roll, the more the fraction $\frac{\text{(number of 3s)}}{\text{(number of rolls)}}$ approaches the **probability** of rolling 3

Probability

Main ideas review

- In practice, **probability** should give us information on how likely it is that a particular event will be the result of a random circumstance
    - It is a measure of the "size" of the event relative to the size of the sample space
- The **probability** of an event is <u>the limit of its relative frequency in infinitely many trials</u>
    - Abstractly, the probability is the relative "size" of the event
    - Practically, doing many trials reveals that relative "size" to us: the long-run relative frequency of the event

$$\lim_{n \to \infty}$$

Probability

PollEv.com/stats8

- Random circumstance: suppose I randomly select one student from this class, where every student has the same probability of being chosen (**simple random sample**, anyone?).
- Define an event "A" to be the event that that person is left-handed. Suppose that in our class of 213, 12 people are left-handed.
- The probability of event A is 12 / 213.
  - Why is the denominator 213?

Probability

PollEv.com/stats8

- Random circumstance: suppose I randomly select one student from this class, where every student has the same probability of being chosen (**simple random sample**, anyone?).
- Define an event “A” to be the event that that person is left-handed. Suppose that in our class of 213, 12 people are left-handed.
- The probability of event A is 12 / 213.
  - Why is the denominator 213?
  - It is the size of the sample space, and probabilities are sizes relative to the sample space

Probability

Simple Random Sample (SRS)

- Then, with simple random sampling, probability is just proportion in the group
- This is exactly what we will do with real data next:
  - We will sample one individual from a group of study participants
    - Our sample space will be all individuals in the study
    - Events will be subsets of those individuals with a particular quality (measurement of a **variable**)
    - Probabilities will be proportions of the people in the study, because each person is equally likely to be selected

Giving Birth with COVID-19 in 2020-21

Running example

- JAMA cohort study of people giving birth between March 2020 and February 2021 at ~500 US hospitals (source)
- The **contingency table** below summarizes the results

	Had COVID during birth	Did not have COVID during birth	Total
Admitted to ICU	977	7943	8920
Not admitted to ICU	17738	842241	859979
Total	18715	850184	868899

Probability

Notation

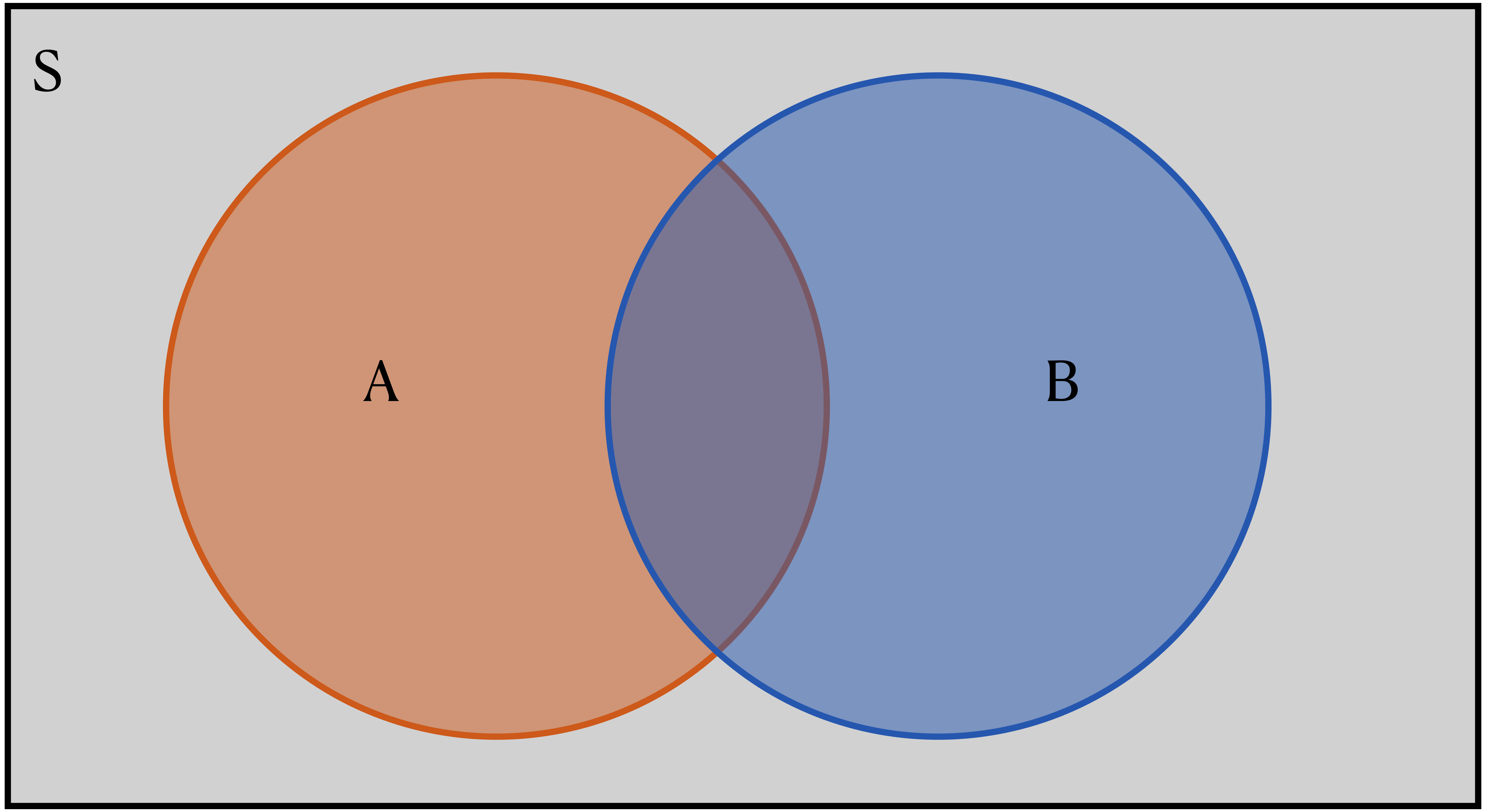
- Notation for "probability of" is $P()$
- Often denote events with capital letters
- E.g.: suppose I randomly select an individual from the COVID/ICU JAMA study
  - Let A = the event that the person I select had COVID during birth
  - $P(A)$ = the probability that the person I select had COVID during birth

	Had COVID during birth	Did not have COVID during birth	Total
Admitted to ICU	977	7943	8920
Not admitted to ICU	17738	842241	859979
Total	18715	850184	868899

Probability Concepts

Venn diagrams for probability

- Areas represent probabilities
- Total area of the box = 1, because P(S) = 1

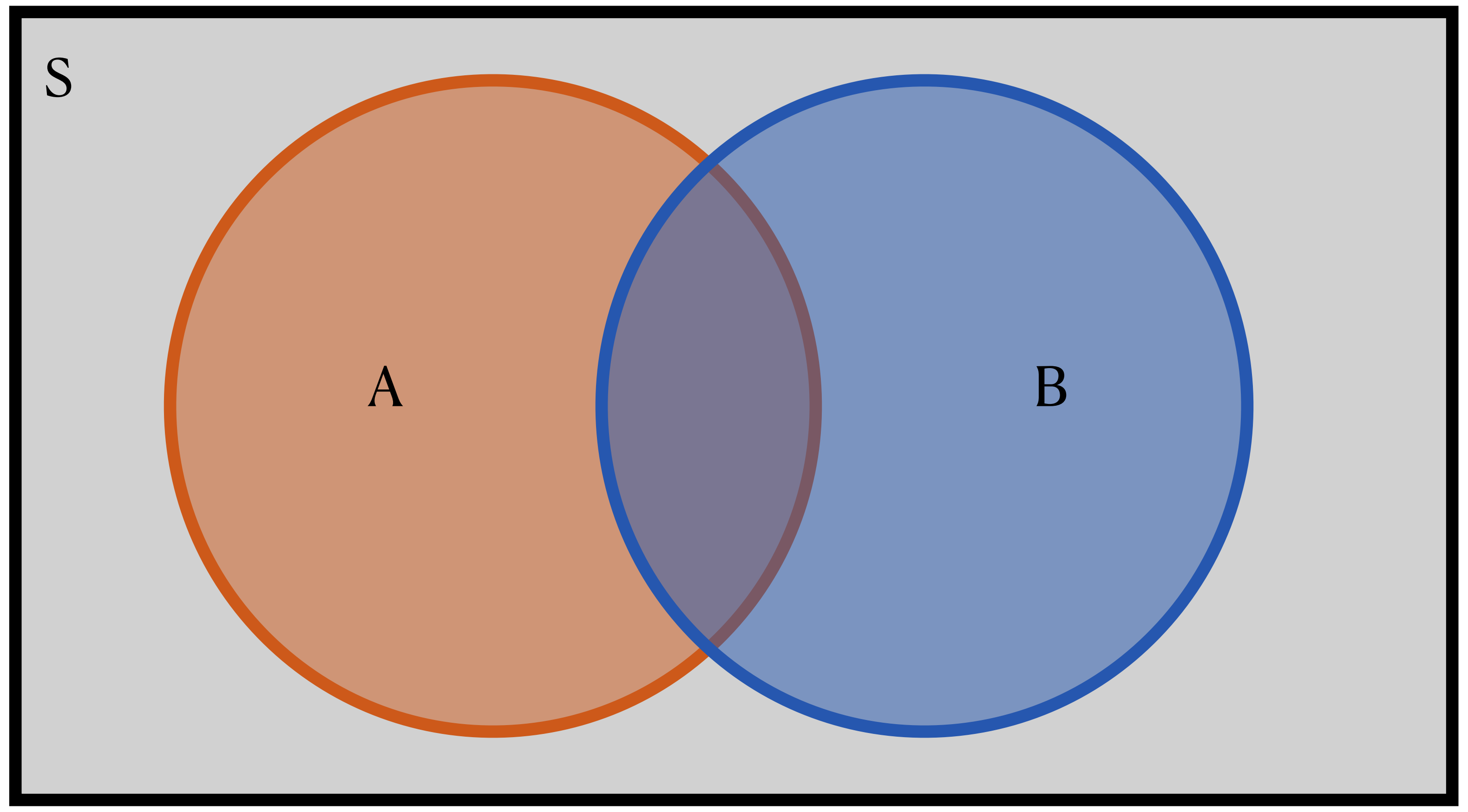


- P(A) = area of the orange circle
- P(B) = area of the blue circle

Probability Concepts

Defining events

- Let the **random** circumstance be me selecting a random person from that study, where everyone has equal probability of being selected
  - Let event A be the event that the randomly selected person had COVID during birth
  - Let event B be the event that the randomly selected patient ended up being admitted to the ICU

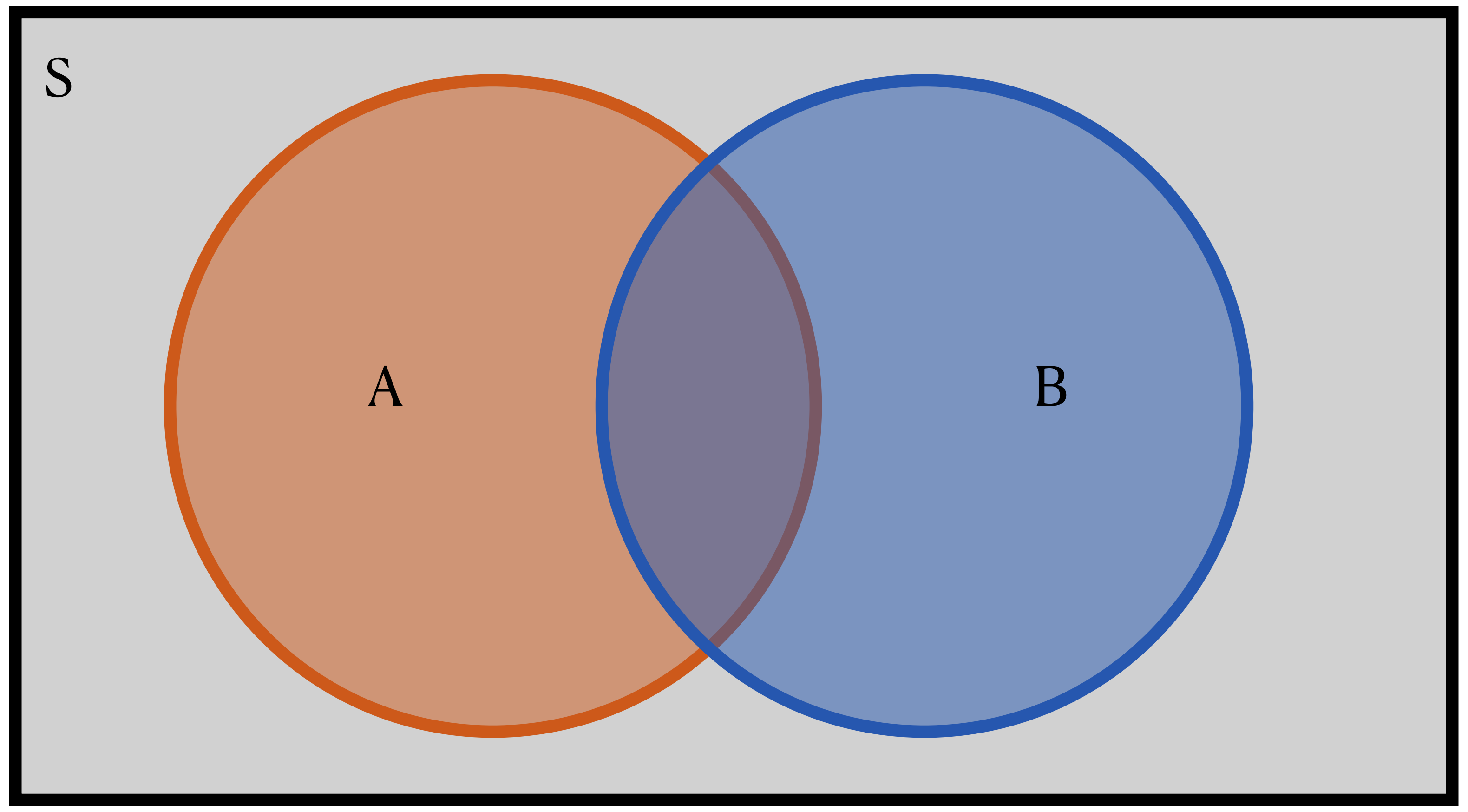


	Had COVID during birth	**Did not have COVID during birth**	**Total**
Admitted to ICU	977	7943	8920
Not admitted to ICU	17738	842241	859979
Total	18715	850184	868899

Probability Concepts

Sample space

- Let the **random** circumstance be me selecting a random person from that study, where everyone has equal probability of being selected
  - What is the **sample space**?

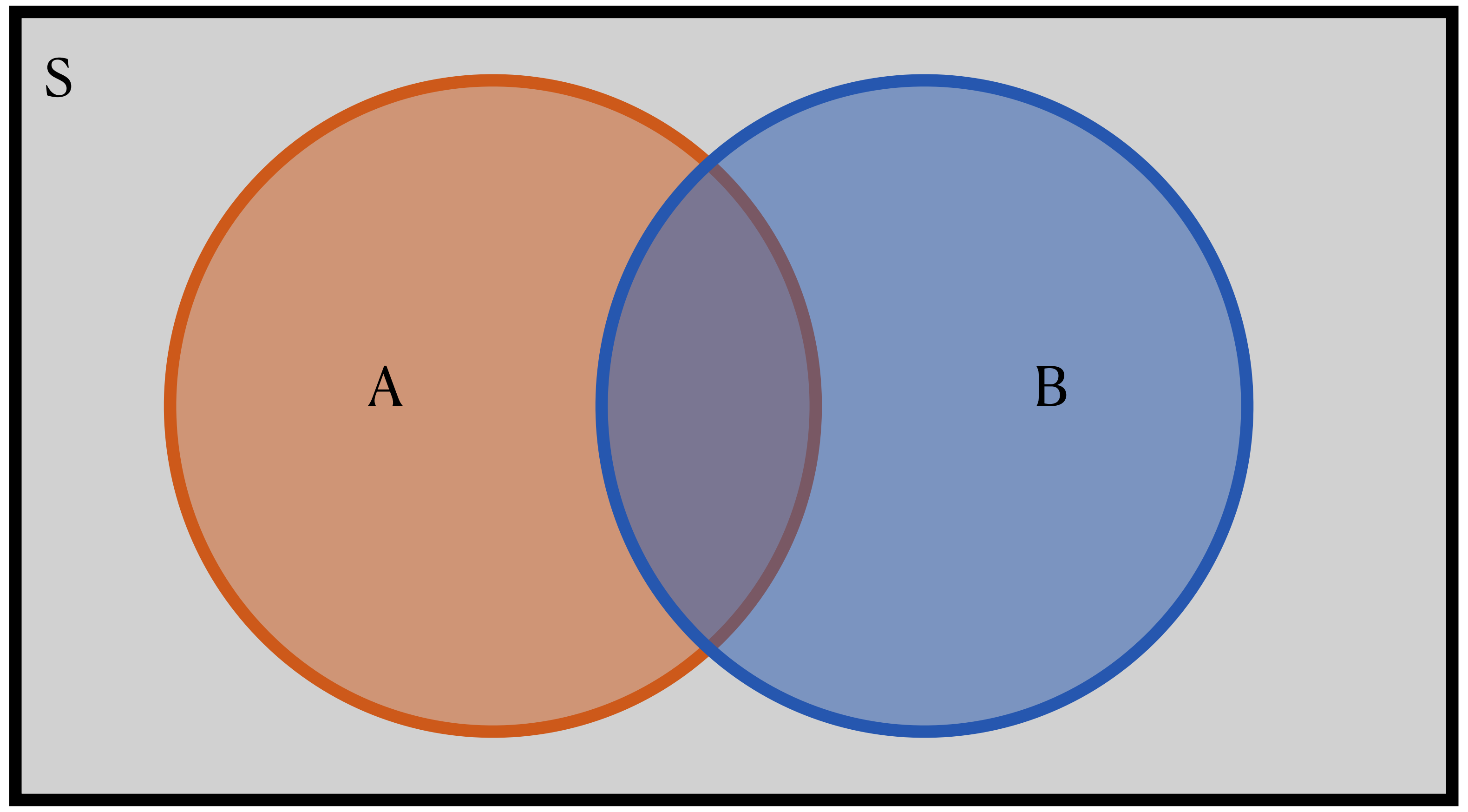


	Had COVID during birth	Did not have COVID during birth	Total
Admitted to ICU	977	7943	8920
Not admitted to ICU	17738	842241	859979
Total	18715	850184	868899

Probability Concepts

Sample space

- Let the **random** circumstance be me selecting a random person from that study, where everyone has equal probability of being selected
  - What is the **sample space**?
    - The set of all people in this study (roughly, the people who gave birth in 2020 in one of 500 hospitals in the US)
  - How big is the **sample space**?

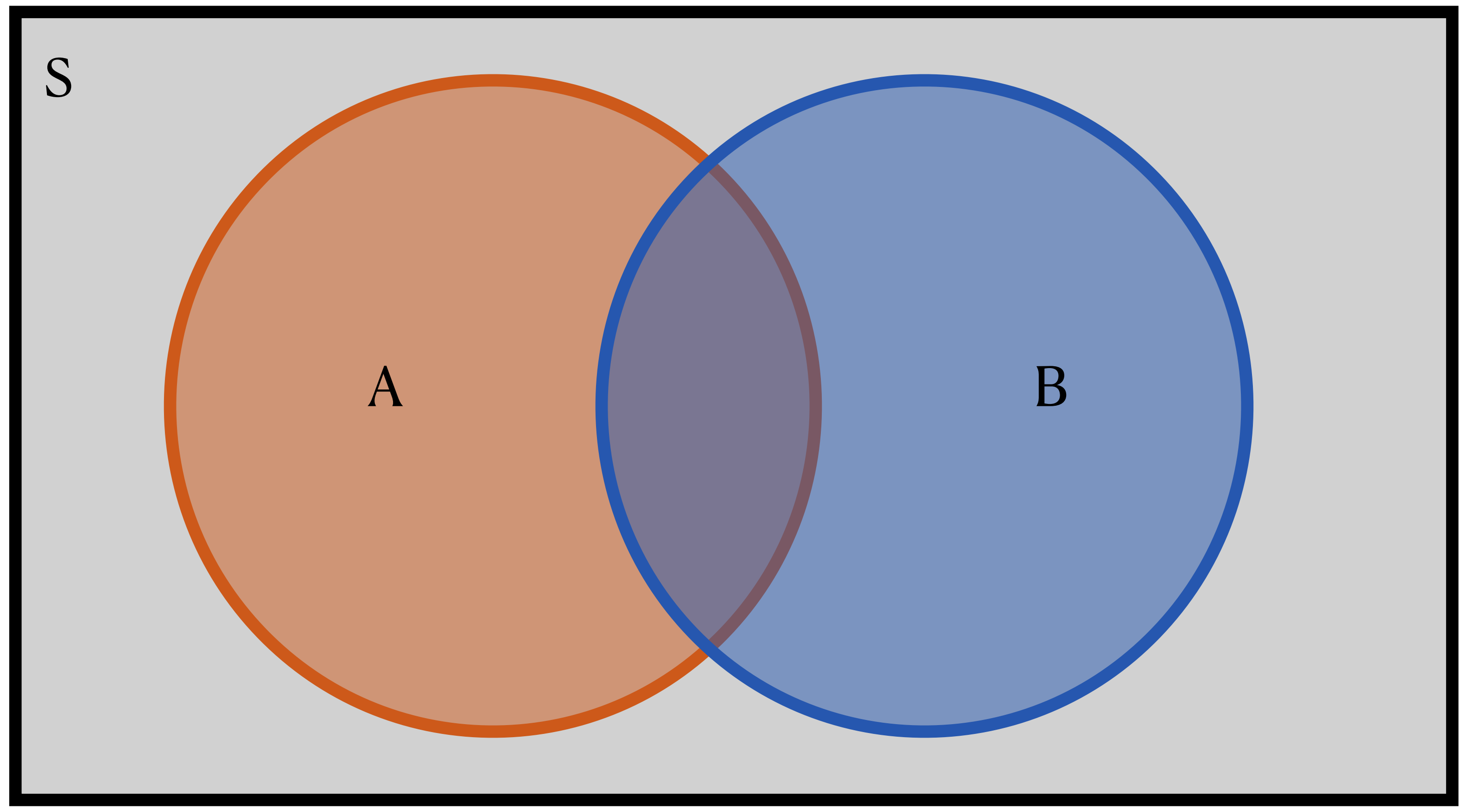


	Had COVID during birth	Did not have COVID during birth	Total
Admitted to ICU	977	7943	8920
Not admitted to ICU	17738	842241	859979
Total	18715	850184	868899

Probability Concepts

Sample space

- Let the **random** circumstance be me selecting a random person from that study, where everyone has equal probability of being selected
  - What is the **sample space**?
    - The set of all people in this study (roughly, the people who gave birth in 2020 in one of 500 hospitals in the US)
  - How big is the **sample space**?
    - 868,899

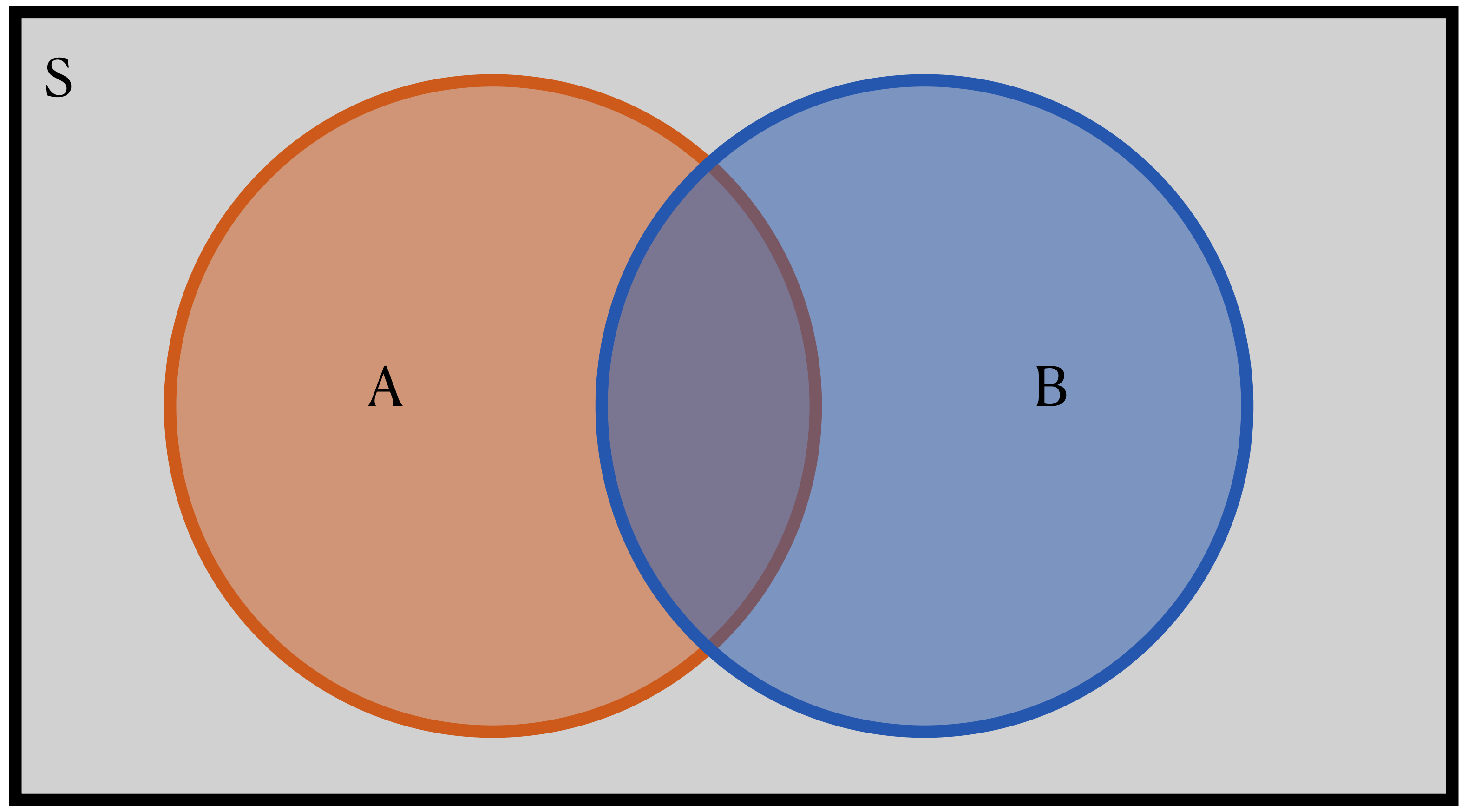


	Had COVID during birth	Did not have COVID during birth	Total
Admitted to ICU	977	7943	8920
Not admitted to ICU	17738	842241	859979
Total	18715	850184	868899

Probability Concepts

Marginal probabilities

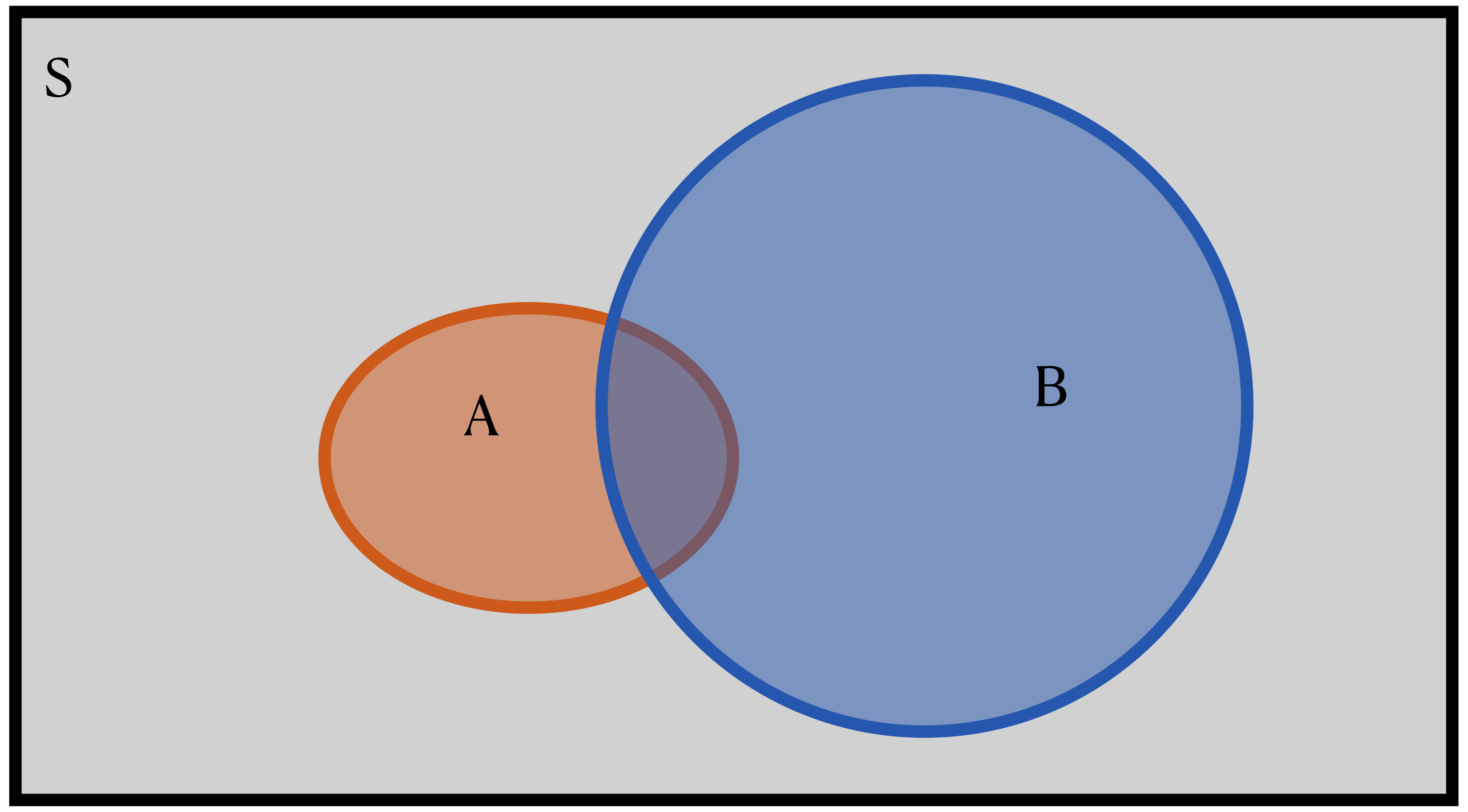
- Let event A be the event that the randomly selected person had COVID during birth
  - What is $P(A)$?
    - (Recall that probability describes the size of the event relative to the size of the sample space)
    - (Recall *also* that in this scenario every person in the study has equal probability of being selected)



	Had COVID during birth	Did not have COVID during birth	Total
Admitted to ICU	977	7943	8920
Not admitted to ICU	17738	842241	859979
Total	18715	850184	868899

Probability Concepts

Marginal probabilities

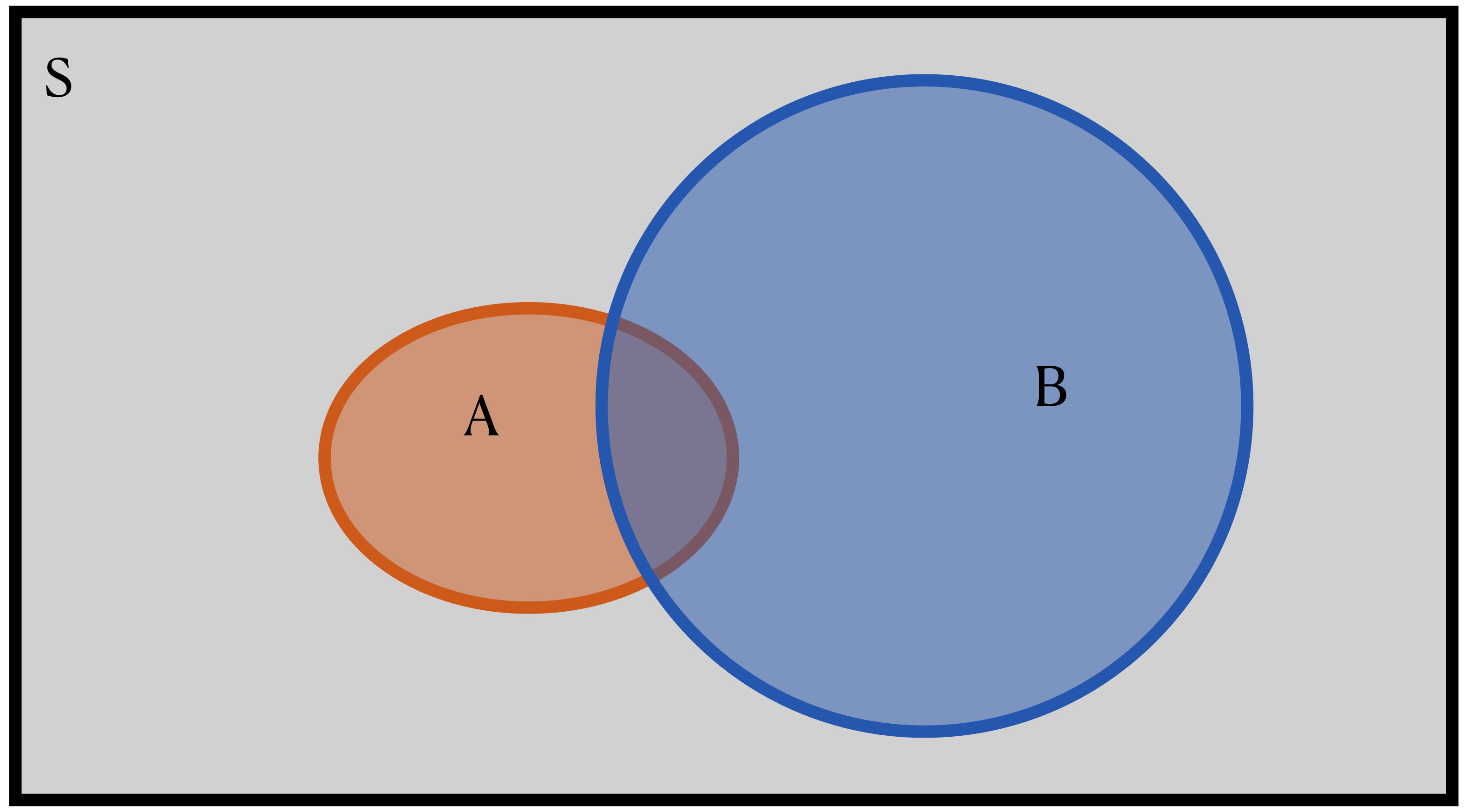


- Let event A be the event that the randomly selected person had COVID during birth
  - What is $P(A)$?
  - $P(A) = \dfrac{\text{size of A}}{\text{size of S}} = \dfrac{18715}{868899} \approx 0.021$
  - Because we only look at A and ignore B, this type of probability is called a **marginal probability**
    - (Ringing any bells from Topic 3?)

	Had COVID during birth	Did not have COVID during birth	Total
Admitted to ICU	977	7943	8920
Not admitted to ICU	17738	842241	859979
Total	18715	850184	868899

Probability Concepts

Marginal probabilities

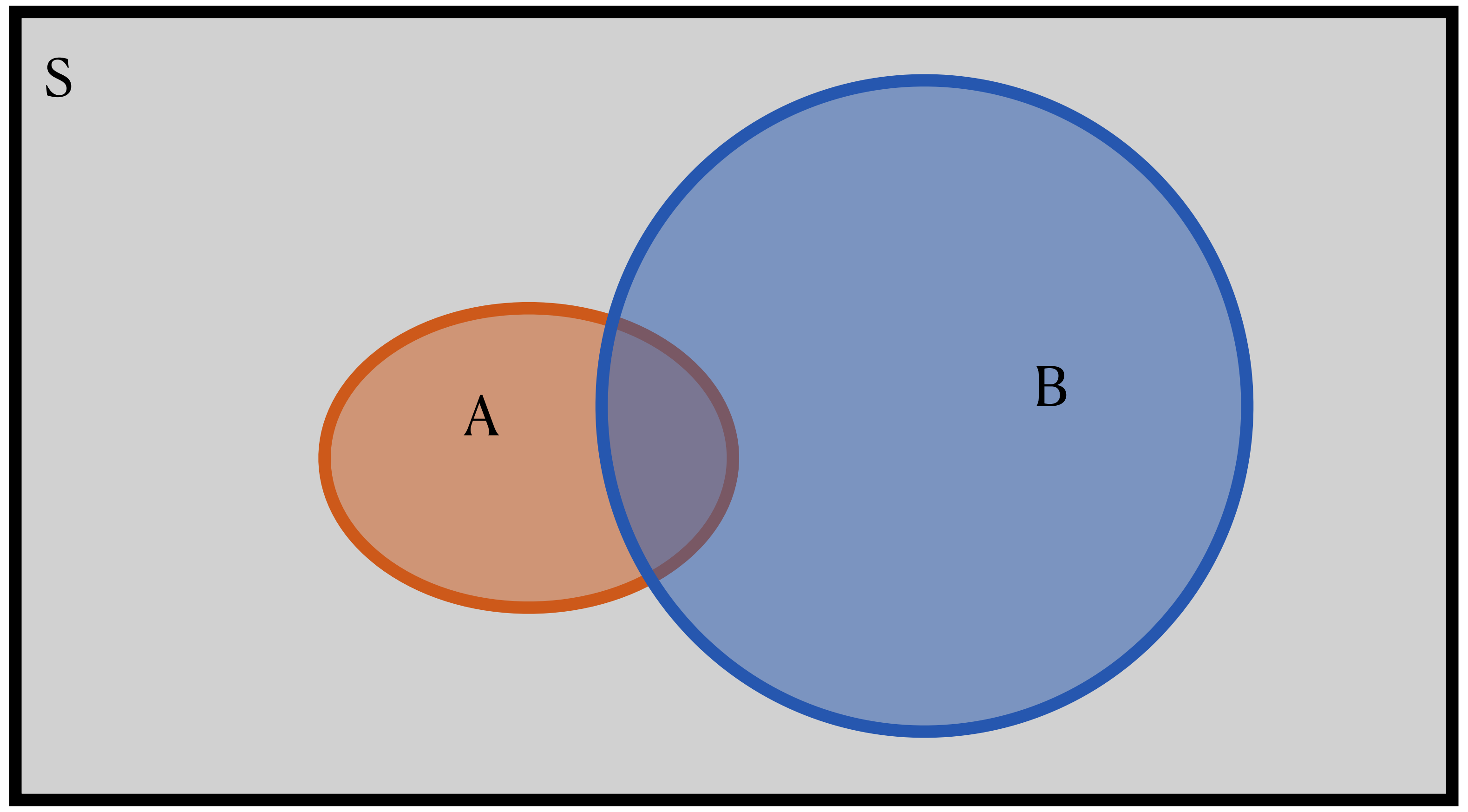


- Let event A be the event that the randomly selected person had COVID during birth
  - What is $P(A)$?
  - $P(A) = \frac{\text{size of A}}{\text{size of S}} = \frac{18715}{868899} \approx 0.021$
  - Note that I made A small in the Venn diagram, why'd I do that?

	Had COVID during birth	Did not have COVID during birth	Total
Admitted to ICU	977	7943	8920
Not admitted to ICU	17738	842241	859979
Total	18715	850184	868899

Probability Concepts

Marginal probabilities



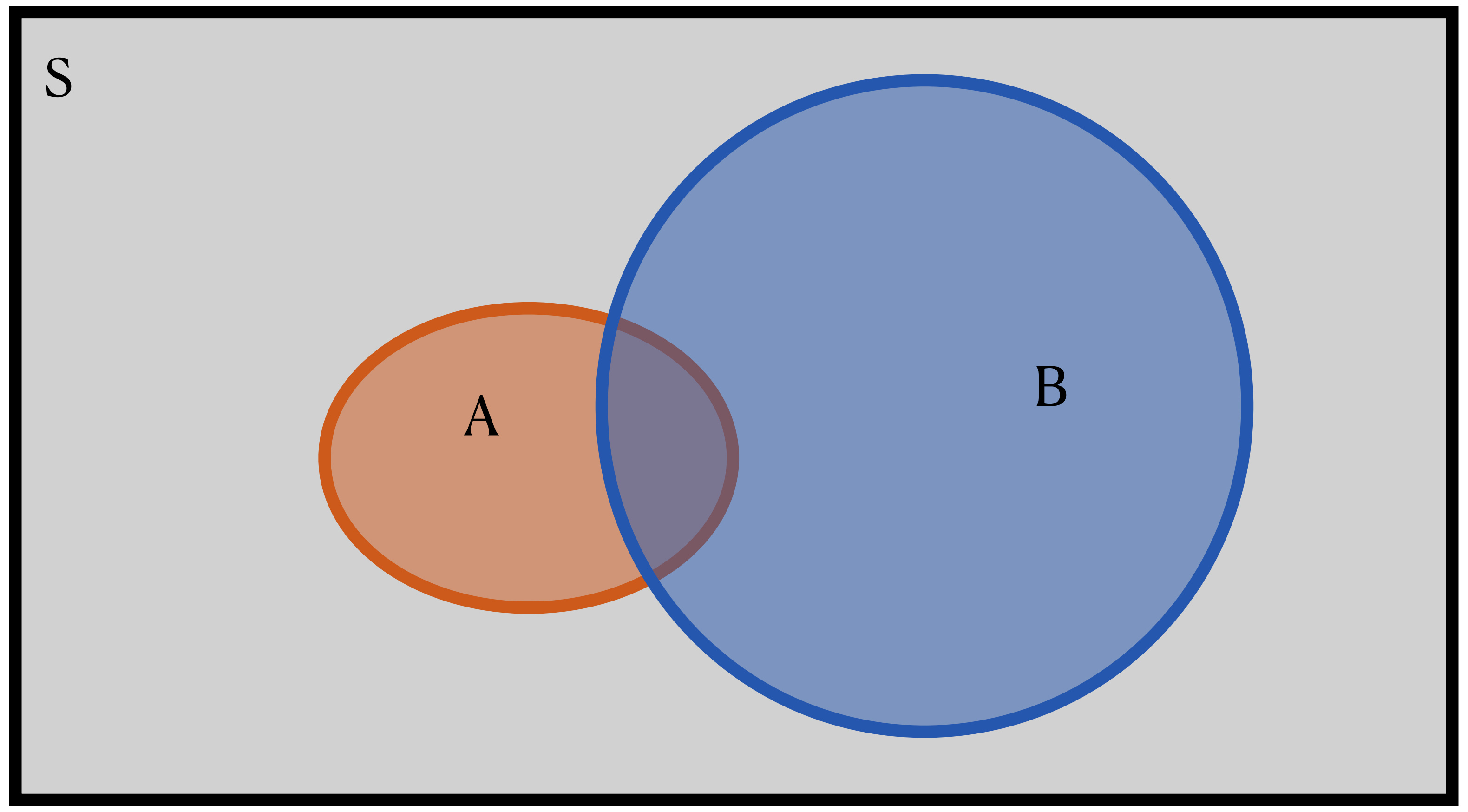
- Let event A be the event that the randomly selected person had COVID during birth
  - What is $P(A)$?
    - $P(A) = \dfrac{\text{size of A}}{\text{size of S}} = \dfrac{18715}{868899} \approx 0.021$
  - Note that I made A small in the Venn diagram, why'd I do that?
    - To remind you that in Venn diagrams, areas are probabilities!
    - Note that this still isn't quite to scale, for visibility purposes

	Had COVID during birth	Did not have COVID during birth	Total
Admitted to ICU	977	7943	8920
Not admitted to ICU	17738	842241	859979
Total	18715	850184	868899

Probability Concepts

Marginal probabilities

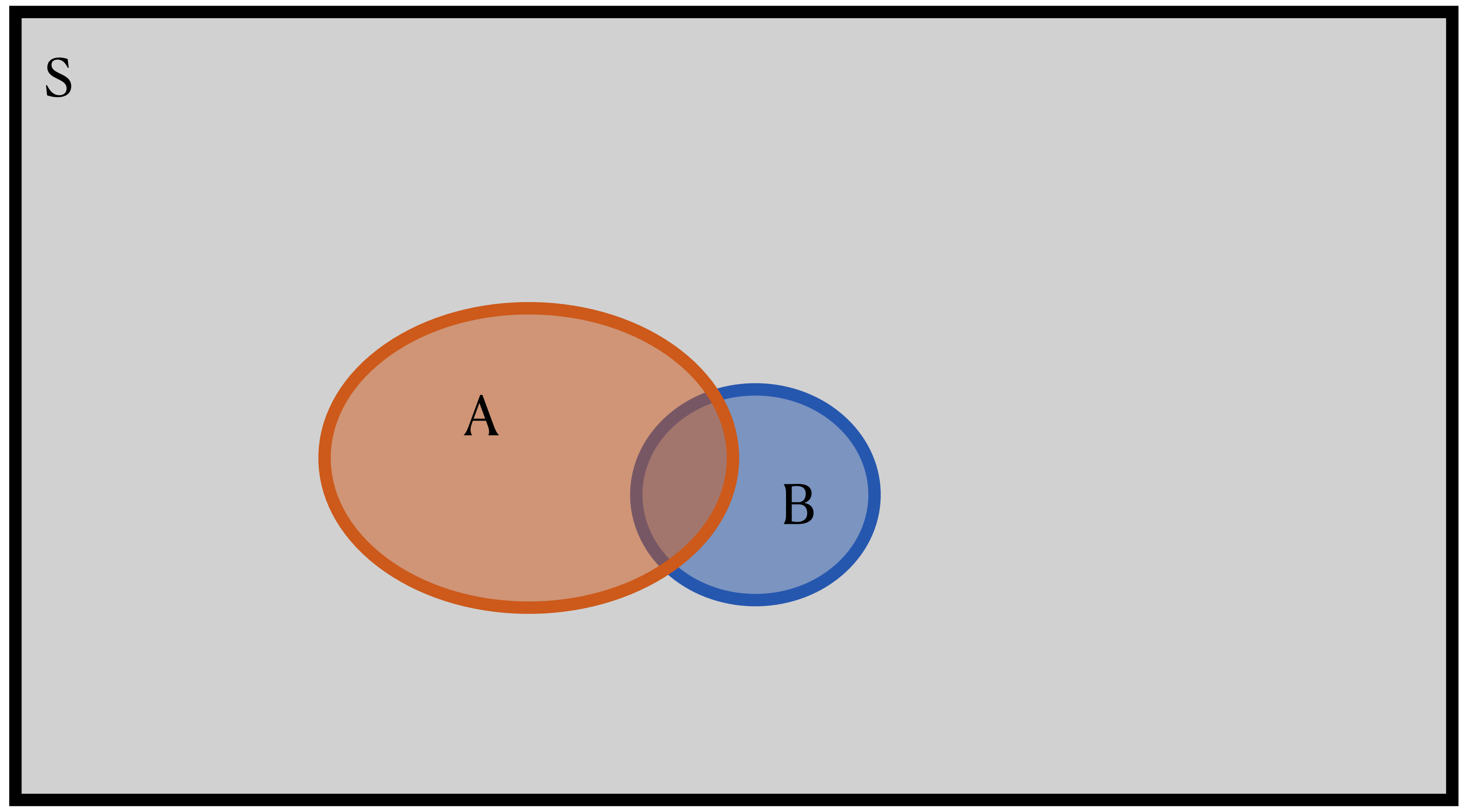
- Let event B be the event that the randomly selected patient ended up being admitted to the ICU
  - What is $P(B)$?



	Had COVID during birth	Did not have COVID during birth	Total
Admitted to ICU	977	7943	8920
Not admitted to ICU	17738	842241	859979
Total	18715	850184	868899

Probability Concepts

Marginal probabilities



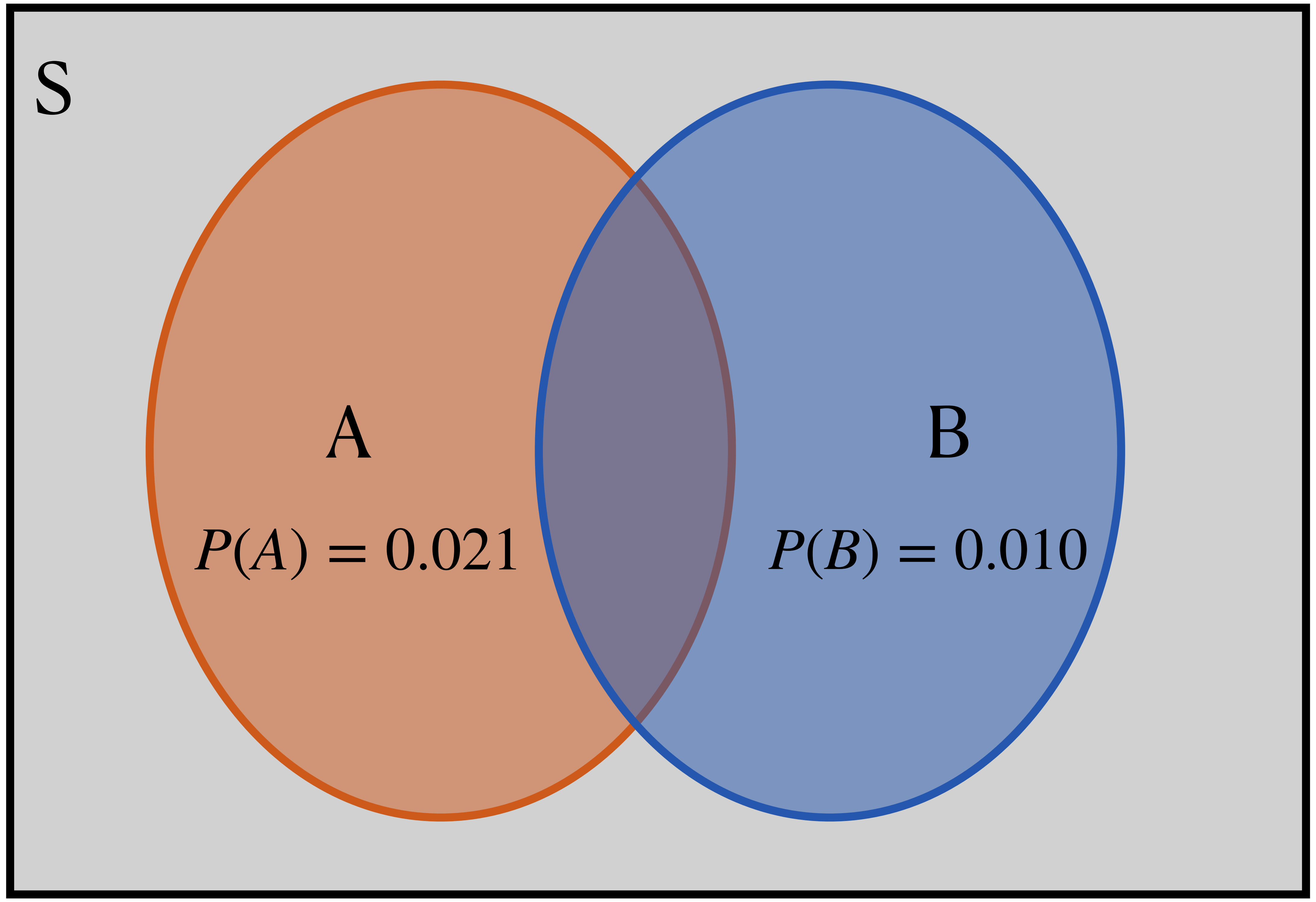
- Let event B be the event that the randomly selected patient ended up being admitted to the ICU
  - What is $P(B)$?
    - $P(B) = \dfrac{\text{size of B}}{\text{size of S}} = \dfrac{8920}{868899} \approx 0.010$

	Had COVID during birth	Did not have COVID during birth	Total
Admitted to ICU	977	7943	8920
Not admitted to ICU	17738	842241	859979
Total	18715	850184	868899

Probability Rules

Venn diagrams

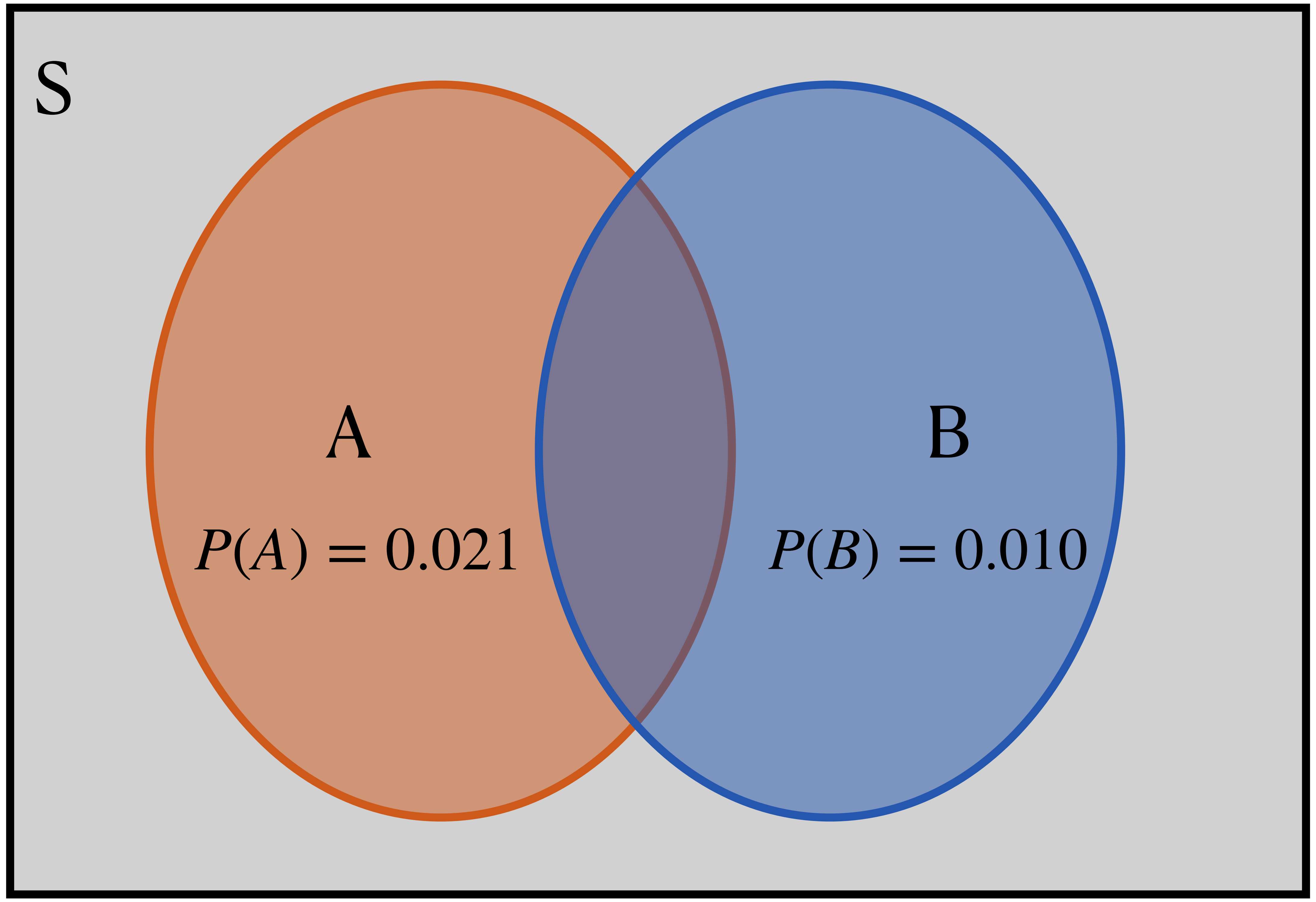
- Now let's use Venn diagrams to learn some probability concepts/rules
- Note that (again, for visibility), the Venn diagram is not to scale *at all*, but I have left the probabilities written in the circles



Probability Rules

Complementary events

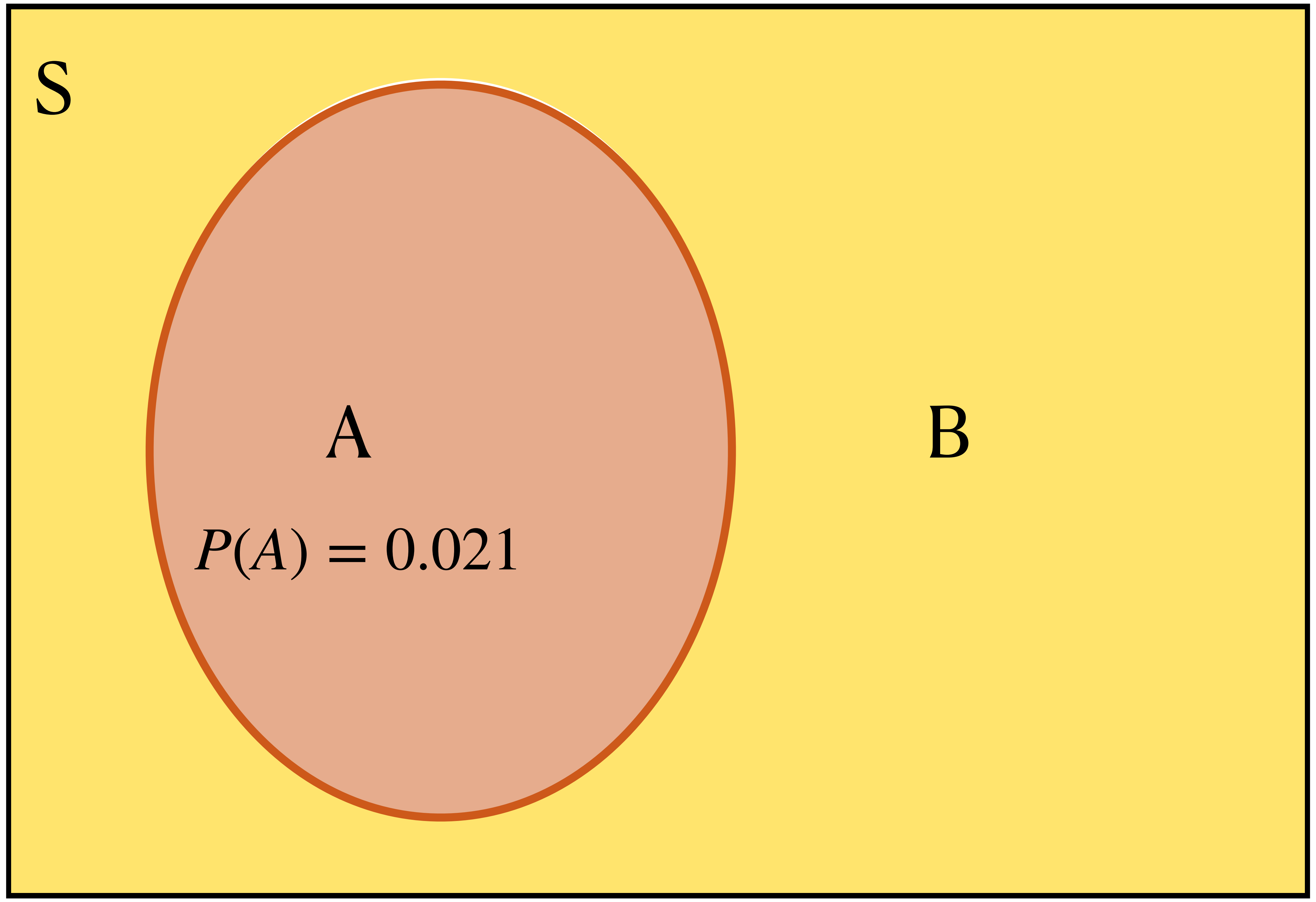
- The **complement** of an event "A" is P(A does not occur) or P(not A)
  - Notation used is sometimes $P(A^c)$
  - Where is it on the Venn diagram?



Probability Rules

Complementary events

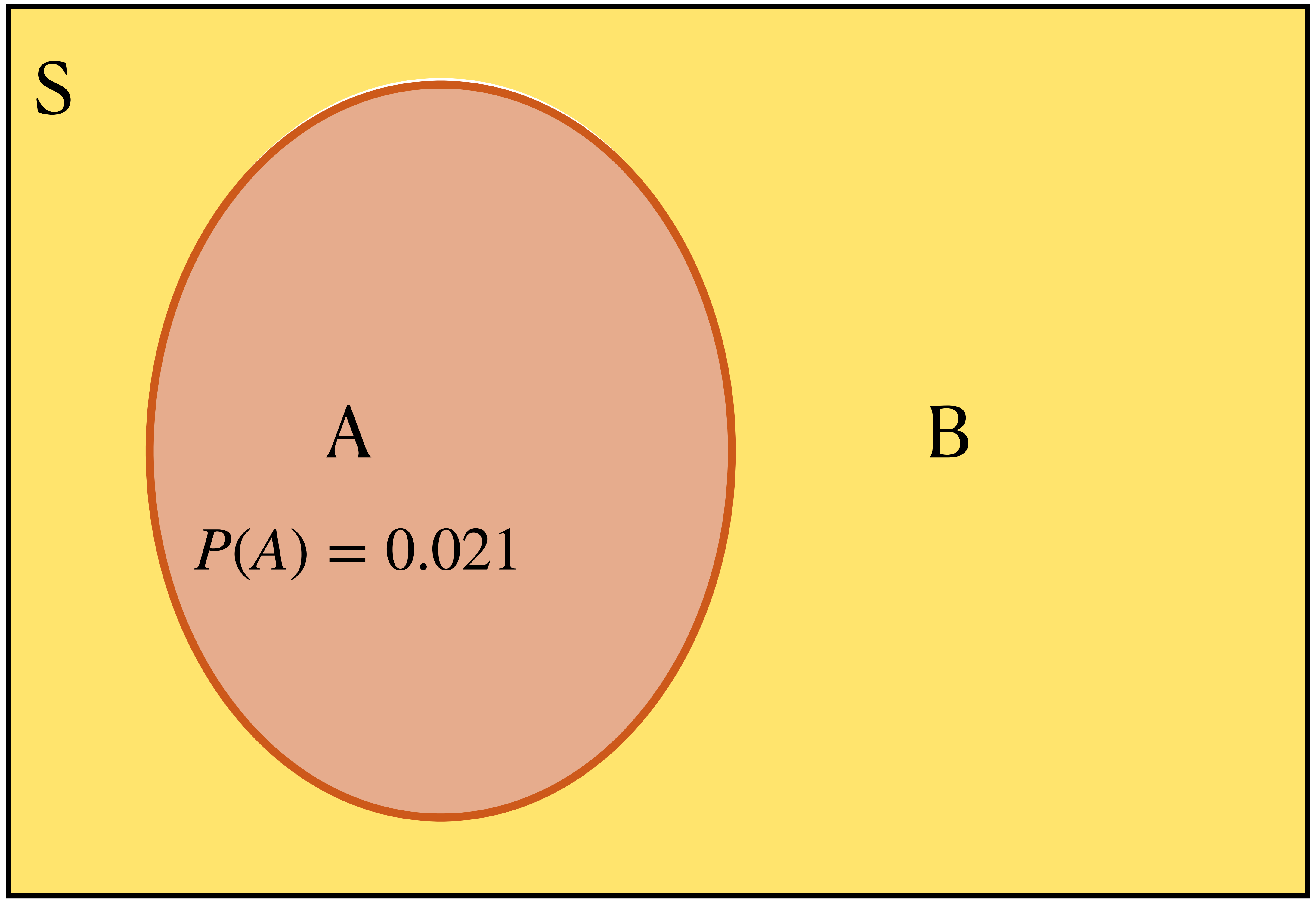
- The **complement** of an event "A" is P(A does not occur) or P(not A)
  - Where is it on the Venn diagram?
    - Everything outside of the circle A
  - What is P(not A) in terms of P(A)?
    - (Remember, probabilities are areas!)



Probability Rules

Complementary events

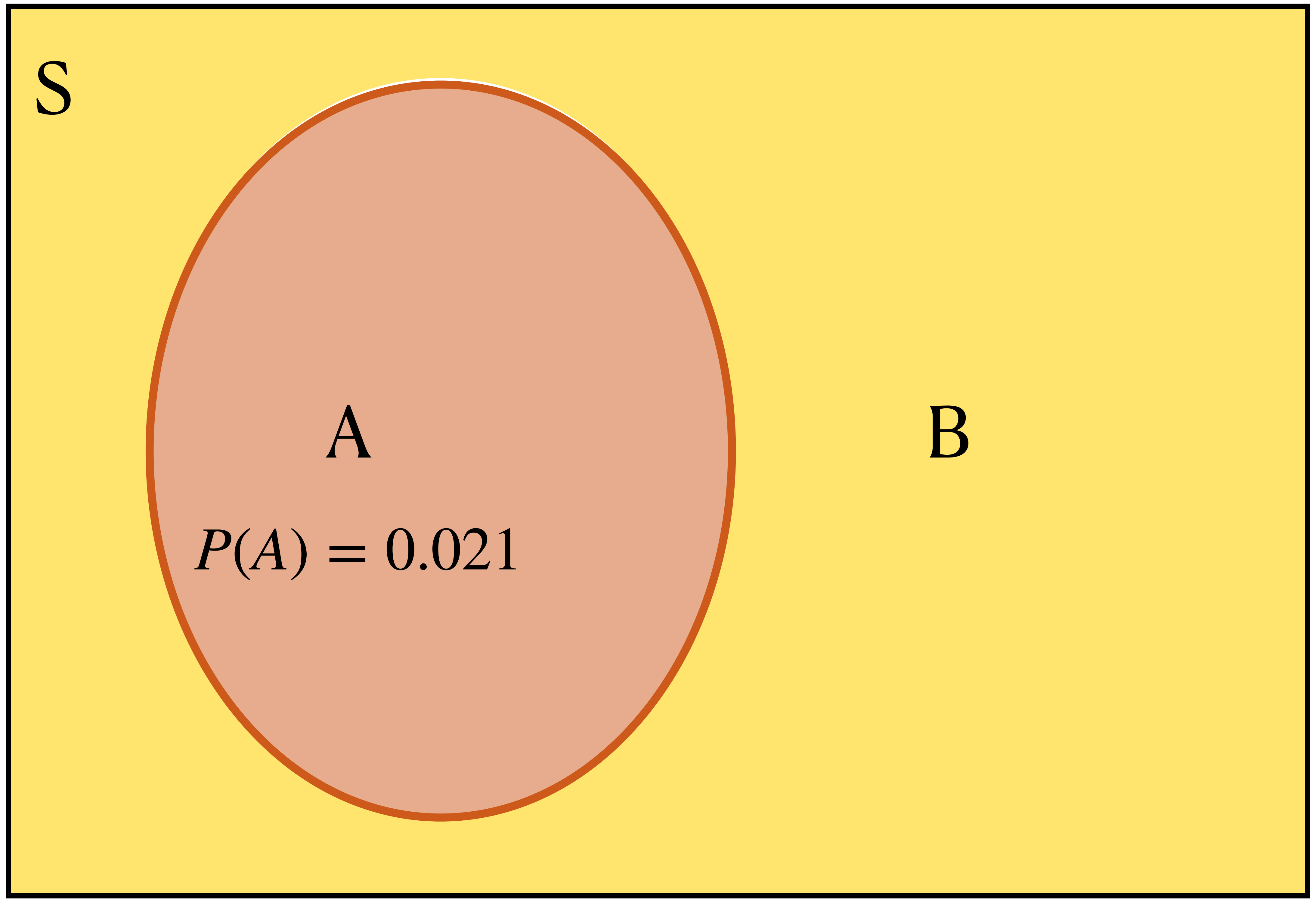
- The **complement** of an event "A" is P(A does not occur) or P(not A)
    - Where is it on the Venn diagram?
        - Everything outside of the circle A
    - What is P(not A) in terms of P(A)?
        - (Remember, probabilities are areas!)
        - $P(\text{not } A) = P(S) - P(A) = 1 - P(A)$



Probability Rules

Complementary events

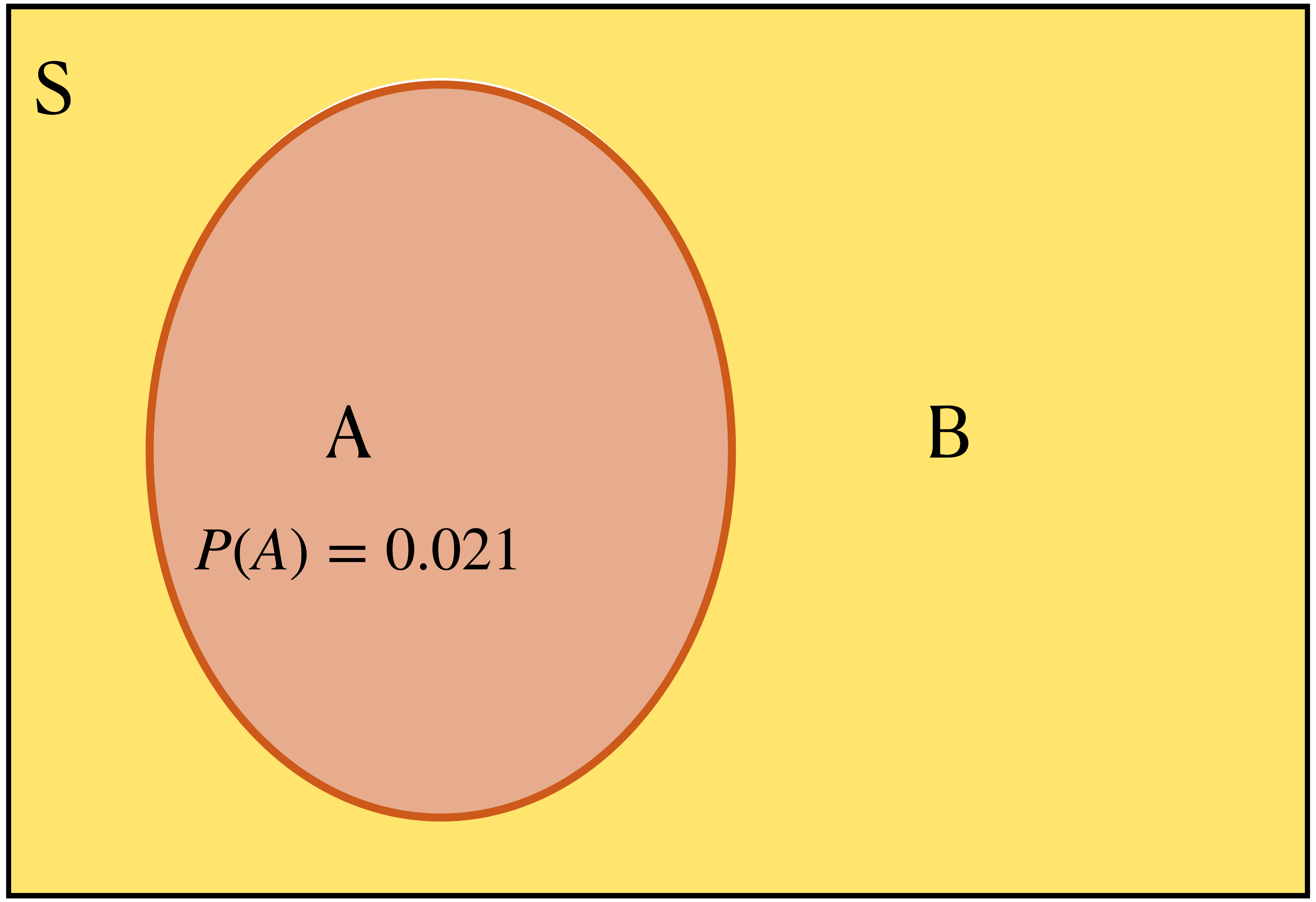
- The **complement** of an event "A" is P(A does not occur) or P(not A)
    - $P(\text{not } A) = P(S) - P(A) = 1 - P(A)$
- *The probability of having COVID while giving birth in 2020 was 0.021.*
    - P(COVID birth) = 0.021
    - P(not COVID birth) = ?



Probability Rules

Complementary events

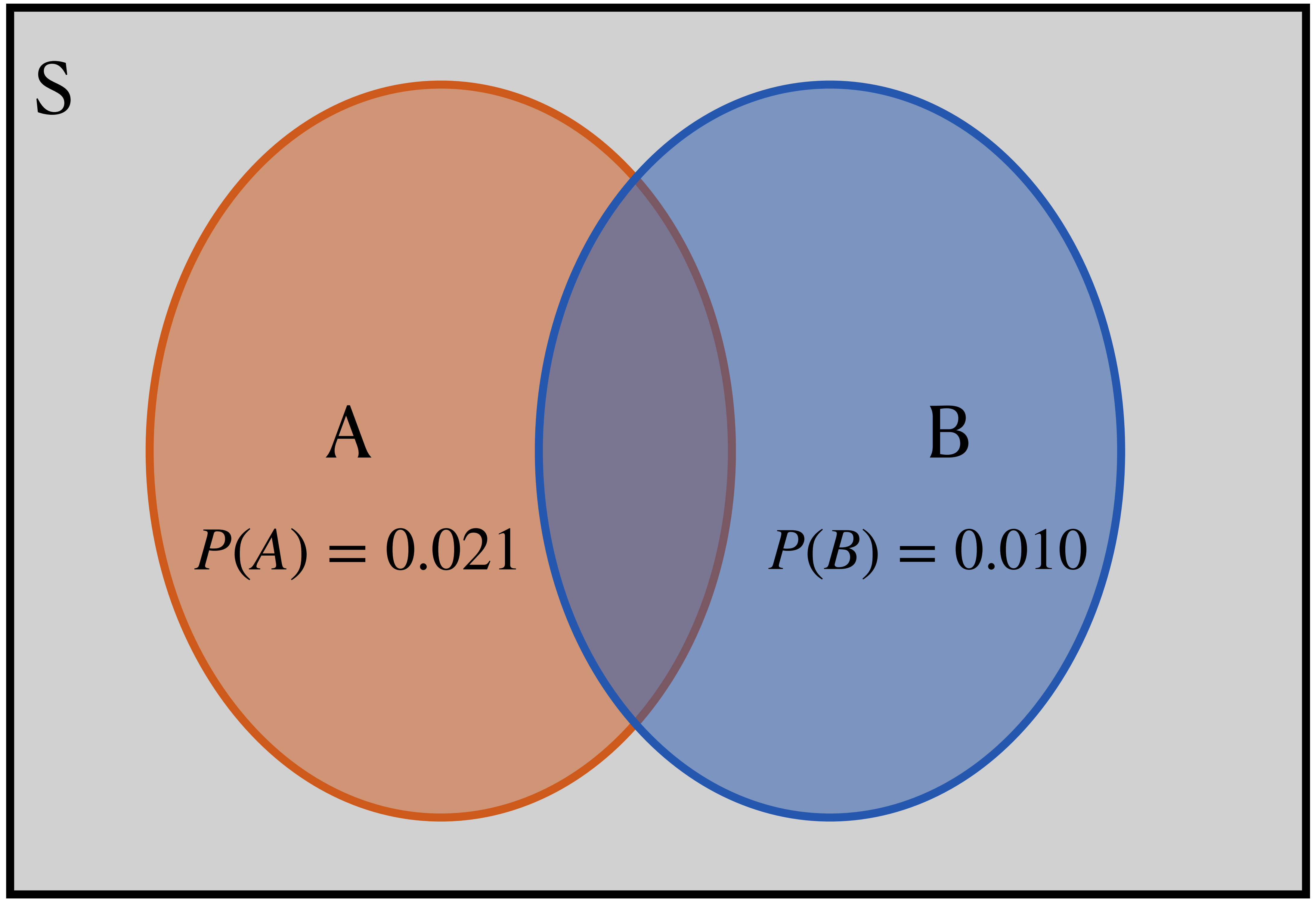
- The **complement** of an event “A” is P(A does not occur) or P(not A)
  - $P(\text{not } A) = P(S) - P(A) = 1 - P(A)$
- *The probability of having COVID while giving birth in 2020 was 0.021.*
  - P(COVID birth) = 0.021
  - P(not COVID birth) = 0.979



Probability Rules

Unions and Intersections

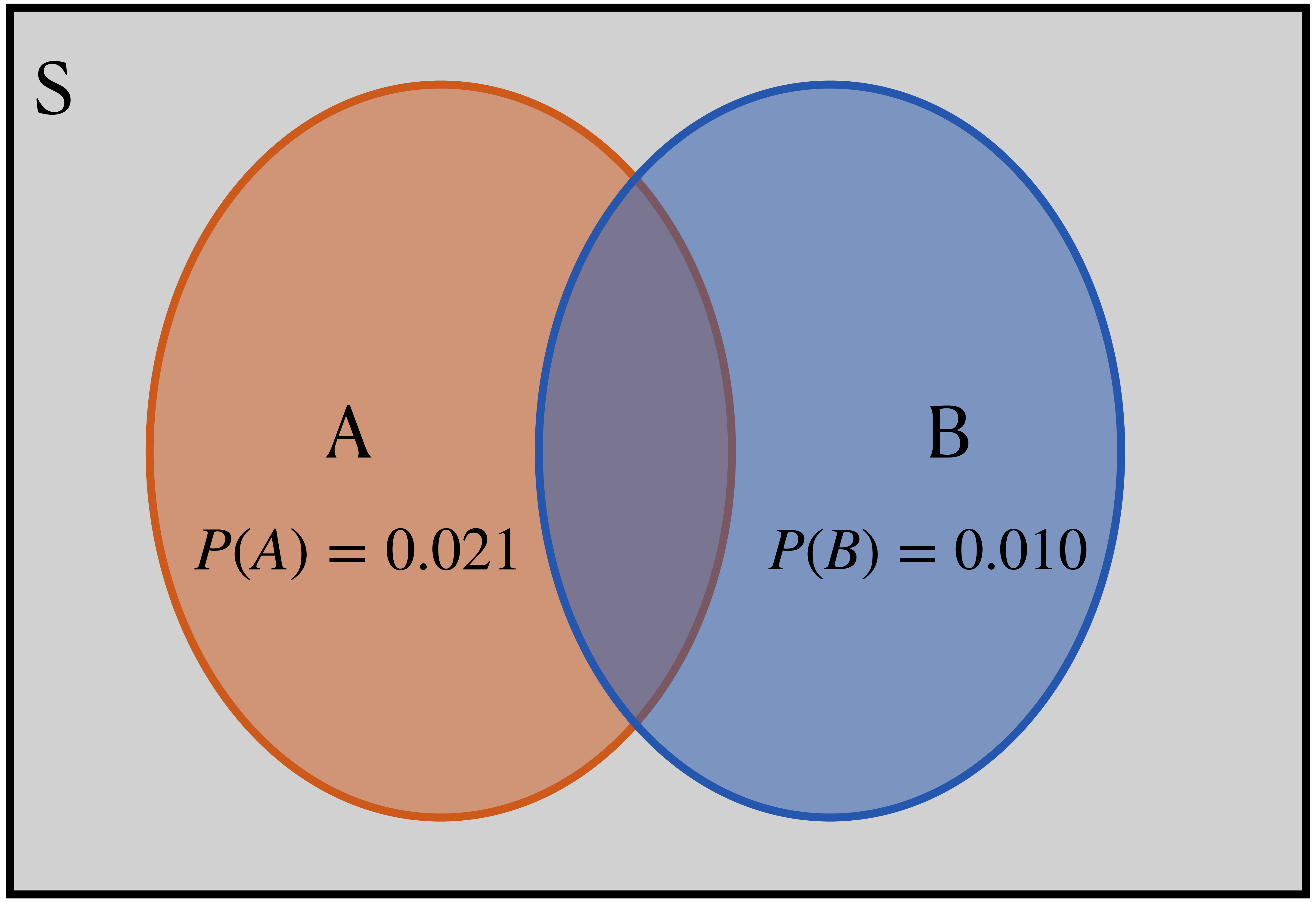
- The **joint probability** that both events A and B occur together is denoted P(A and B)
  - The probability of the **intersection**
  - Event A = *the event that someone giving birth in 2020 had COVID*
  - Event B = *the event that someone giving birth in 2020 ended up in the ICU*
  - A and B is the event that…



Probability Rules

Unions and Intersections

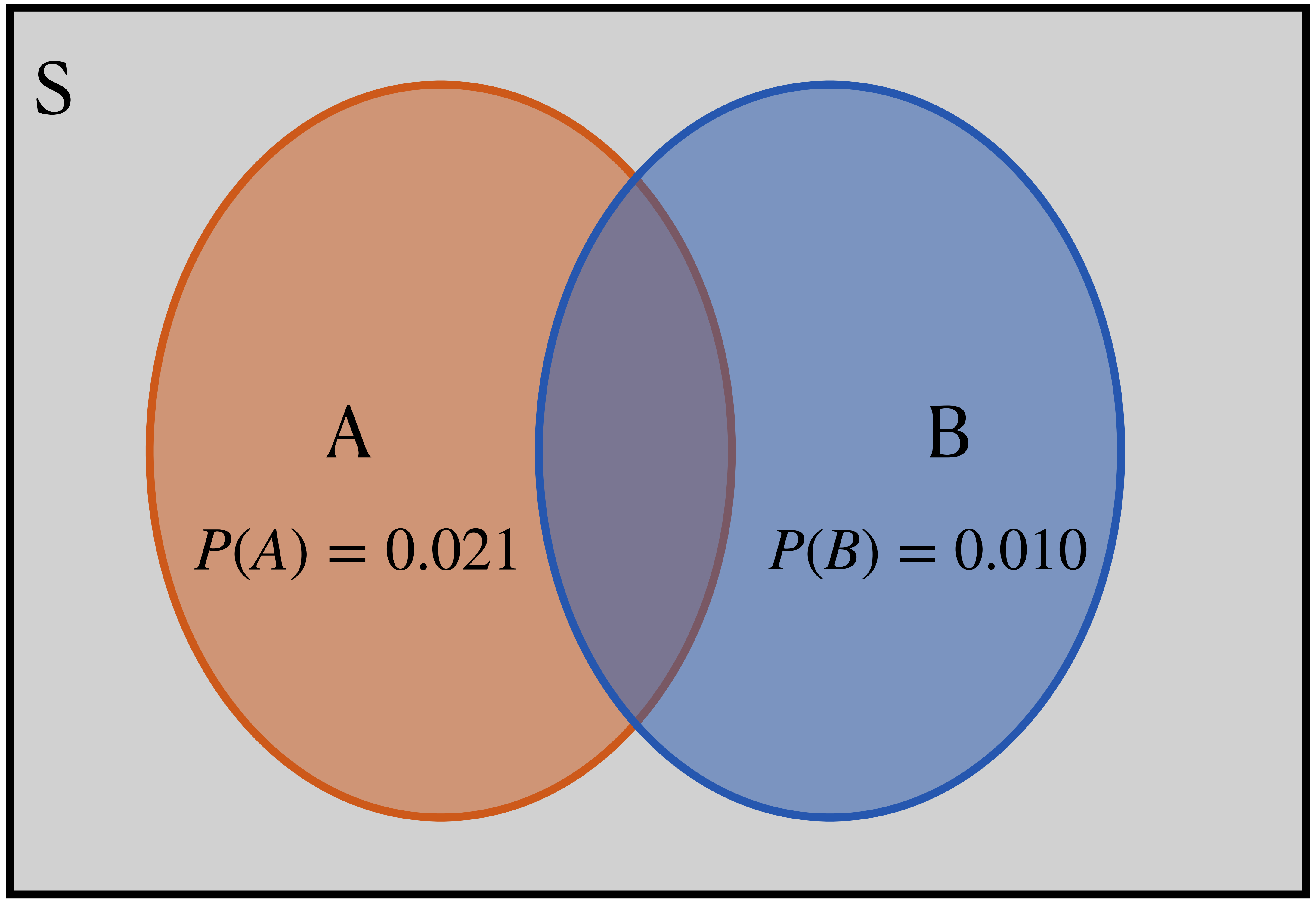
- The **joint probability** that both events A and B occur together is denoted P(A and B)
  - The probability of the **intersection**
  - Event A = *the event that someone giving birth in 2020 had COVID*
  - Event B = *the event that someone giving birth in 2020 ended up in the ICU*
  - A and B is the event that someone giving birth in 2020 had COVID *and* ended up in the ICU



Probability Rules

Unions and Intersections

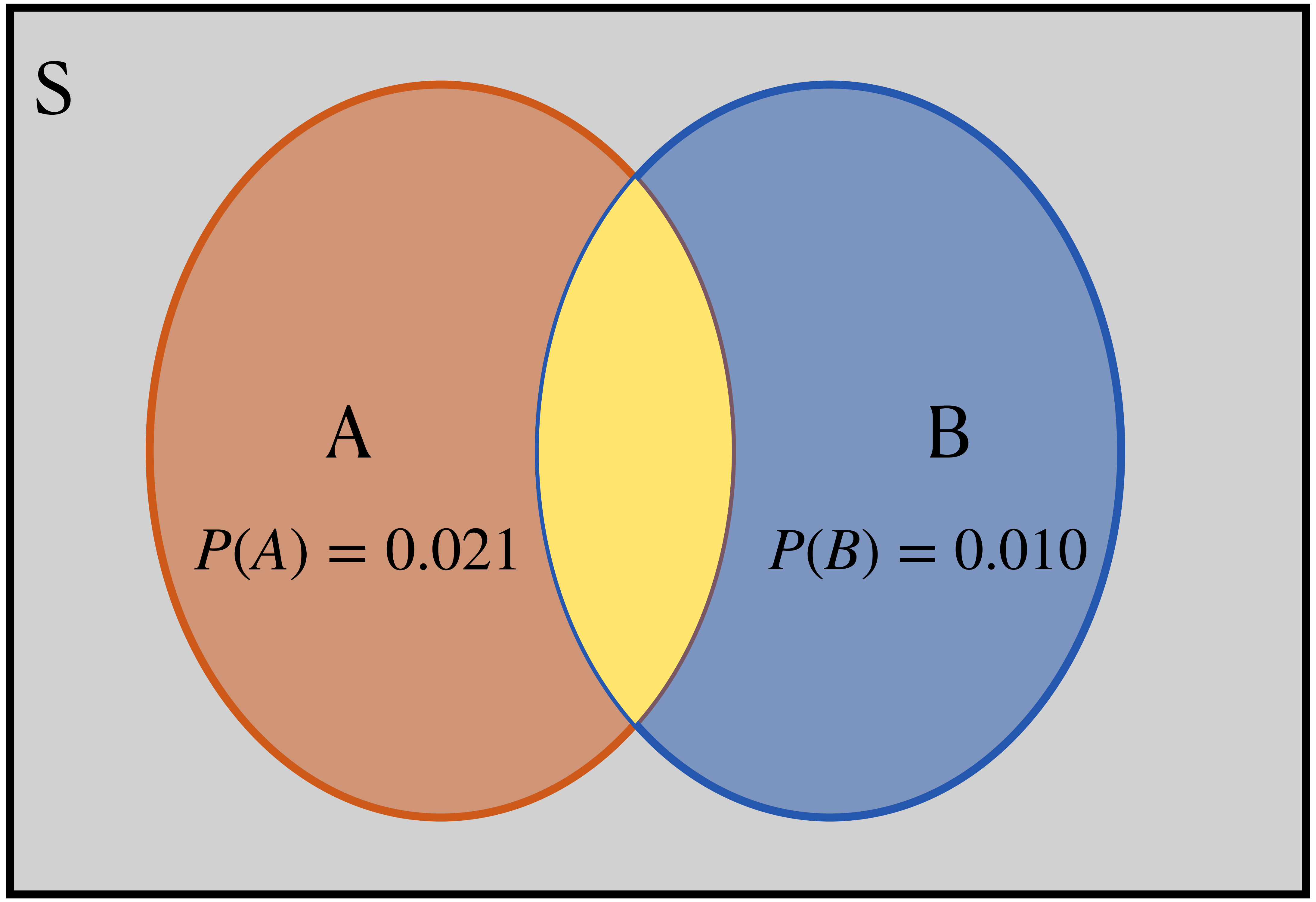
- The **joint probability** that both events A and B occur together is denoted P(A and B)
  - The probability of the **intersection**
  - Where is this on the Venn diagram?



Probability Rules

Unions and Intersections

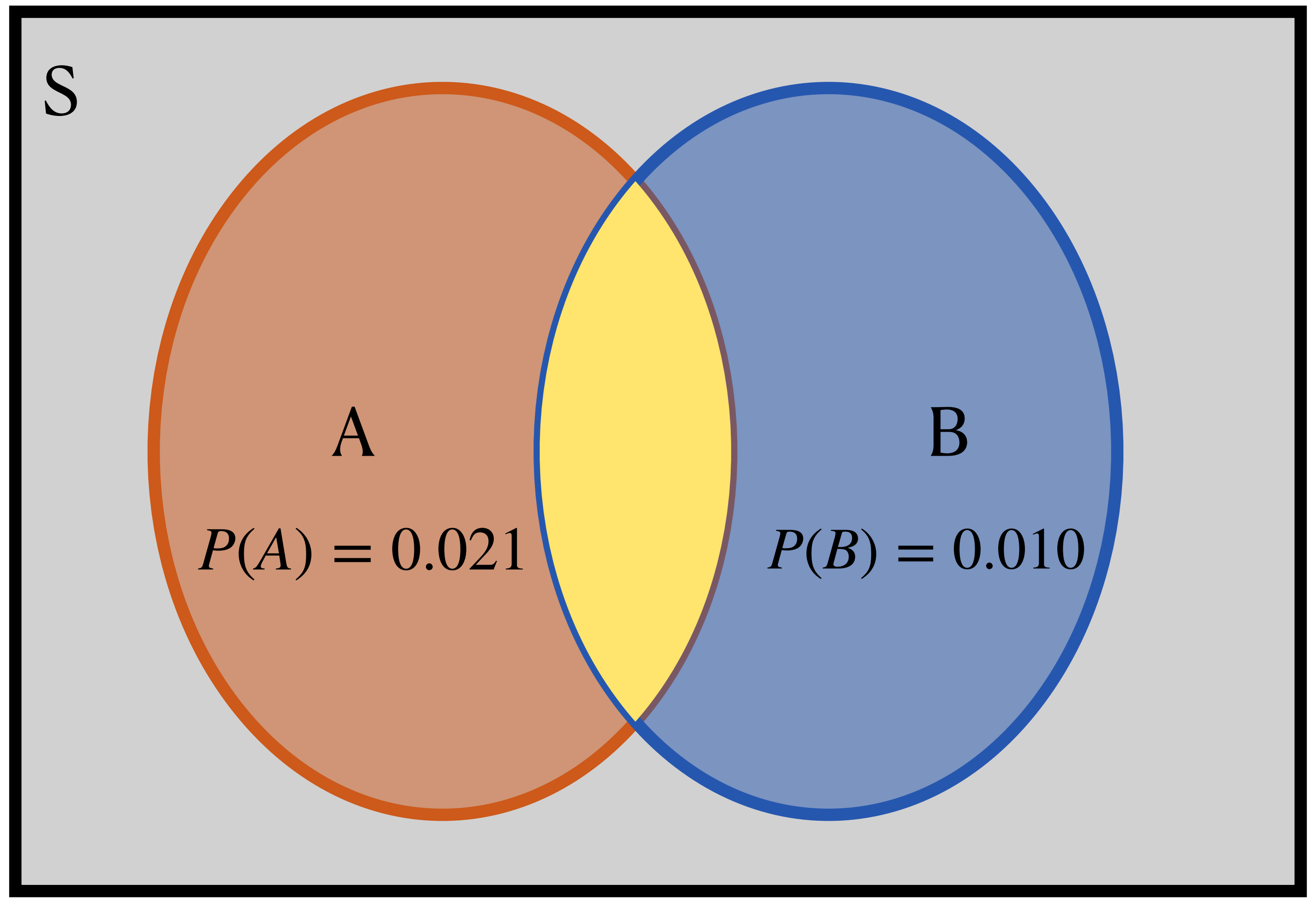
- The **joint probability** that both events A and B occur together is denoted P(A and B)
  - The probability of the **intersection**
  - Where is this on the Venn diagram?
    - P(A and B) is the overlap between A and B



Probability Rules

Unions and Intersections

- The **joint probability** that both events A and B occur together is denoted P(A and B)
  - The probability of the **intersection**
- What is it? Let's go back to the table…
- What is P(A and B)?
  - PollEv.com/stats8
  - Round to 3 decimal places



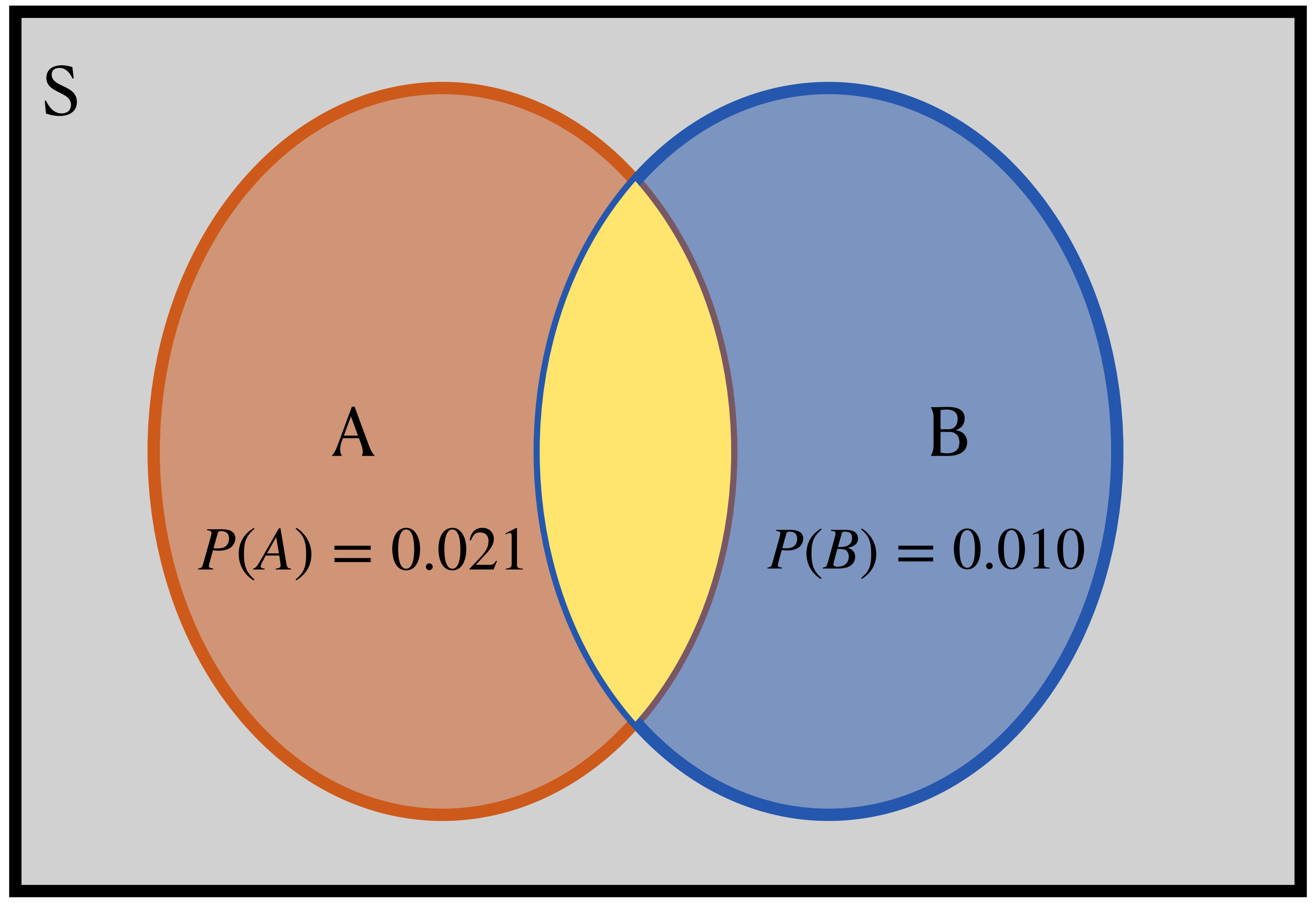
	Had COVID during birth	Did not have COVID during birth	Total
Admitted to ICU	977	7943	8920
Not admitted to ICU	17738	842241	859979
Total	18715	850184	868899

Probability Rules

Unions and Intersections

- The **joint probability** that both events A and B occur together is denoted P(A and B)
  - The probability of the **intersection**
- What is it? Let's go back to the table…
- What is P(A and B)?
  - What is the size of the event A and B, relative to the size of the whole sample space?

0.001

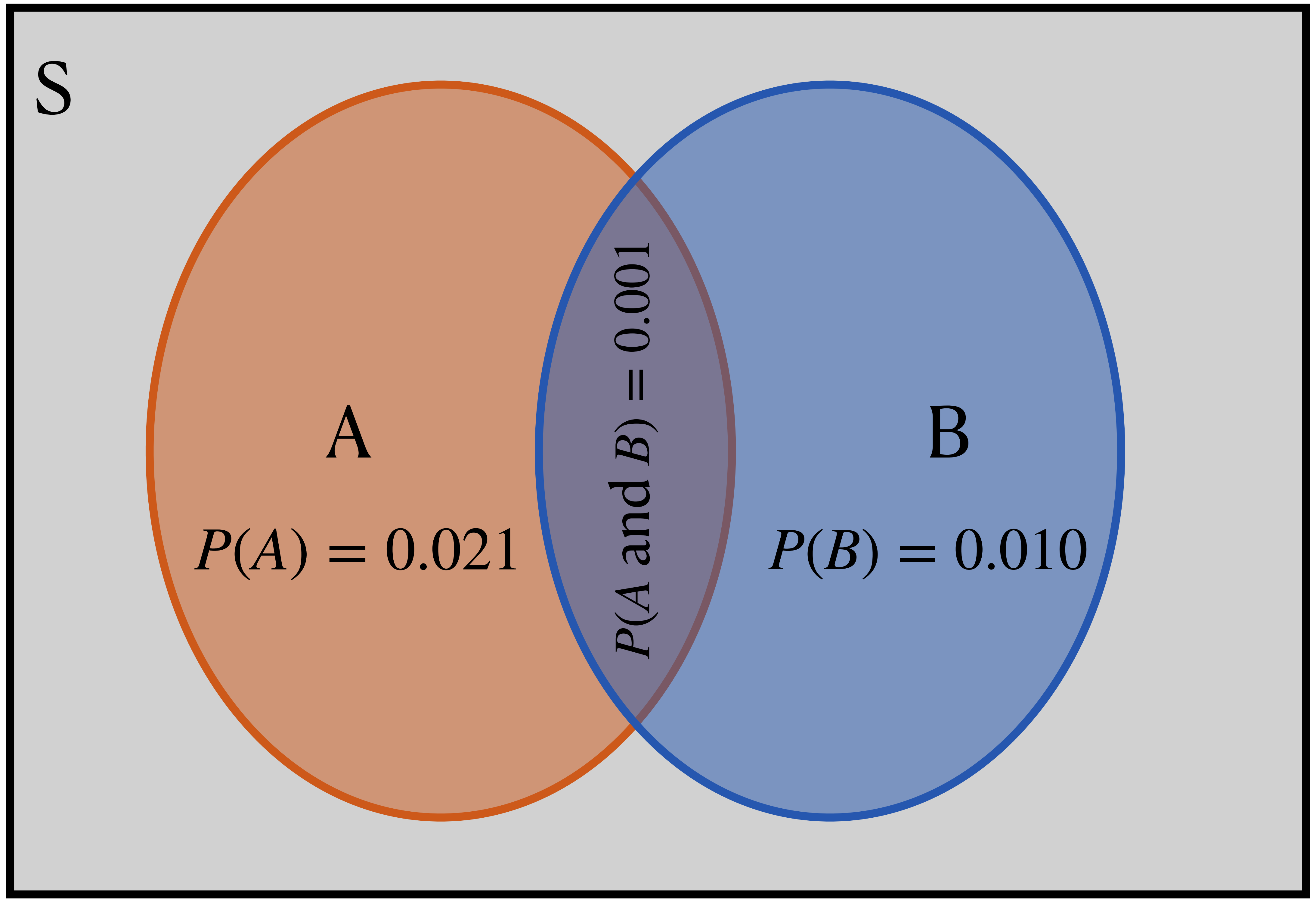


	Had COVID during birth	Did not have COVID during birth	Total
Admitted to ICU	977	7943	8920
Not admitted to ICU	17738	842241	859979
Total	18715	850184	868899

Probability Rules

Unions and Intersections

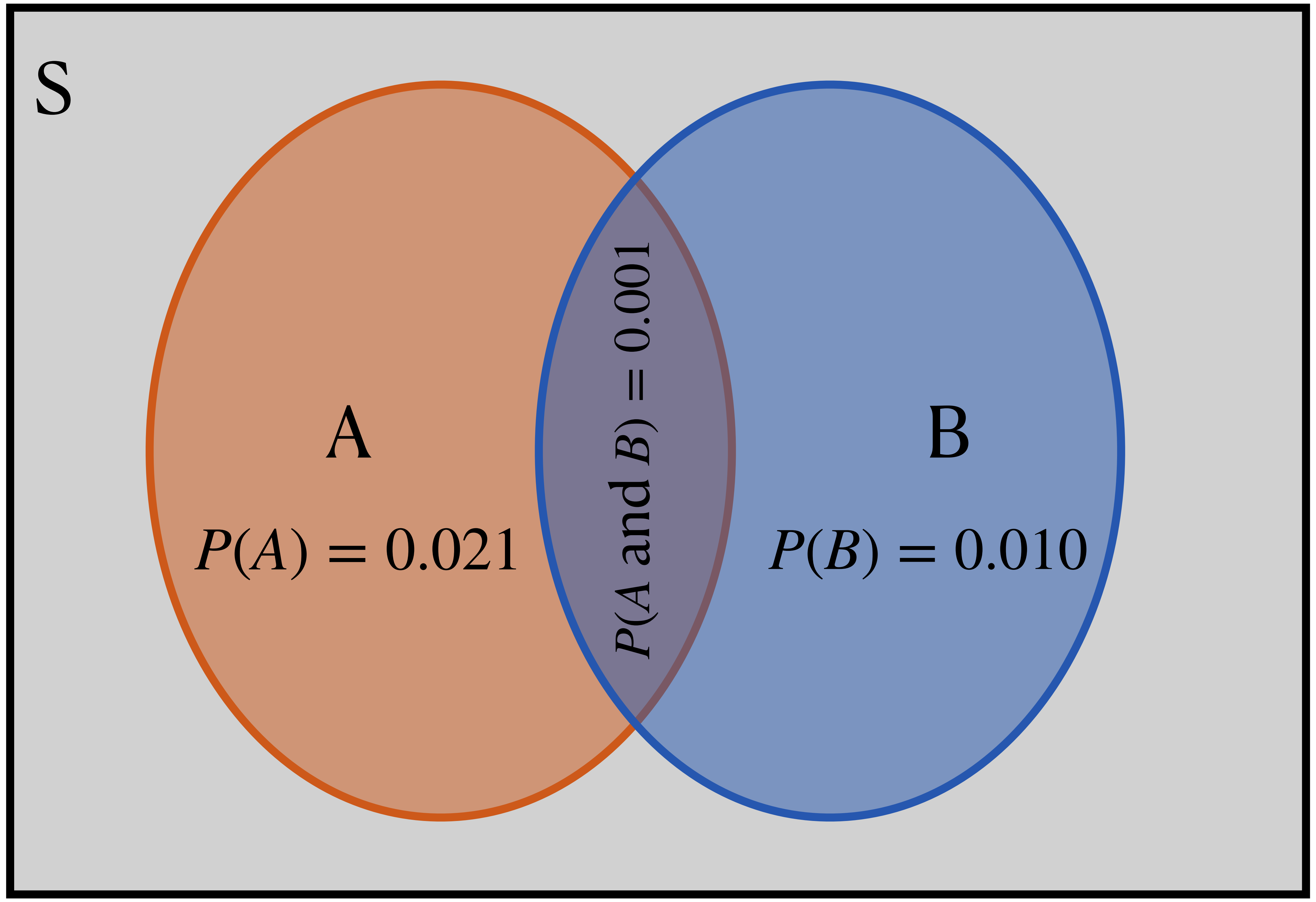
- The probability that at least one of events A or B occurs is denoted P(A or B)
  - The probability of the **union**
  - Event A = *the event that someone giving birth in 2020 had COVID*
  - Event B = *the event that someone giving birth in 2020 ended up in the ICU*
  - A or B is the event that someone giving birth in 2020 had COVID *or* ended up in the ICU (or both!)



Probability Rules

Unions and Intersections

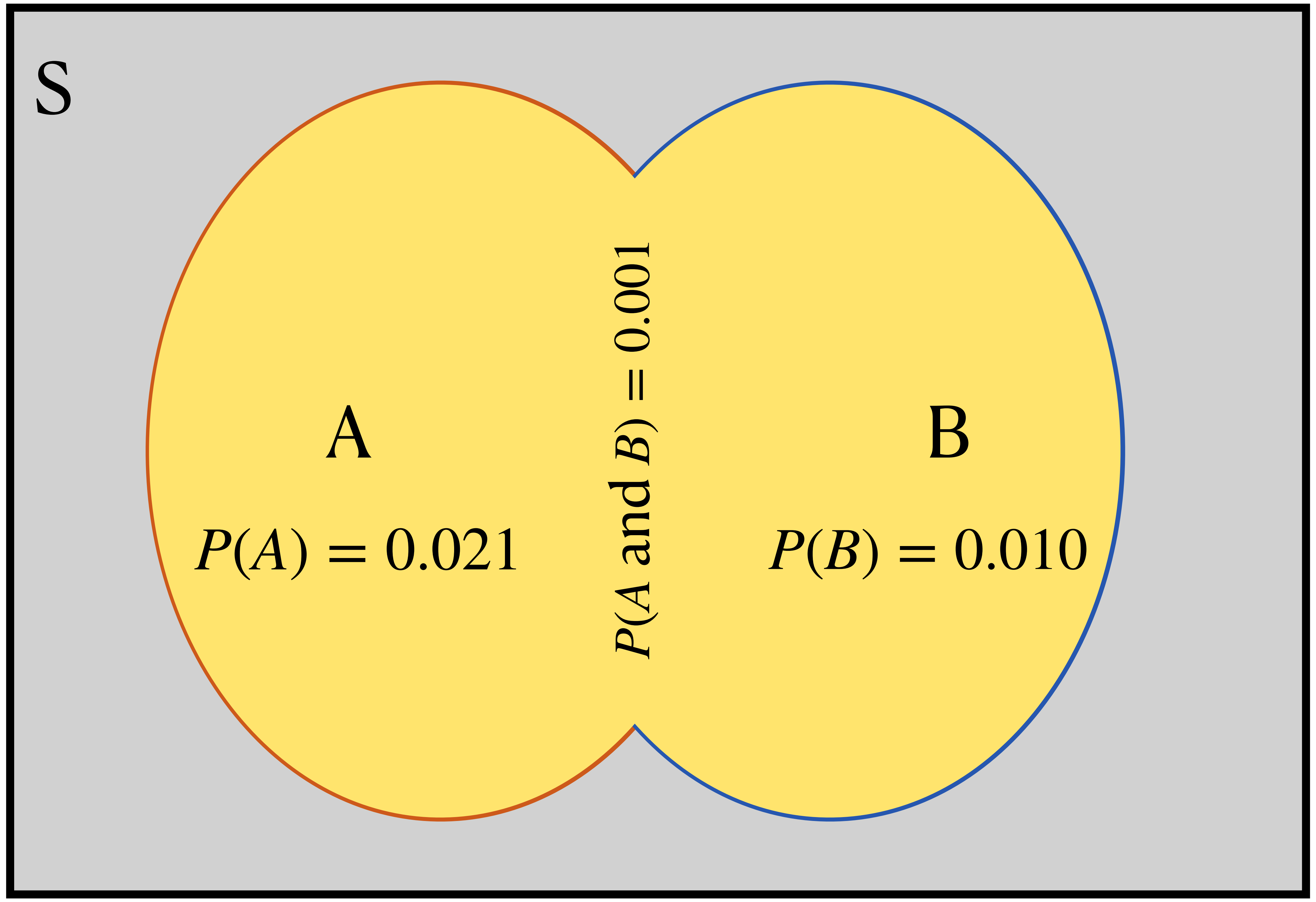
- The probability that at least one of events A or B occurs is denoted P(A or B)
  - The probability of the **union**
  - Where is this on the Venn diagram?



Probability Rules

Unions and Intersections

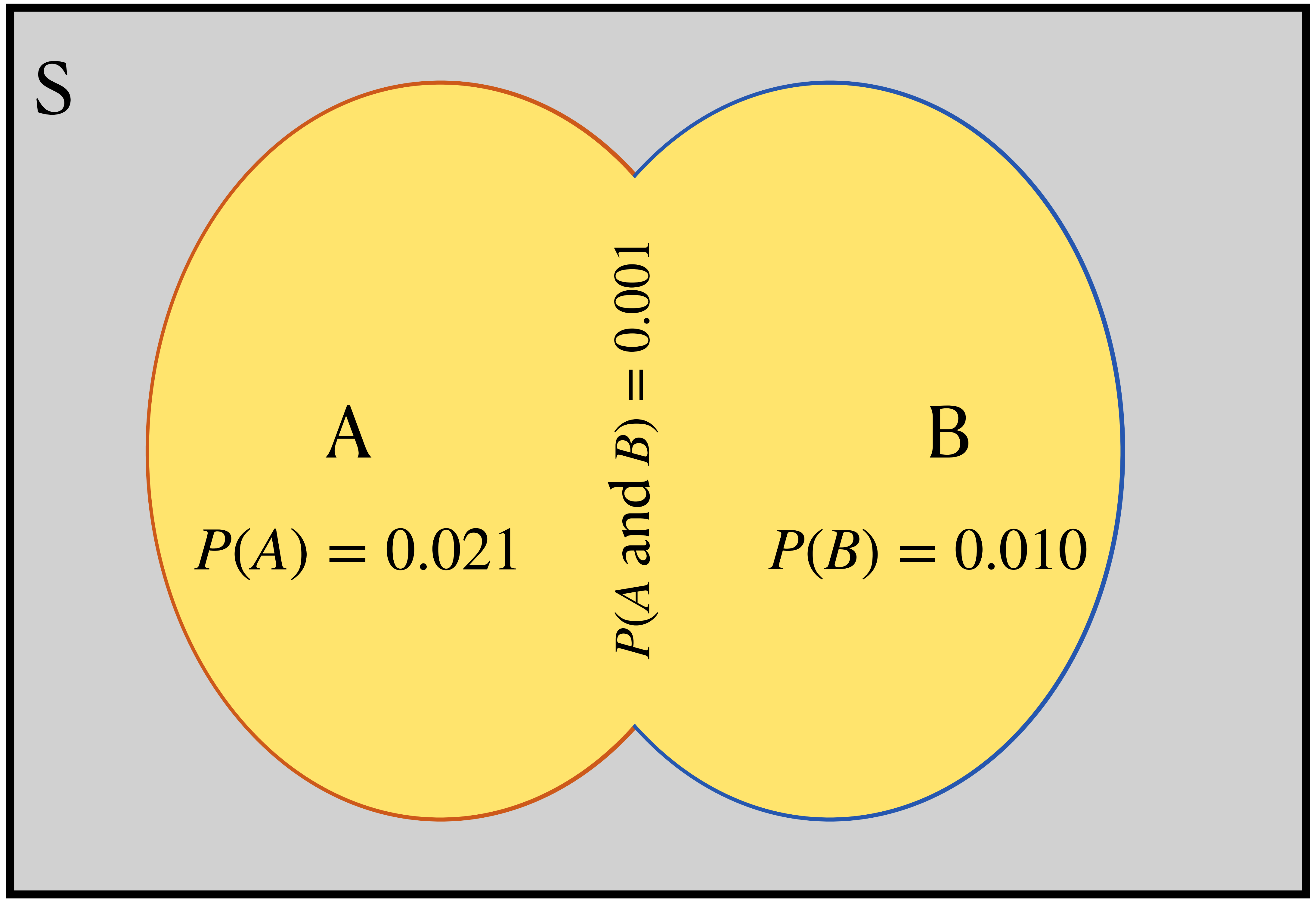
- The probability that at least one of events A or B occurs is denoted P(A or B)
  - The probability of the **union**
  - Where is this on the Venn diagram?
    - P(A or B) is all of the area of A and B combined



Probability Rules

Addition rule

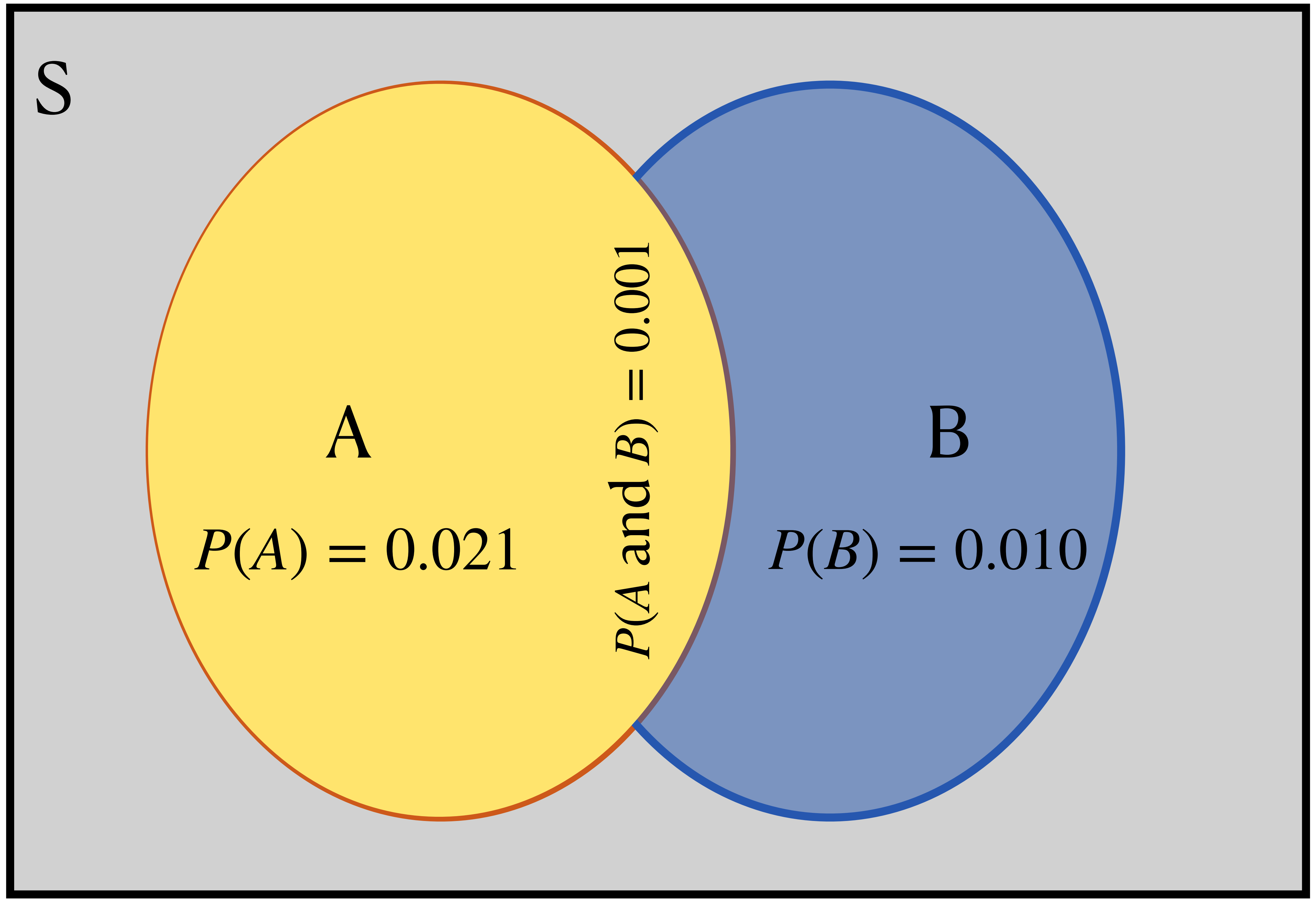
- Event A = *the event that someone giving birth in 2020 had COVID*
- Event B = *the event that someone giving birth in 2020 ended up in the ICU*
- What is P(A or B)?
  - We want the total area of A and B together, so how do we get it…?



Probability Rules

Addition rule

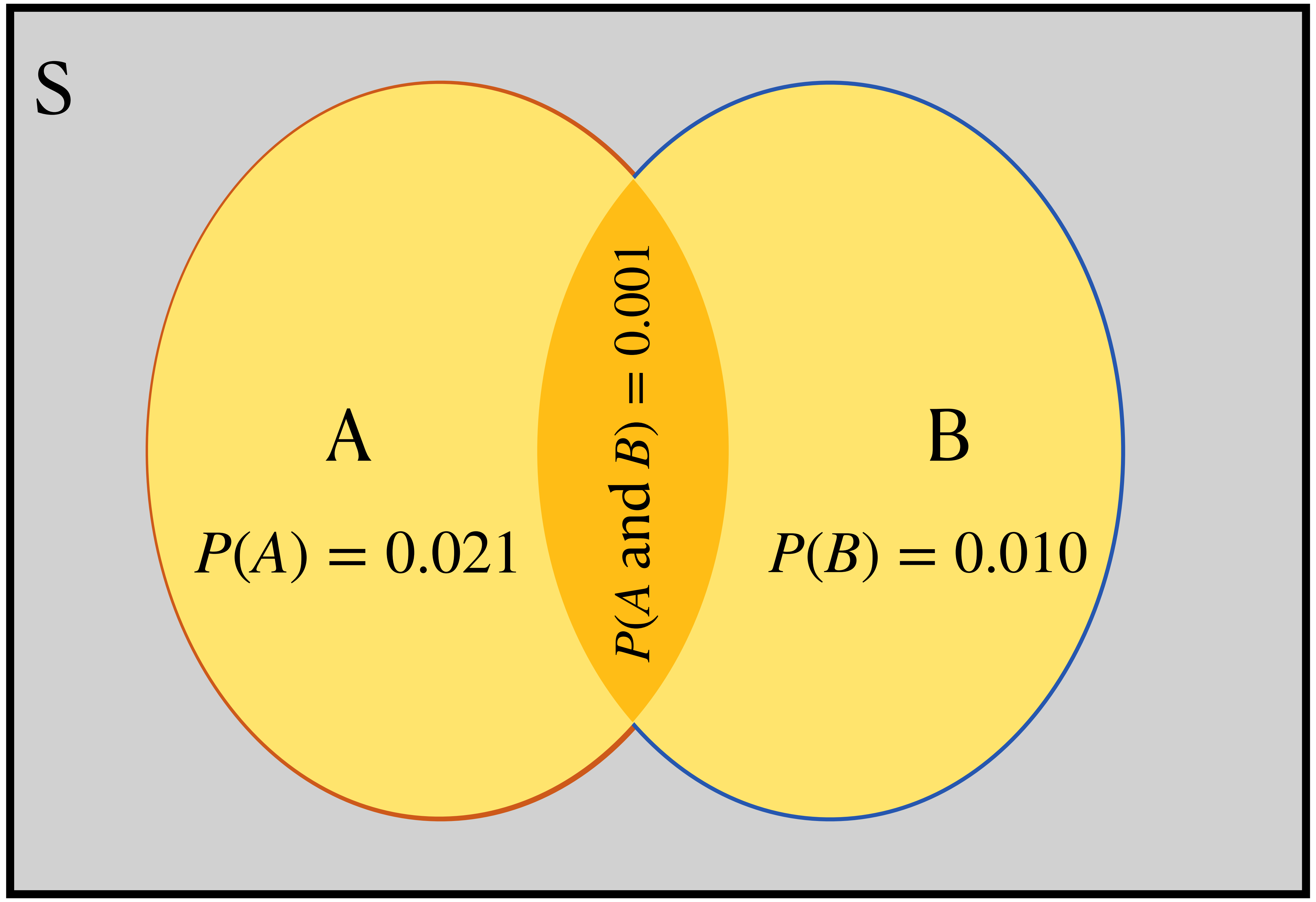
- Event A = *the event that someone giving birth in 2020 had COVID*
- Event B = *the event that someone giving birth in 2020 ended up in the ICU*
- What is P(A or B)?
  - P(A) + …



Probability Rules

Addition rule

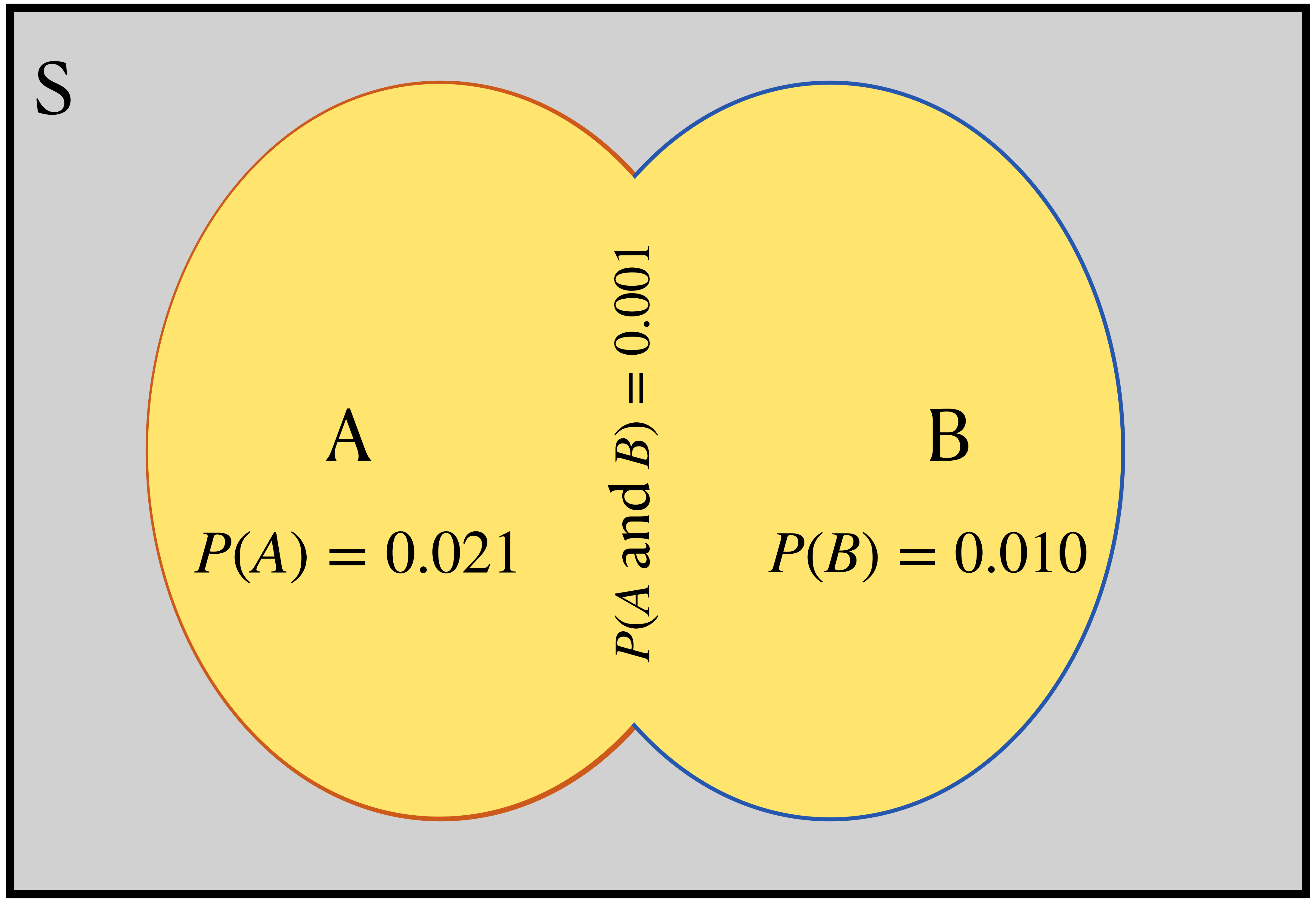
- Event A = *the event that someone giving birth in 2020 had COVID*
- Event B = *the event that someone giving birth in 2020 ended up in the ICU*
- What is P(A or B)?
    - P(A) + P(B)?



Probability Rules

Addition rule

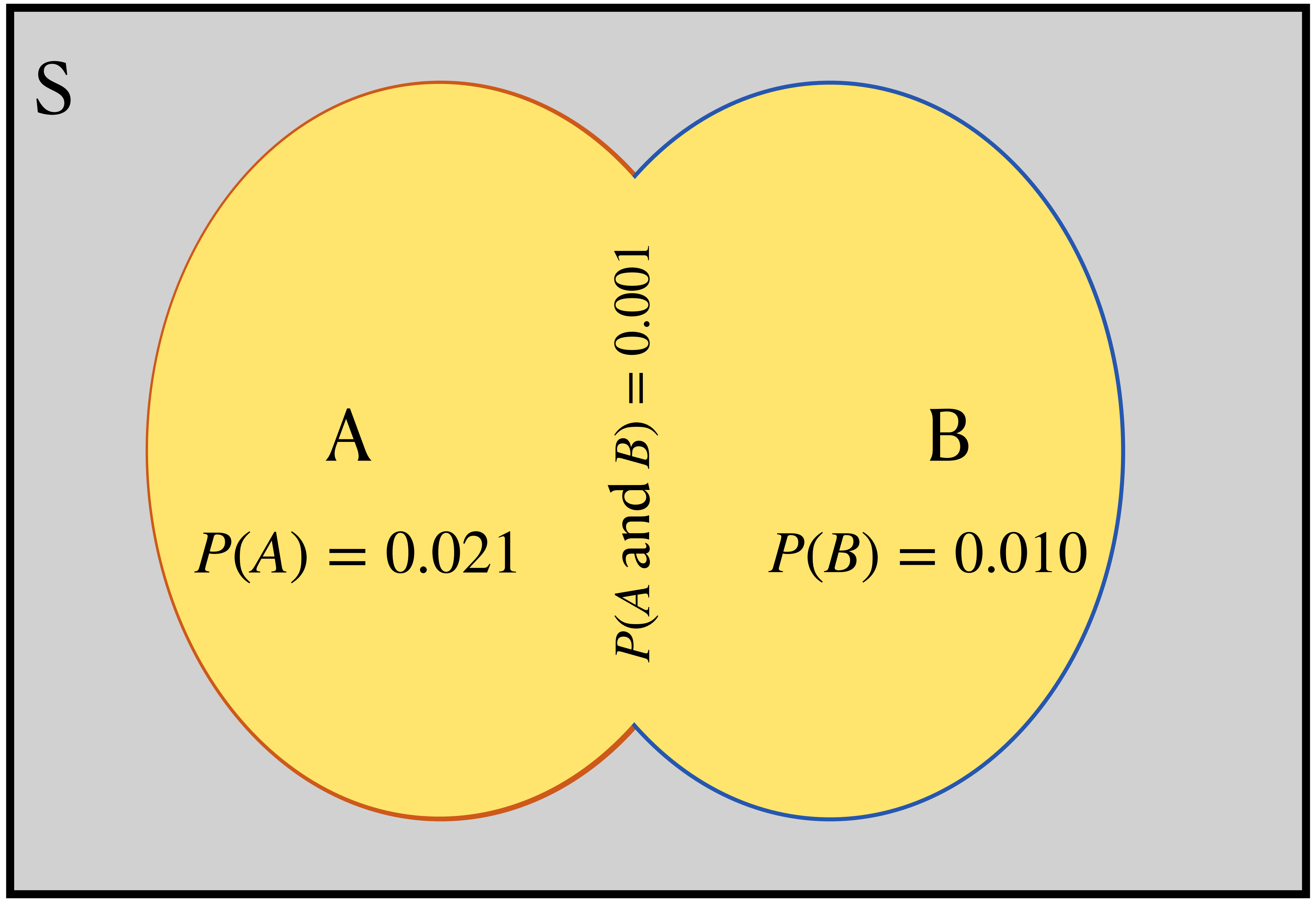
- Event A = *the event that someone giving birth in 2020 had COVID*
- Event B = *the event that someone giving birth in 2020 ended up in the ICU*
- What is P(A or B)?
    - P(A) + P(B) - P(A and B) ✓



Probability Rules

Addition rule

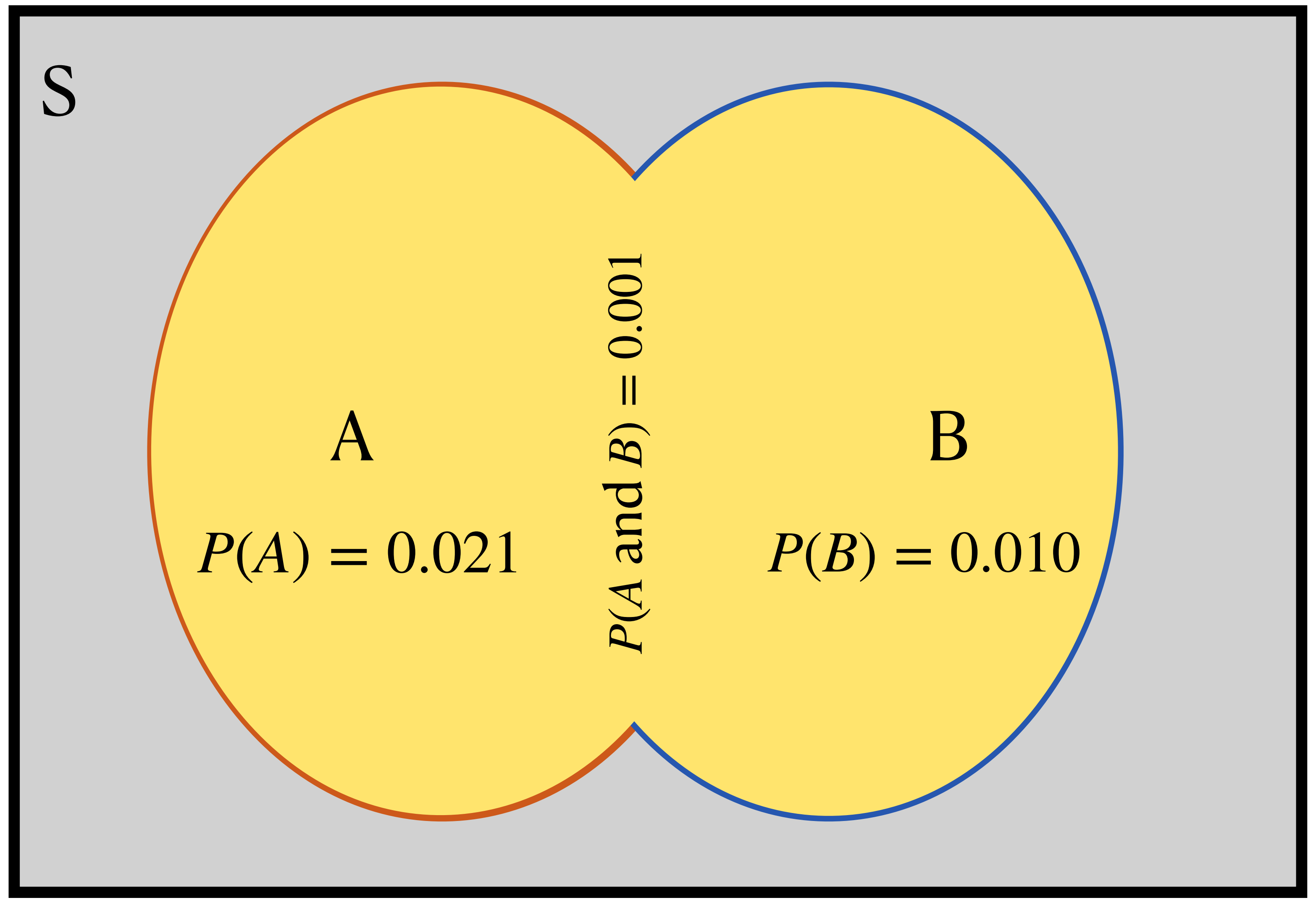
- Event A = *the event that someone giving birth in 2020 had COVID*
- Event B = *the event that someone giving birth in 2020 ended up in the ICU*
  - P(A or B) = P(A) + P(B) - P(A and B) ✓
    - This is called the **addition rule**
  - P(A or B) = 0.021 + 0.010 - 0.001 = 0.03



Probability Rules

Addition rule

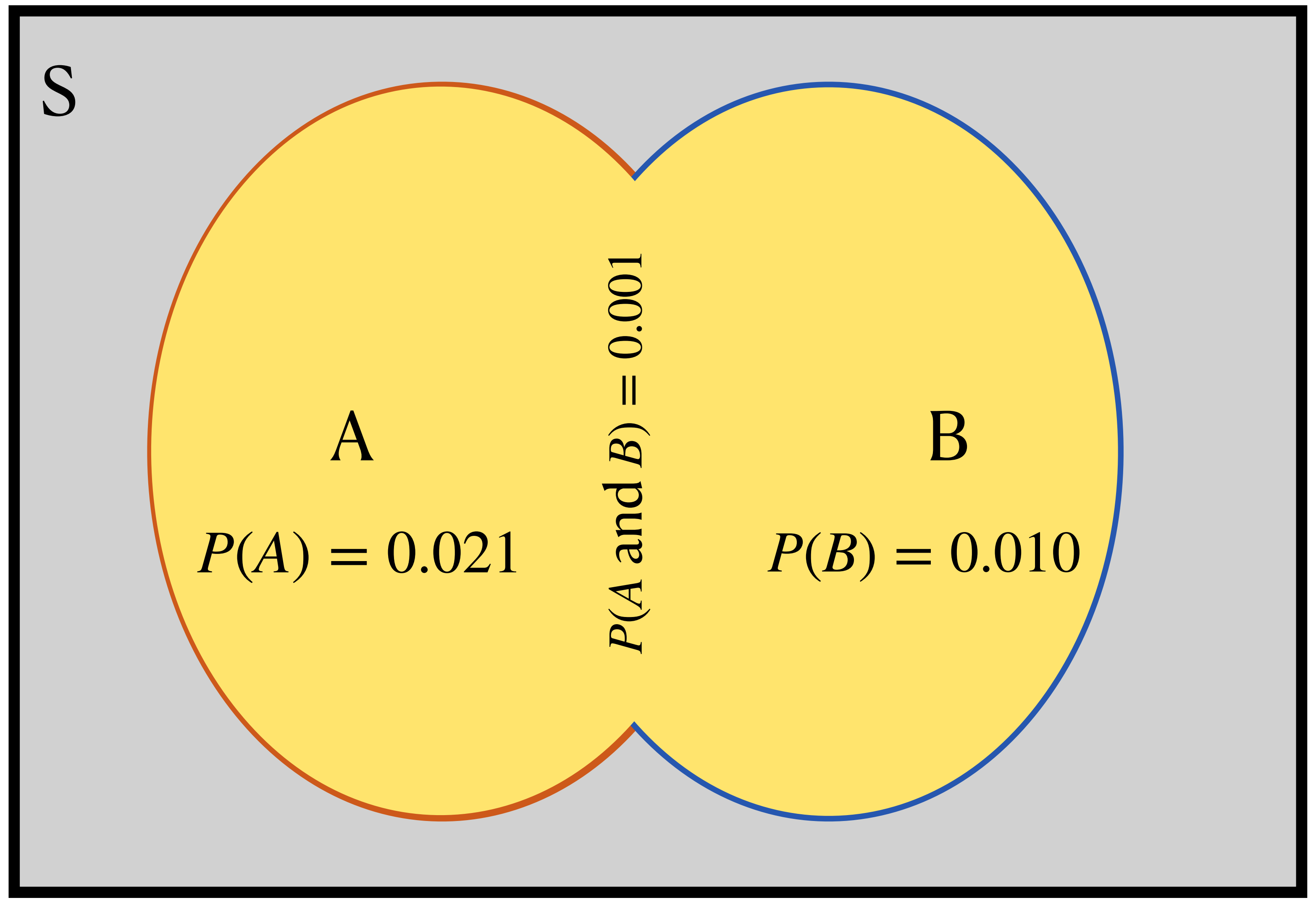
- Event A = *the event that someone giving birth in 2020 had COVID*
- Event B = *the event that someone giving birth in 2020 ended up in the ICU*
- P(A or B) = 0.021 + 0.010 - 0.001 = 0.03
- But couldn't we do this with the contingency table too?



	Had COVID during birth	Did not have COVID during birth	Total
Admitted to ICU	977	7943	8920
Not admitted to ICU	17738	842241	859979
Total	18715	850184	868899

Probability Rules

Addition rule



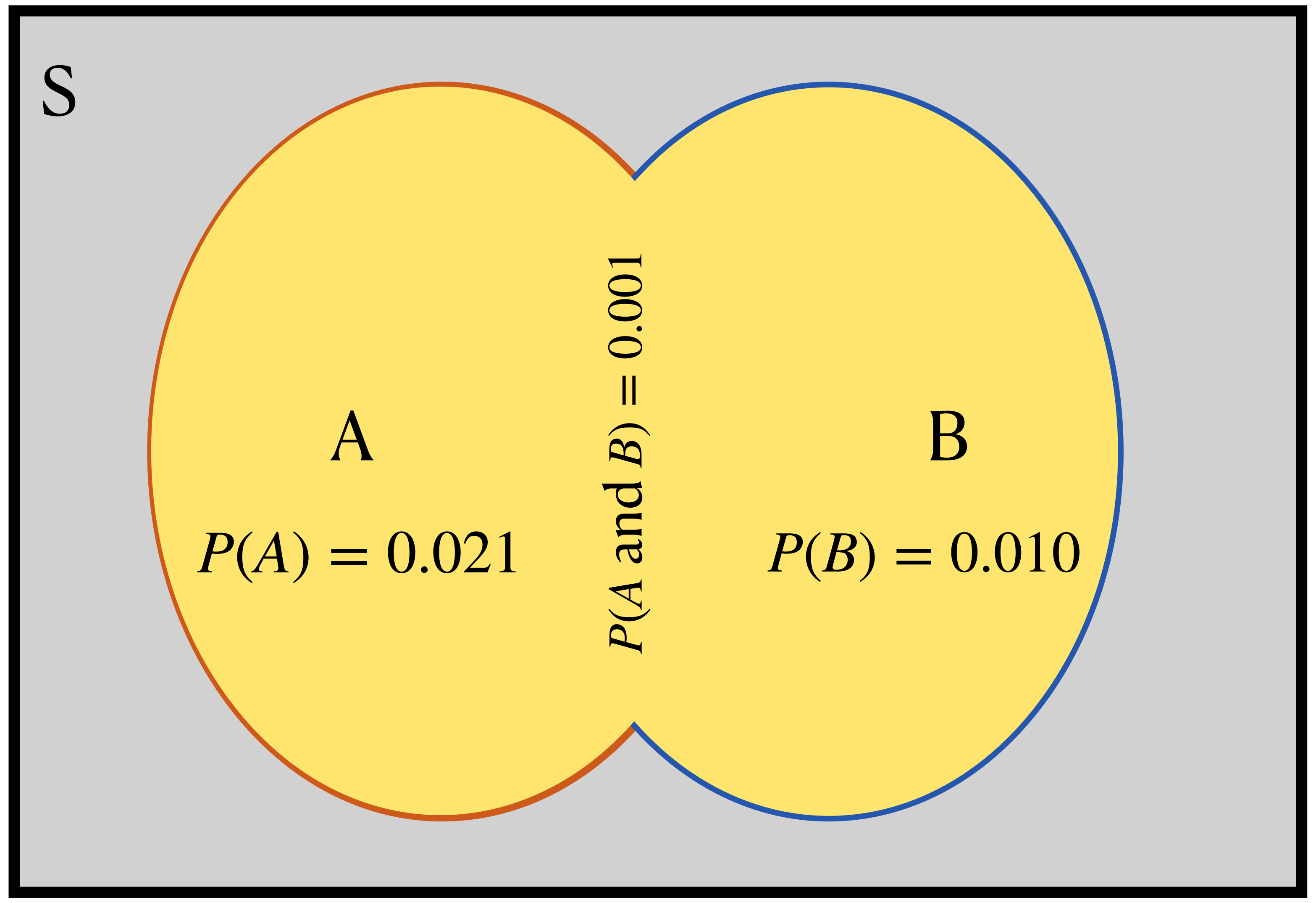
- Event A = *the event that someone giving birth in 2020 had COVID*
- Event B = *the event that someone giving birth in 2020 ended up in the ICU*
- P(A or B) = 0.021 + 0.010 - 0.001 = 0.03
- But couldn't we do this with the contingency table too?
  - Size of (A or B) is…
  - PollEv.com/stats8

	Had COVID during birth	Did not have COVID during birth	Total
Admitted to ICU	977	7943	8920
Not admitted to ICU	17738	842241	859979
Total	18715	850184	868899

Probability Rules

Addition rule

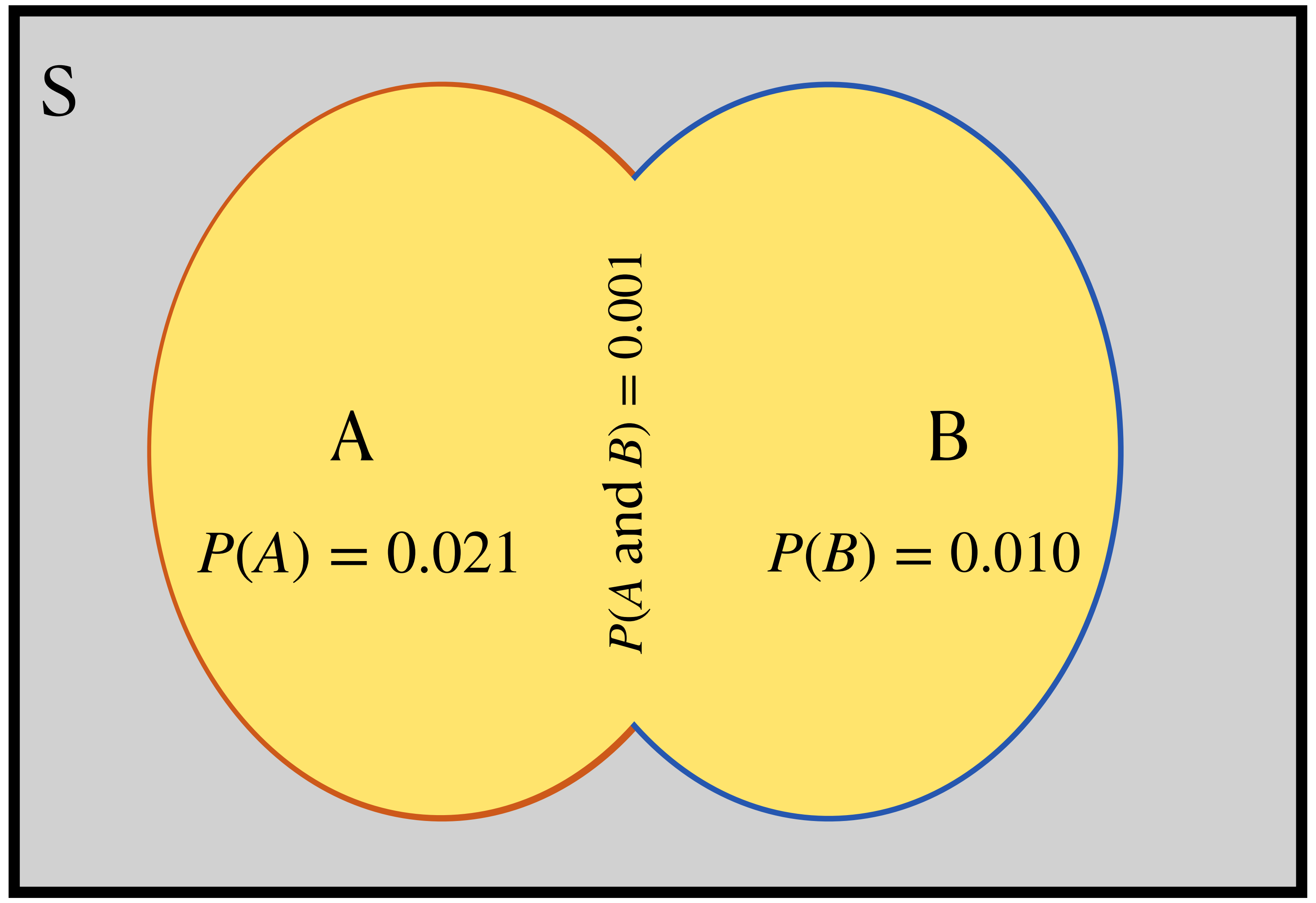
- Event A = *the event that someone giving birth in 2020 had COVID*
- Event B = *the event that someone giving birth in 2020 ended up in the ICU*
- P(A or B) = 0.021 + 0.010 - 0.001 = 0.03
- But couldn't we do this with the contingency table too?
  - Size of (A or B) = (number of people who had COVID during birth) + (number who were admitted to ICU) - (the overlap we double-counted)



	Had COVID during birth	**Did not have COVID during birth**	**Total**
Admitted to ICU	977	7943	8920
Not admitted to ICU	17738	842241	859979
Total	18715	850184	868899

Probability Rules

Addition rule



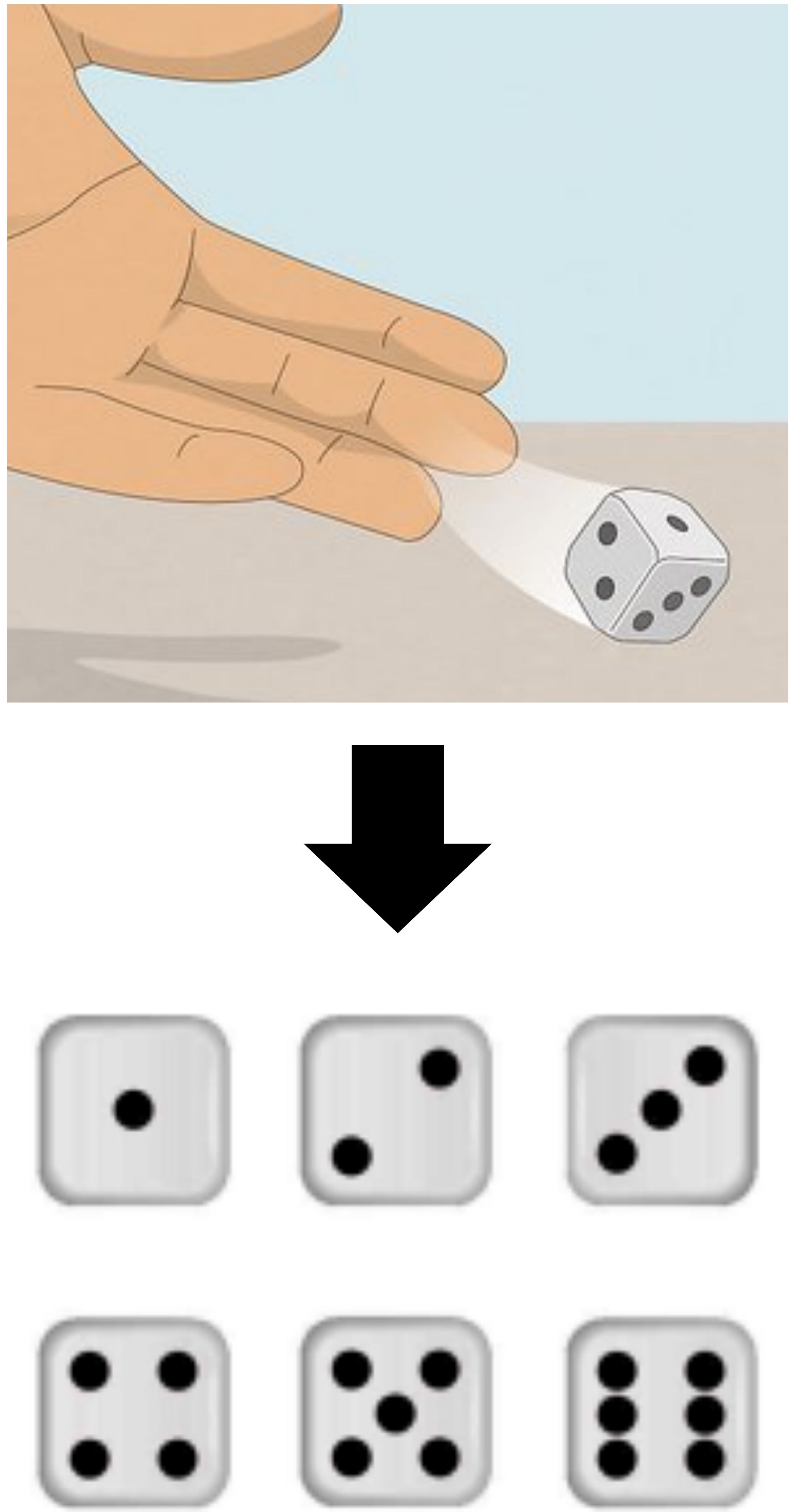
- Event A = *the event that someone giving birth in 2020 had COVID*
- Event B = *the event that someone giving birth in 2020 ended up in the ICU*
- P(A or B) = 0.021 + 0.010 - 0.001 = 0.03
- But couldn't we do this with the contingency table too?
  - Size of (A or B) = 8920 + 18715 - 977 = 26658
  - 26658 / 868899 ≈ 0.03

	Had COVID during birth	Did not have COVID during birth	Total
Admitted to ICU	977	7943	8920
Not admitted to ICU	17738	842241	859979
Total	18715	850184	868899

Probability Rules

Dice example

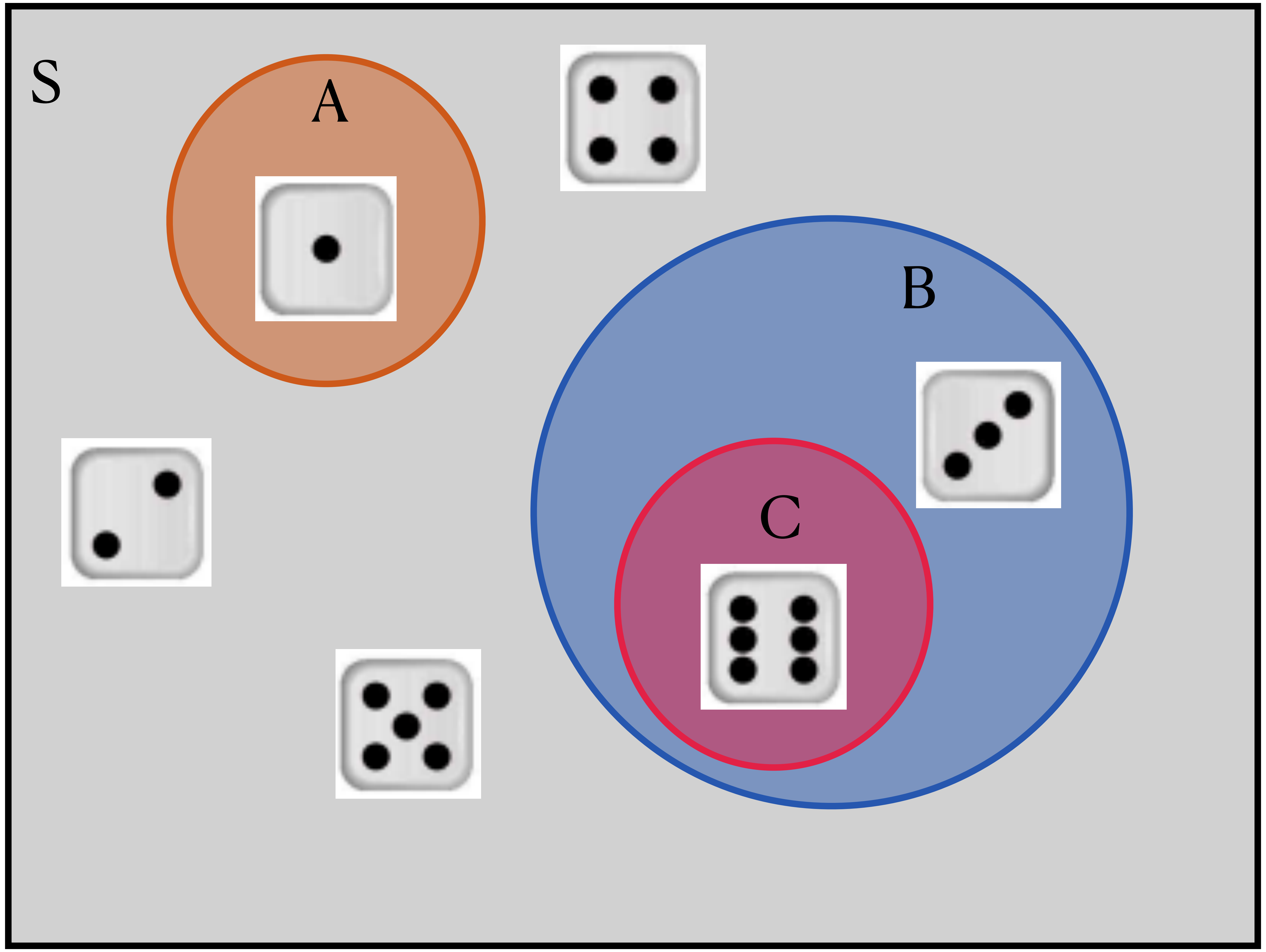
- One fair, six-sided die is rolled (so the probability of seeing any number 1-6 is 1/6)
- Event A = *the event that a rolled die yields a 1*
- Event B = *the event that a rolled die yields a number divisible by 3*
- Event C = *the event that a rolled die yields a 6*



Probability Rules

Dice example

- One fair, six-sided die is rolled (so the probability of seeing any number 1-6 is 1/6)
- Event A = *the event that a rolled die yields a 1*
- Event B = *the event that a rolled die yields a number divisible by 3*
- Event C = *the event that a rolled die yields a 6*

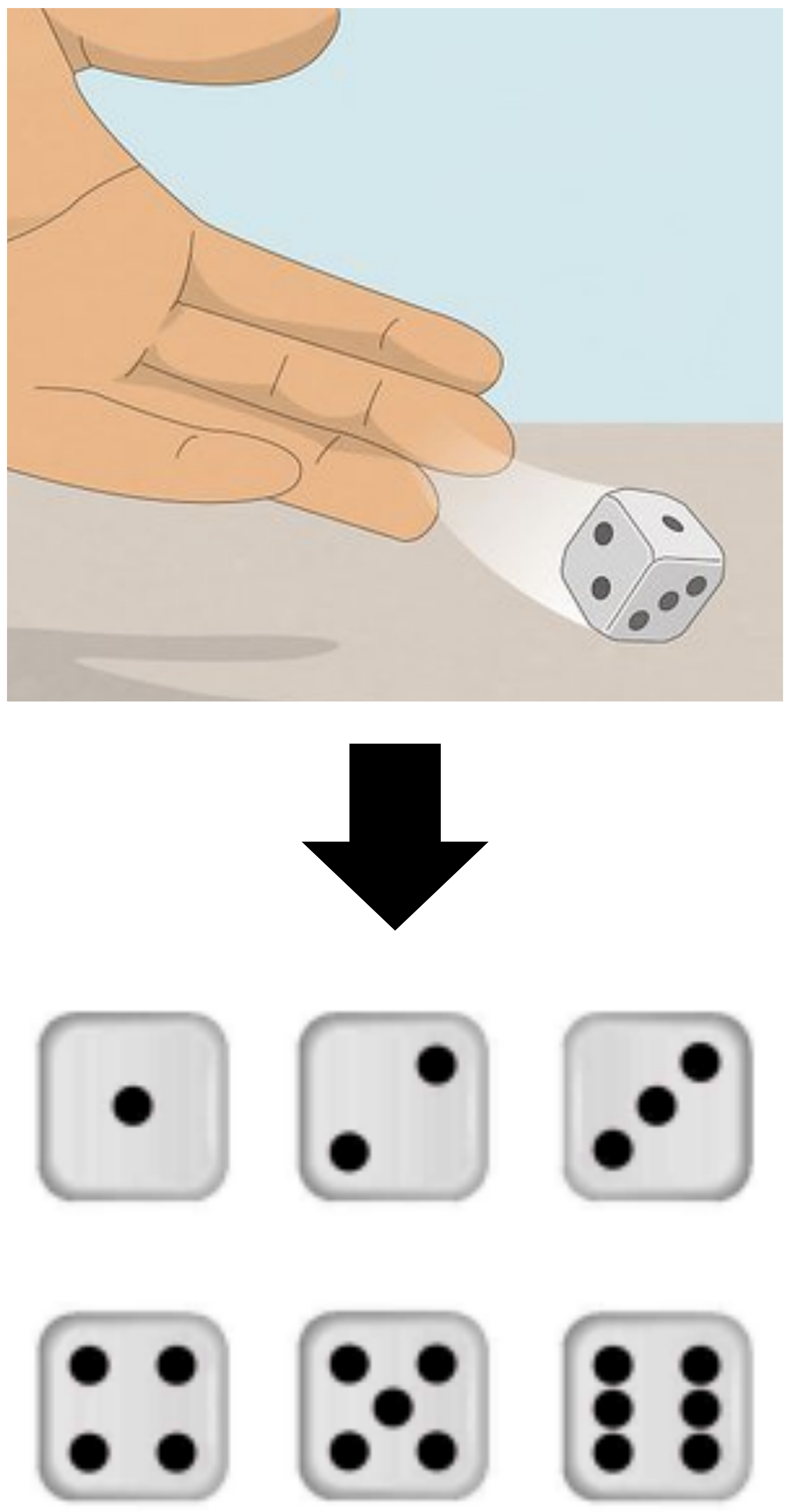


Probability Rules

Dice example

- One fair, six-sided die is rolled (so the probability of seeing any number 1-6 is 1/6)
- Event A = *the event that a rolled die yields a 1*
- Event B = *the event that a rolled die yields a number divisible by 3*
- Event C = *the event that a rolled die yields a 6*

- What is P(A)? PollEv.com/stats8
    - Answer in fraction form (i.e. 1/2)

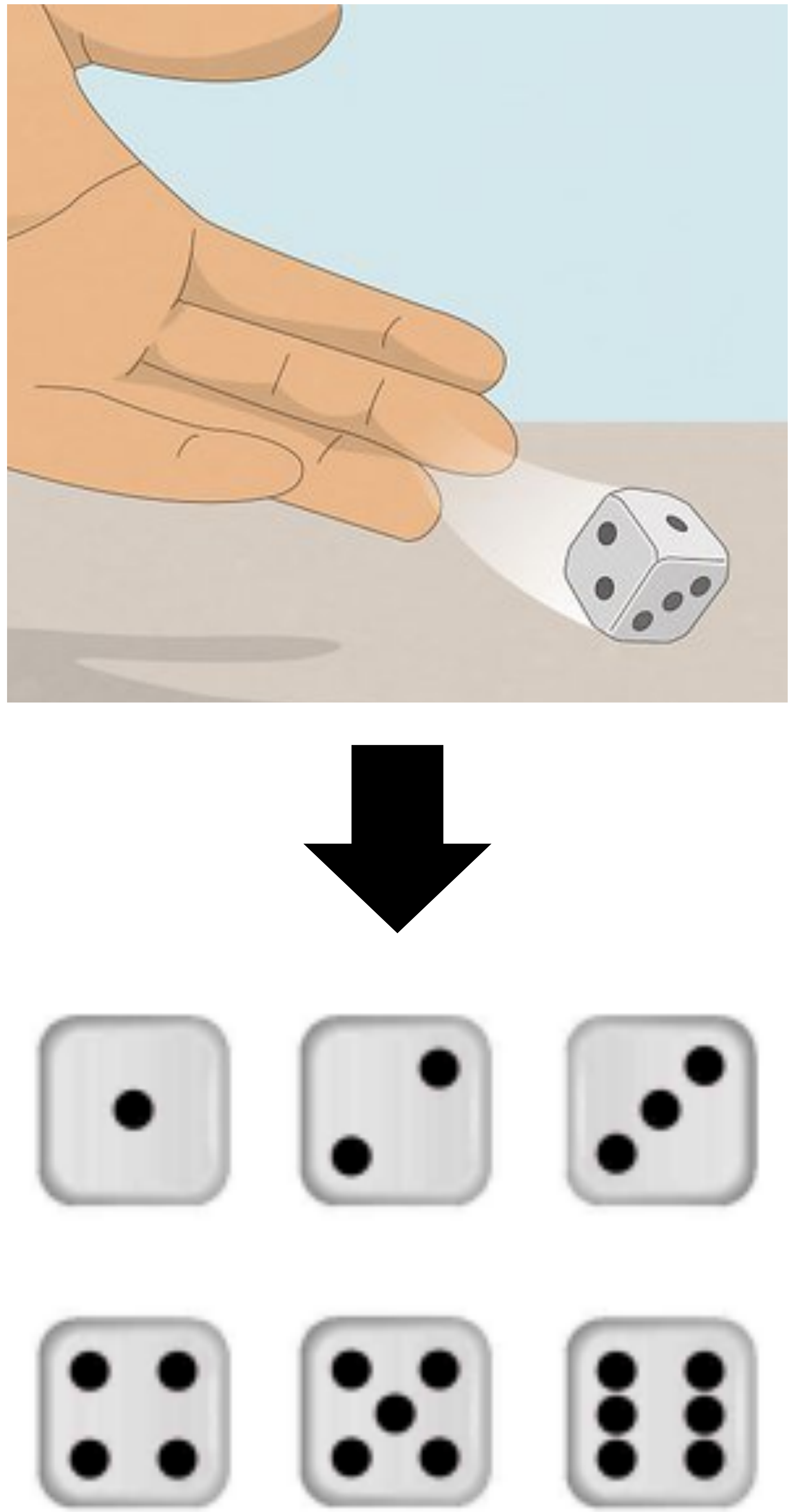


Probability Rules

Dice example

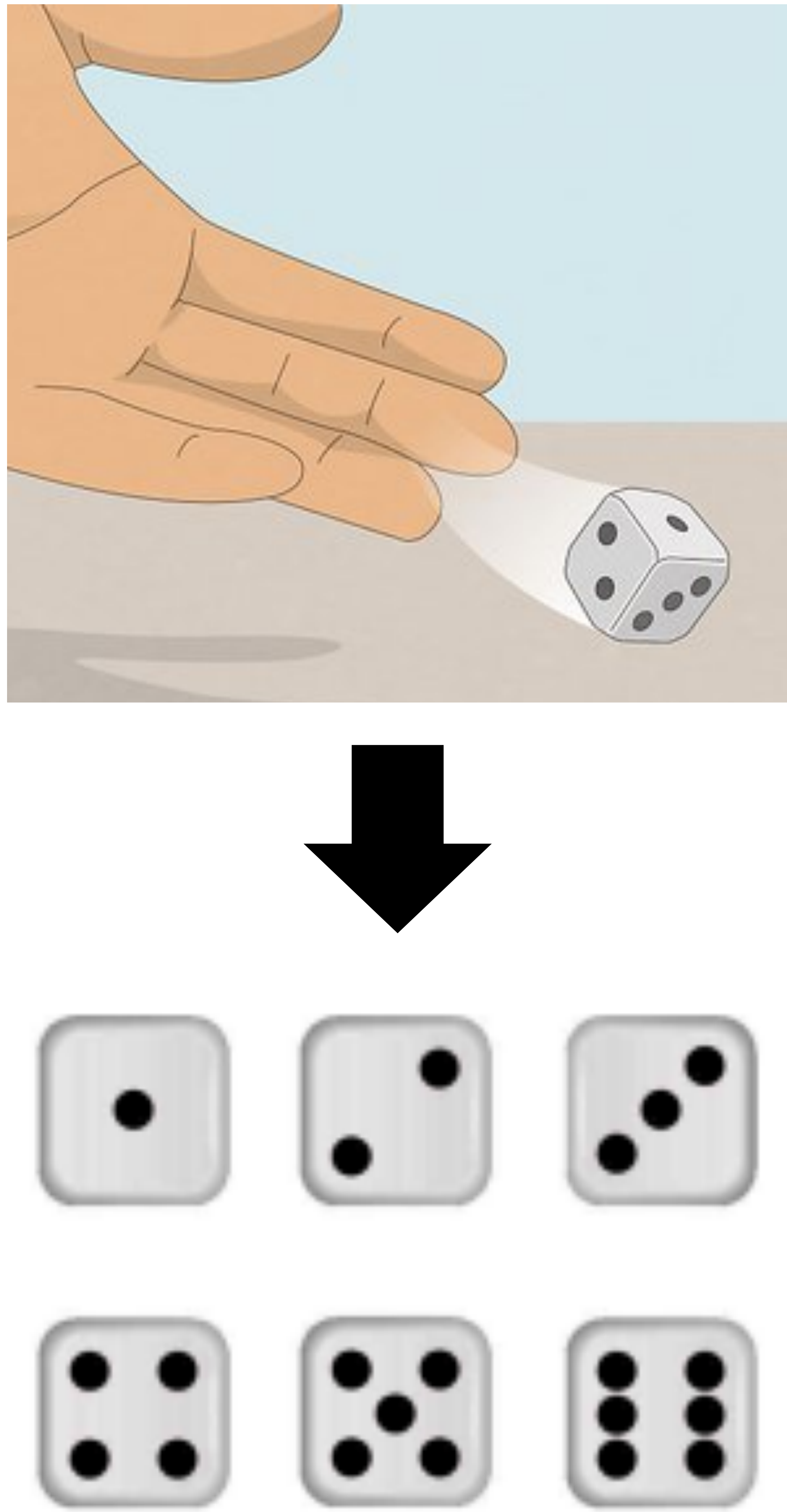
- One fair, six-sided die is rolled (so the probability of seeing any number 1-6 is 1/6)
- Event A = *the event that a rolled die yields a 1* = **1/6**
- Event B = *the event that a rolled die yields a number divisible by 3*
- Event C = *the event that a rolled die yields a 6*

- What is P(A)?
  - P(A) = P({1})= 1/6
  - (rolling a 1 is one out of the 6 equally likely options in the sample space)



Probability Rules

Dice example



- One fair, six-sided die is rolled (so the probability of seeing any number 1-6 is 1/6)
- Event A = *the event that a rolled die yields a 1* = **1/6**
- Event B = *the event that a rolled die yields a number divisible by 3*
- Event C = *the event that a rolled die yields a 6*

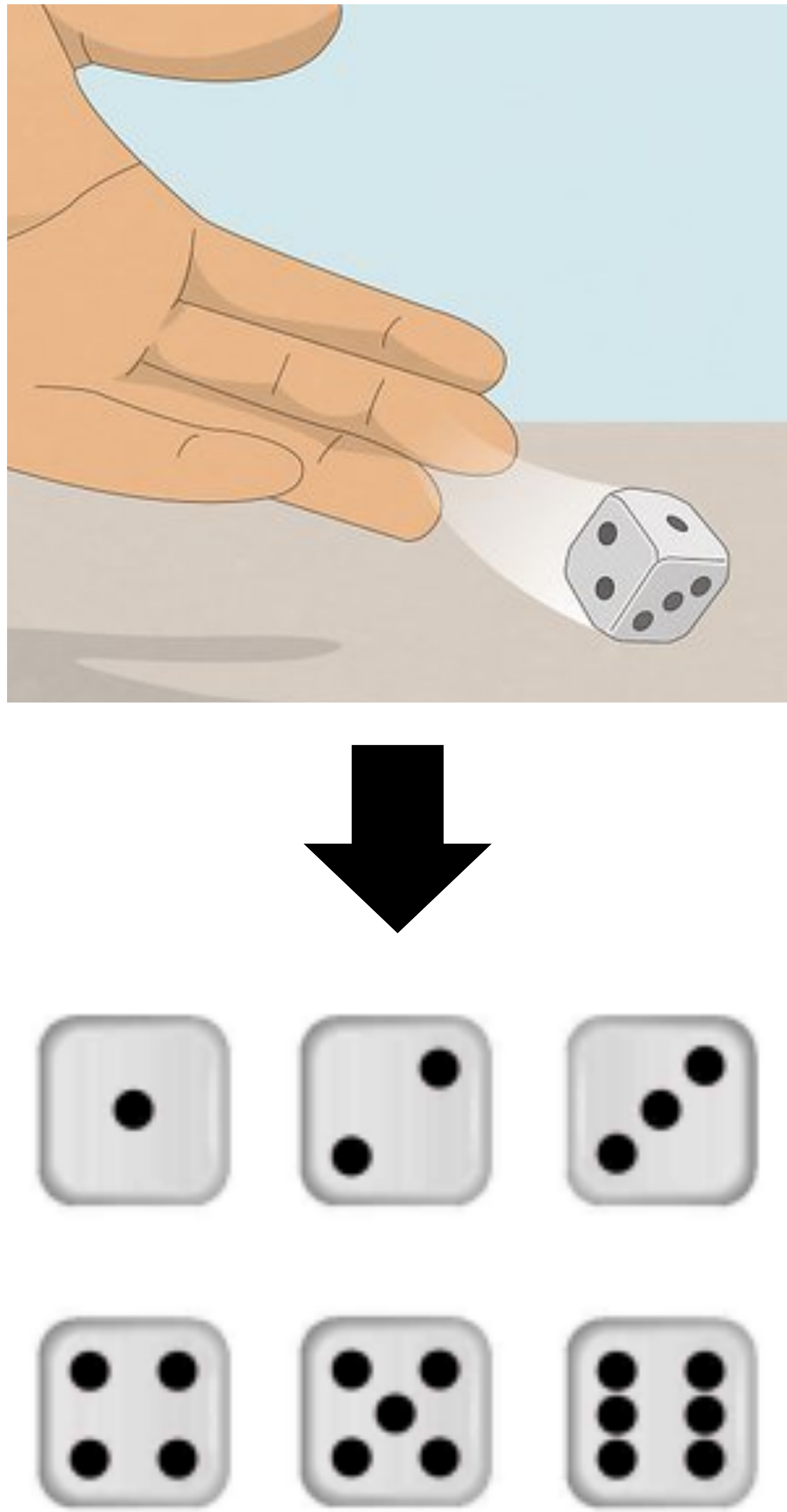
- What is P(B)? PollEv.com/stats8
  - Answer in fraction form (i.e. 1/2)

Probability Rules

Dice example

- One fair, six-sided die is rolled (so the probability of seeing any number 1-6 is 1/6)
- Event A = *the event that a rolled die yields a 1* = **1/6**
- Event B = *the event that a rolled die yields a number divisible by 3* = **2/6**
- Event C = *the event that a rolled die yields a 6*

- What is P(B)?
  - P(B) = P({3, 6})
    - **2/6** because 2 equally likely options out of 6 total
    - P(3) + P(6) = 1/6 + 1/6 = **2/6** by addition rule

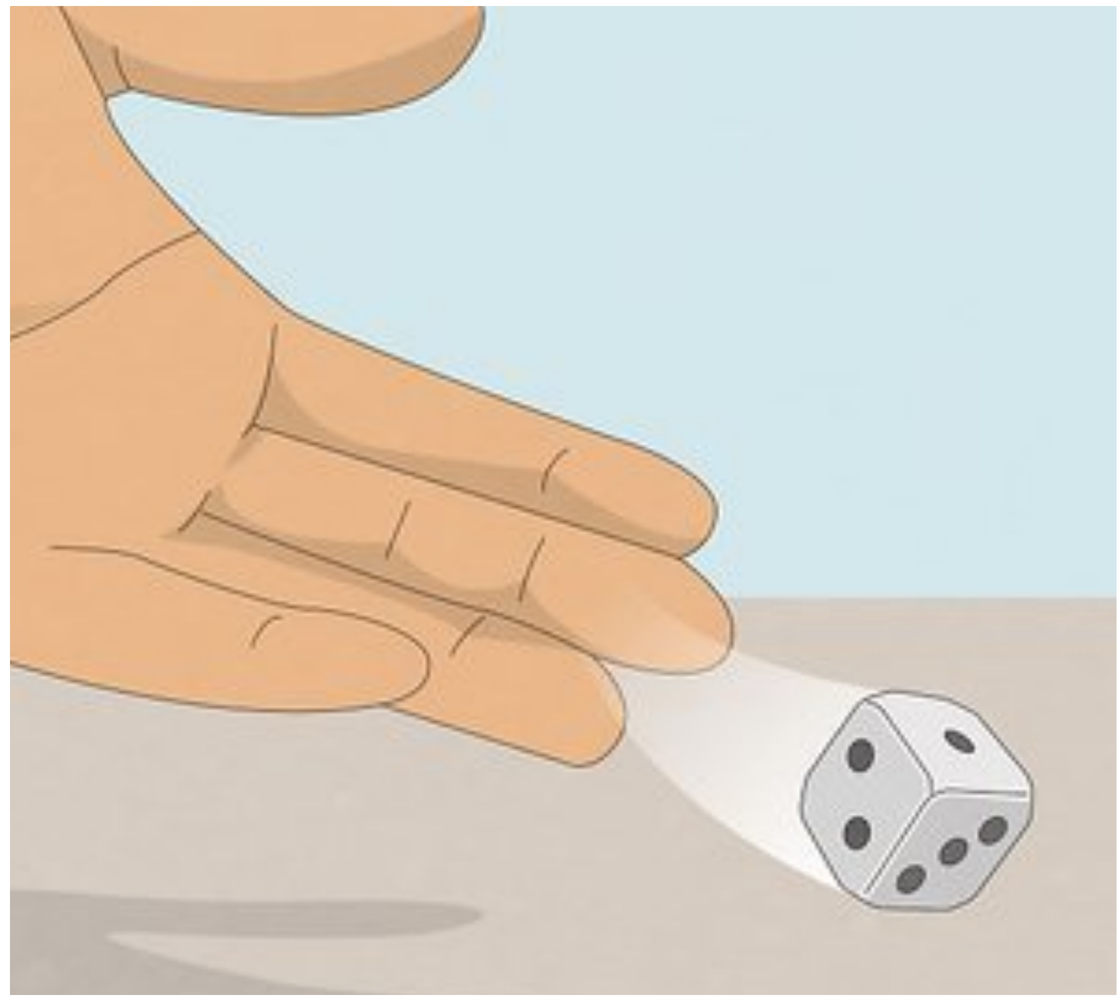


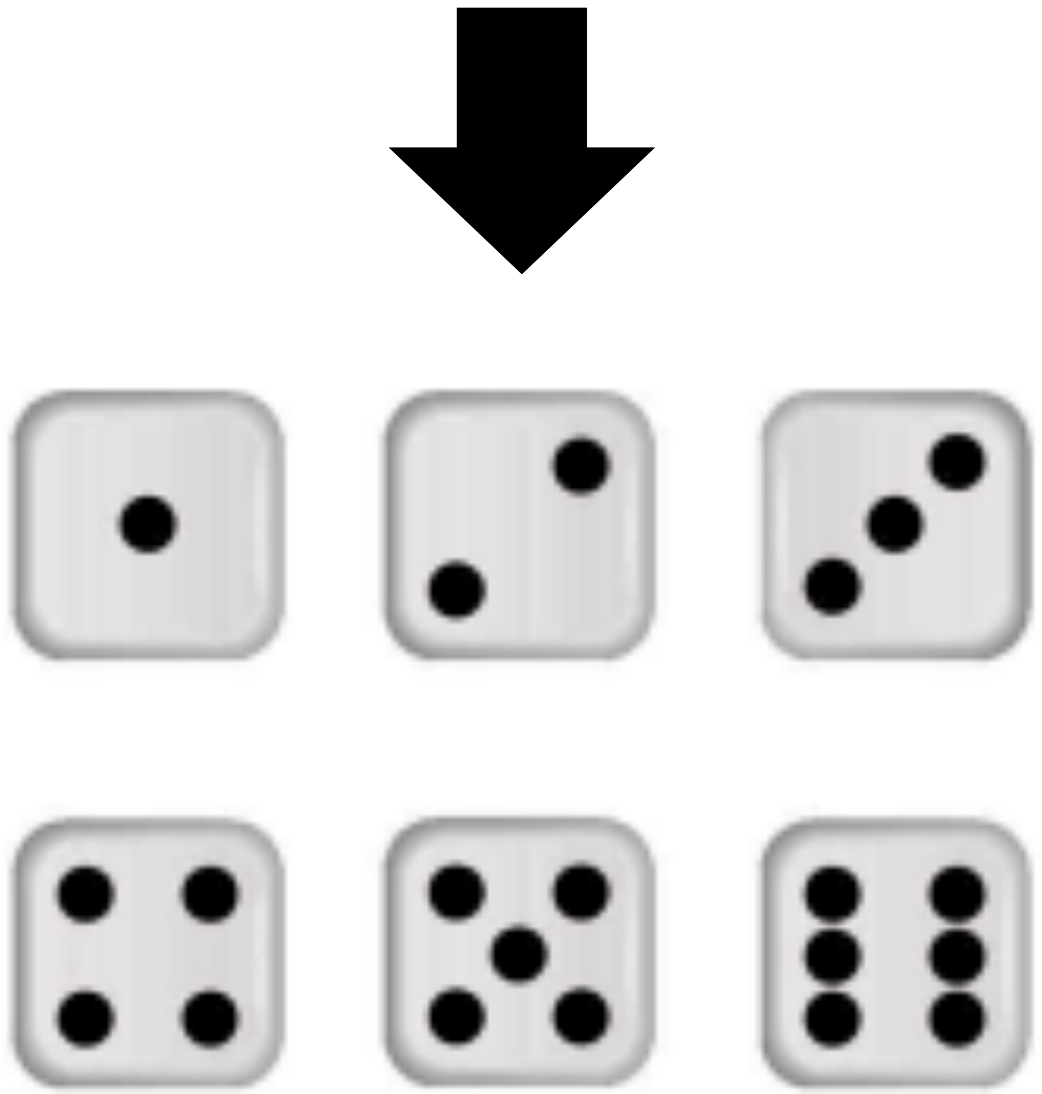
Probability Rules

Dice example

- One fair, six-sided die is rolled (so the probability of seeing any number 1-6 is 1/6)
- Event A = *the event that a rolled die yields a 1* = **1/6**
- Event B = *the event that a rolled die yields a number divisible by 3* = **2/6**
- Event C = *the event that a rolled die yields a 6*

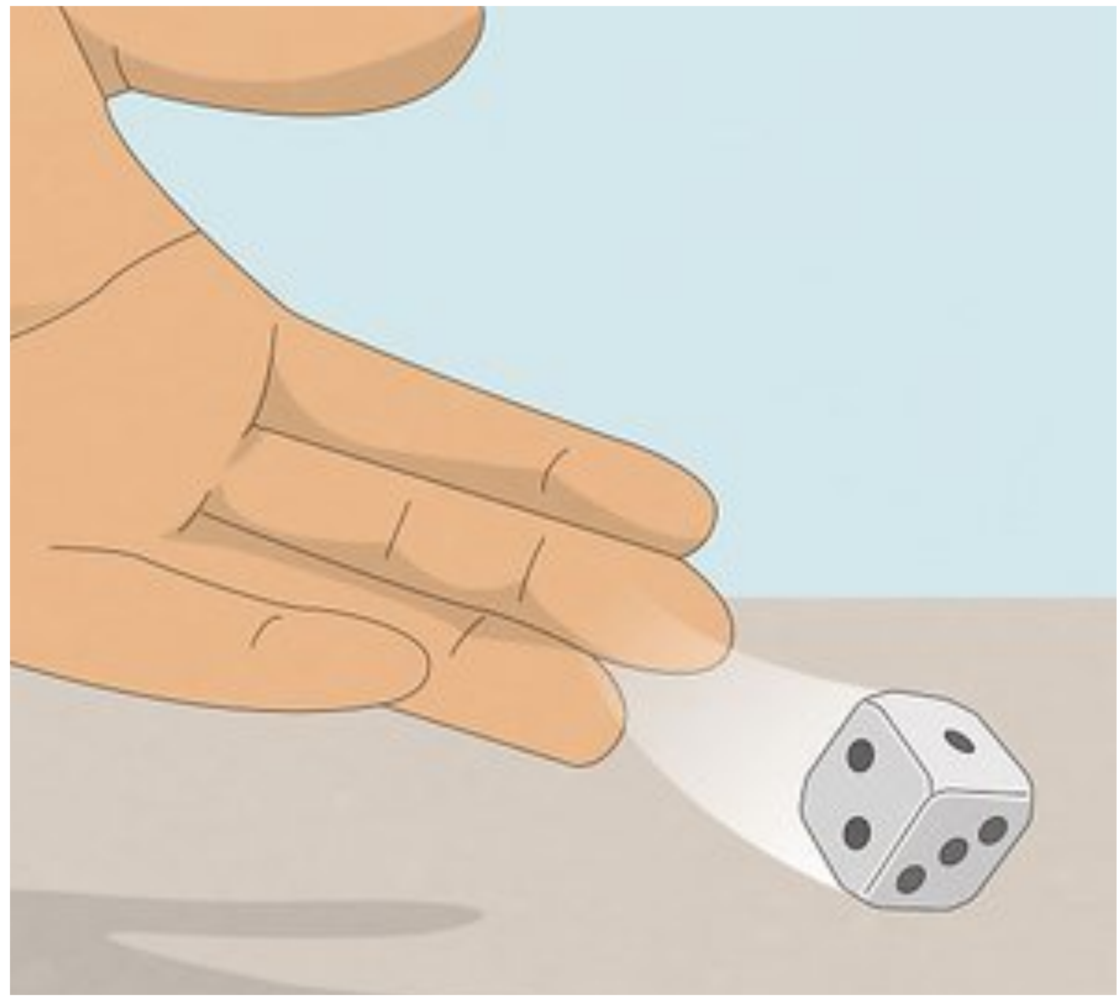
- What is P(C)? PollEv.com/stats8
  - Answer in fraction form (i.e. 1/2)



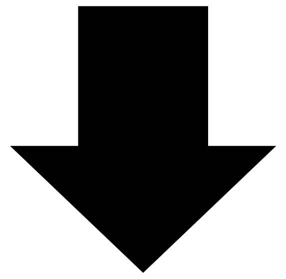


Probability Rules

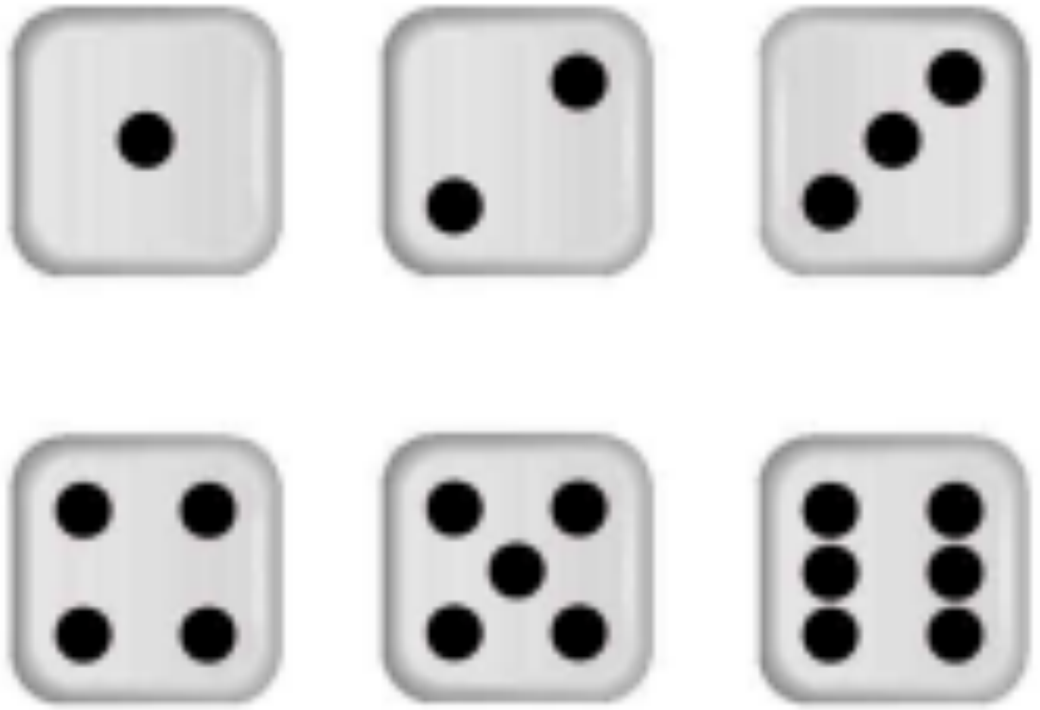
Dice example



- One fair, six-sided die is rolled (so the probability of seeing any number 1-6 is 1/6)
- Event A = *the event that a rolled die yields a 1* = **1/6**
- Event B = *the event that a rolled die yields a number divisible by 3* = **2/6**
- Event C = *the event that a rolled die yields a 6* = **1/6**



- What is P(C)?
  - P(C) = P({6}) = 1/6

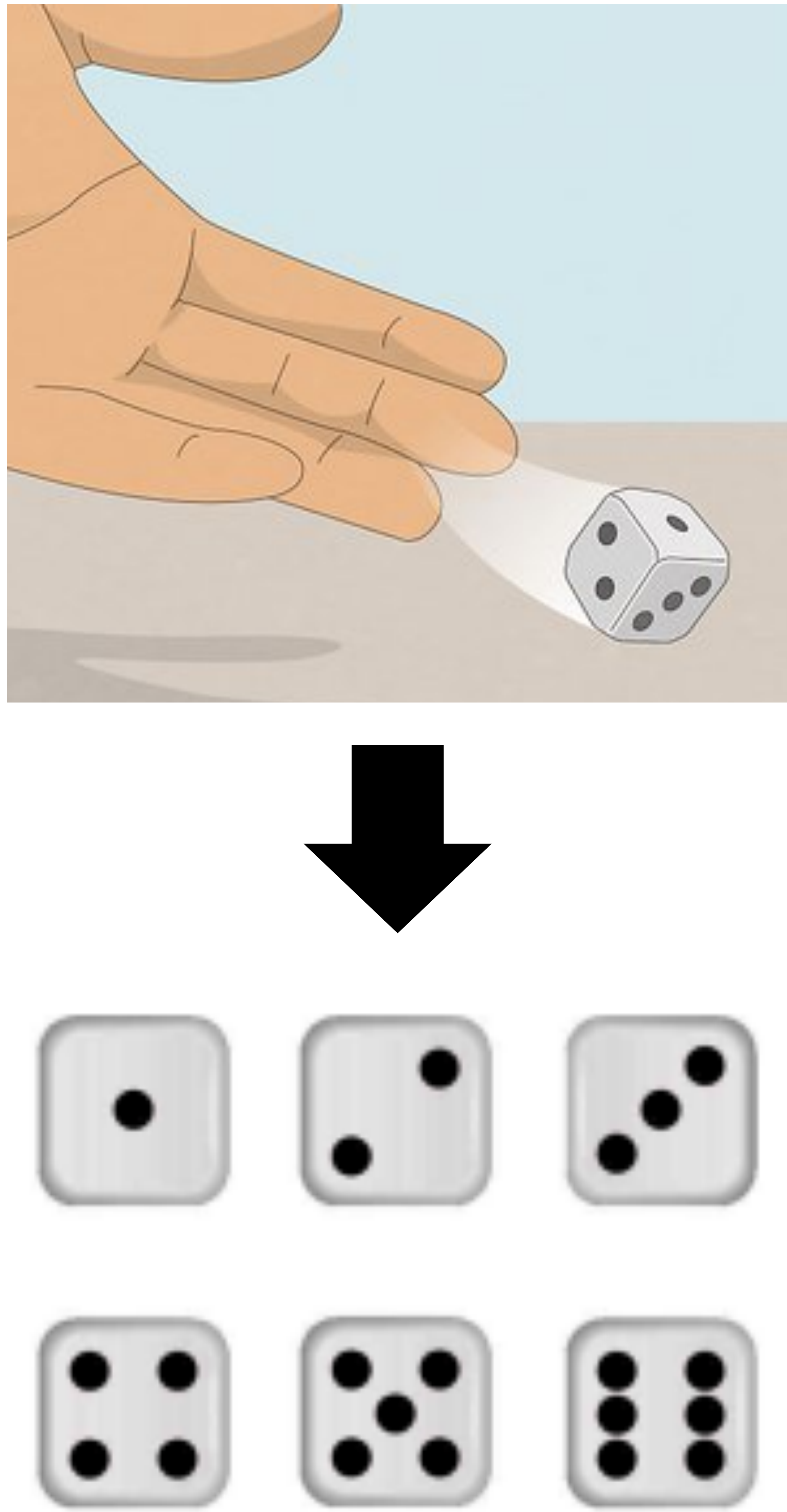


Probability Rules

Dice example

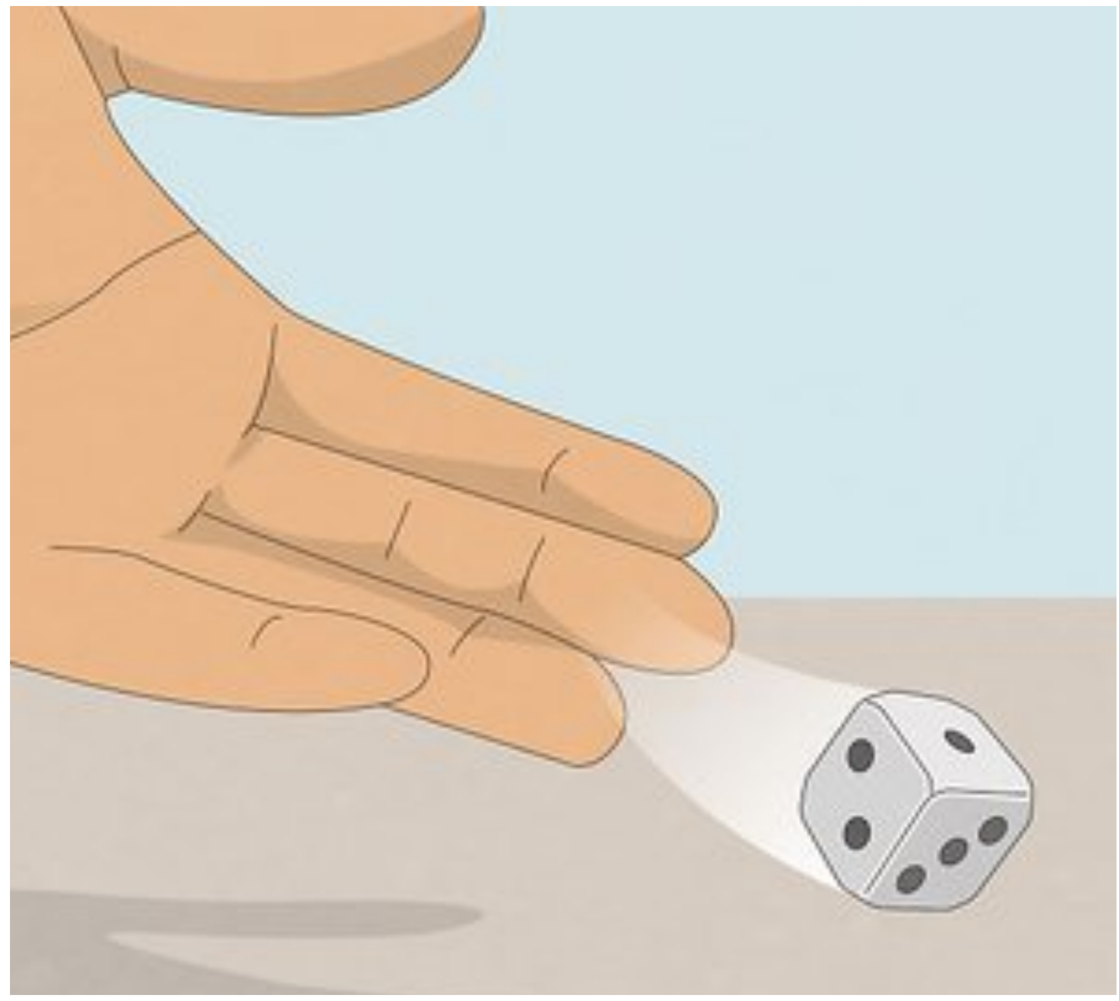
- One fair, six-sided die is rolled (so the probability of seeing any number 1-6 is 1/6)
- Event A = *the event that a rolled die yields a 1* = **1/6**
- Event B = *the event that a rolled die yields a number divisible by 3* = **2/6**
- Event C = *the event that a rolled die yields a 6* = **1/6**

- What is P(A or B)? PollEv.com/stats8
  - Answer in fraction form (i.e. 1/2)

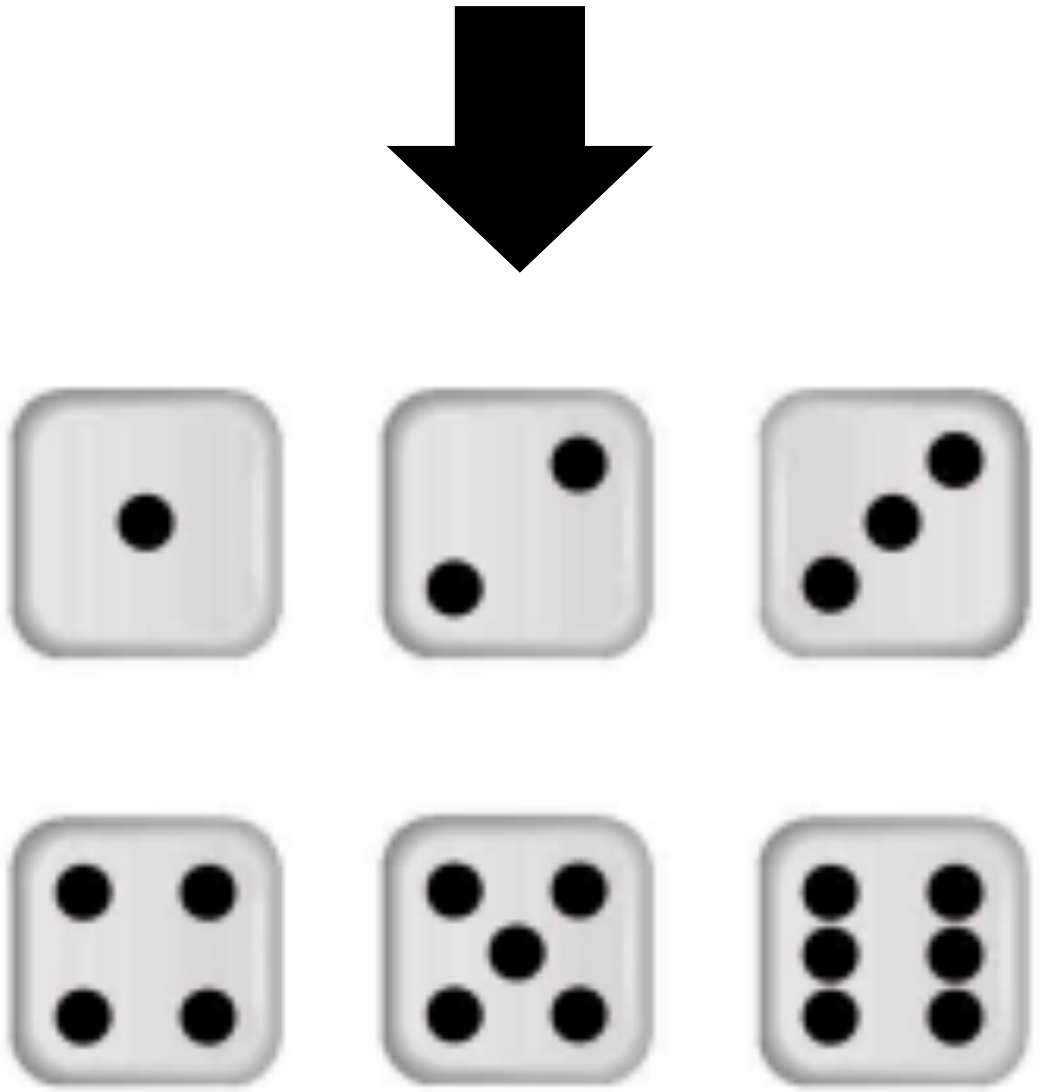


Probability Rules

Dice example



- One fair, six-sided die is rolled (so the probability of seeing any number 1-6 is 1/6)
- Event A = *the event that a rolled die yields a 1* = **1/6**
- Event B = *the event that a rolled die yields a number divisible by 3* = **2/6**
- Event C = *the event that a rolled die yields a 6* = **1/6**

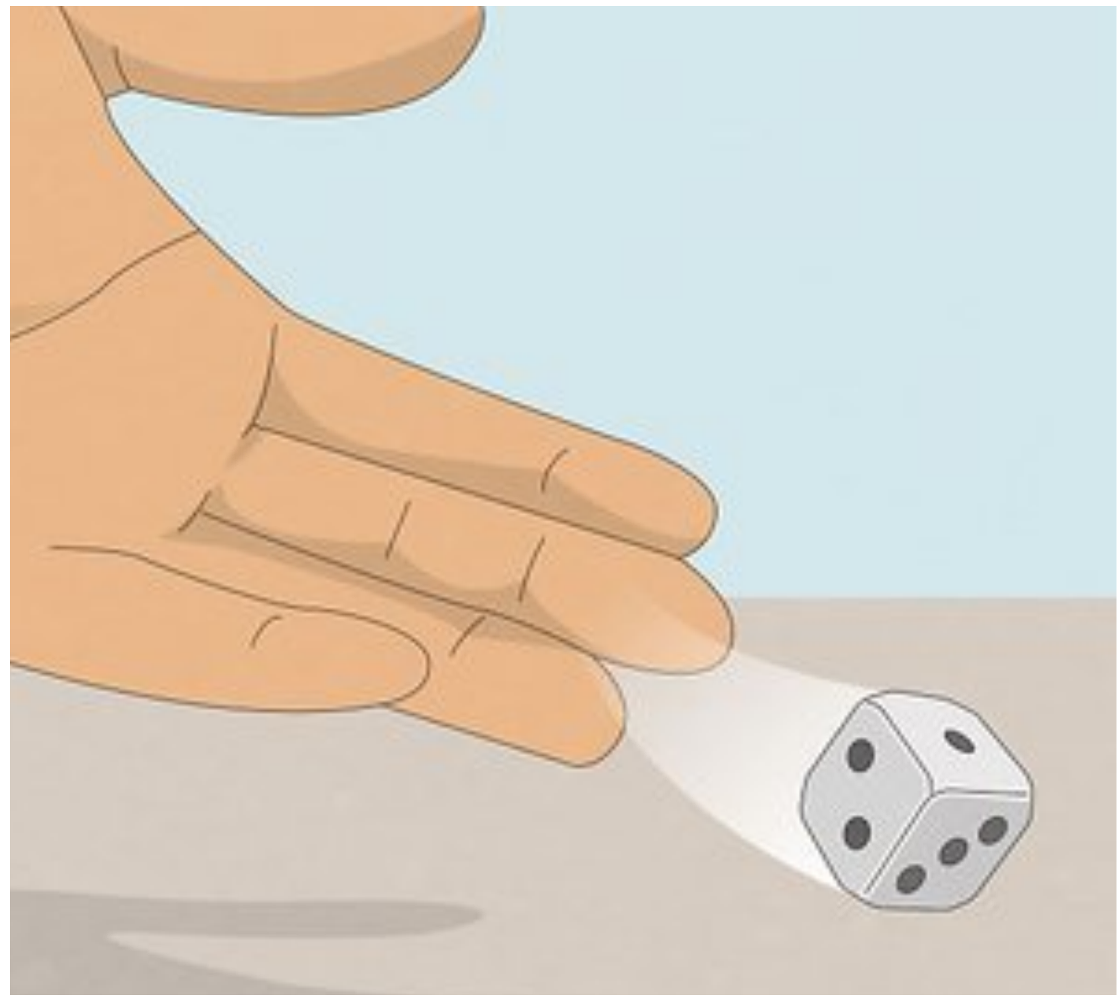


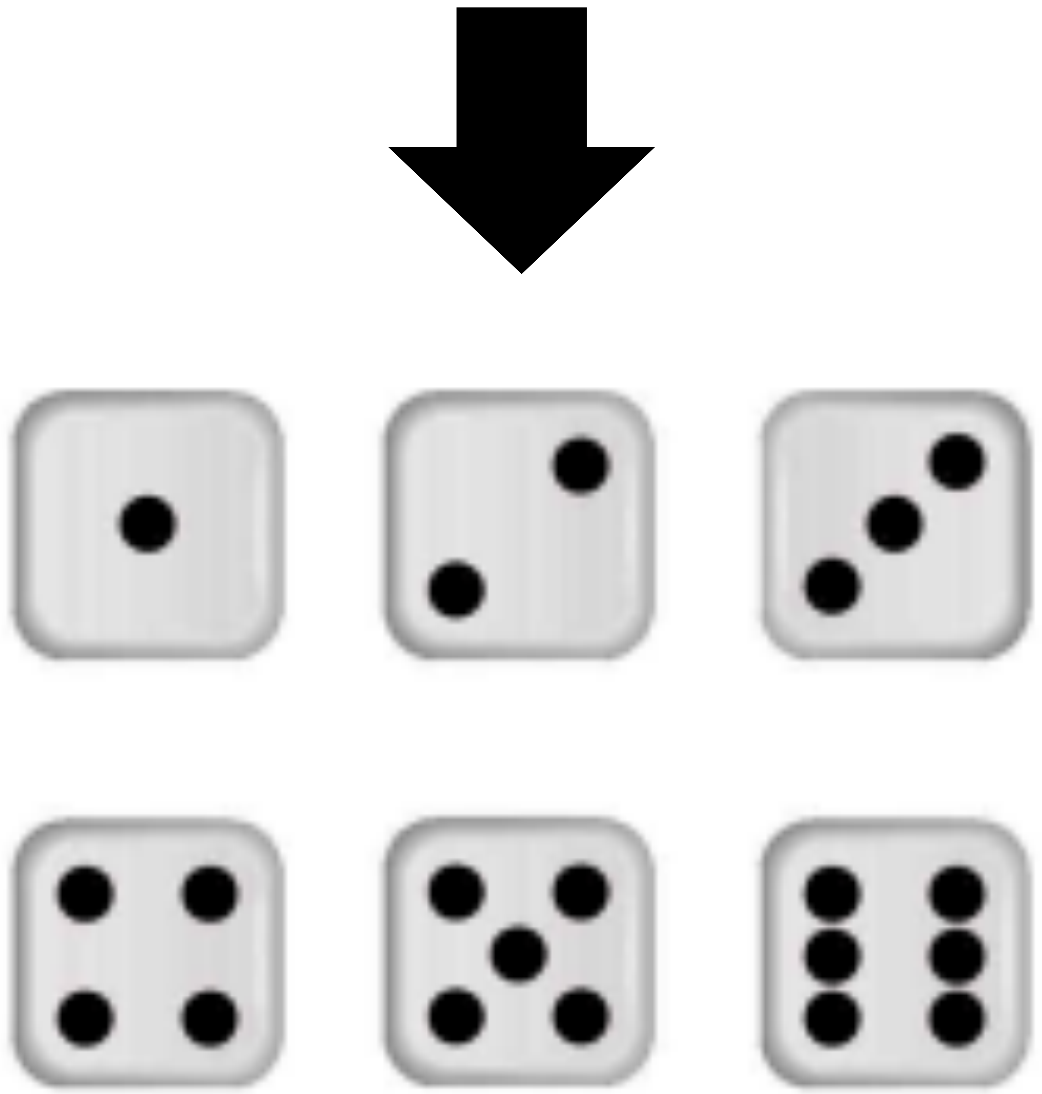
- What is P(A or B)?
  - Addition rule: P(A or B) = P(A) + P(B) - P(A and B)
    - P(A or B) = 1/6 + 2/6 - ??

what's P(A and B) here?

Probability Rules

Dice example





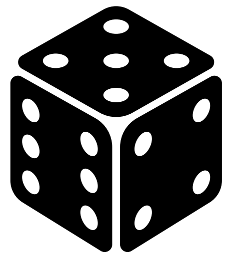
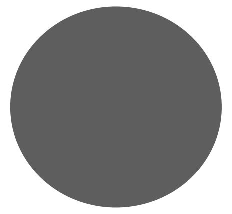
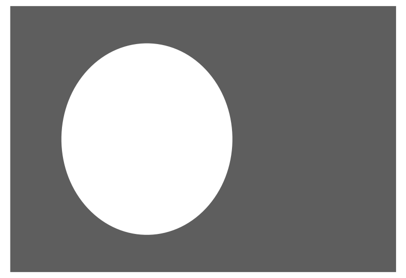
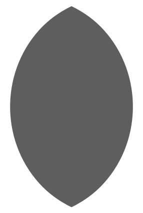
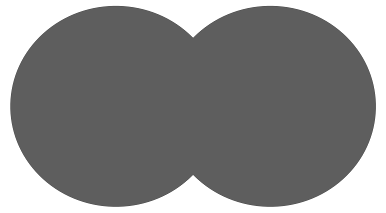
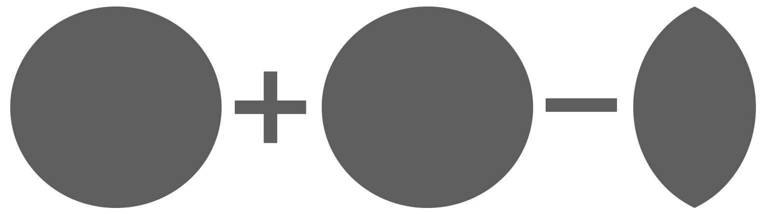
- One fair, six-sided die is rolled (so the probability of seeing any number 1-6 is 1/6)
- Event A = *the event that a rolled die yields a 1* = **1/6**
- Event B = *the event that a rolled die yields a number divisible by 3* = **2/6**
- Event C = *the event that a rolled die yields a 6* = **1/6**

- What is P(A or B)?
  - Addition rule: P(A or B) = P(A) + P(B) - P(A and B)
    - P(A or B) = 1/6 + 2/6 - 0 = 3/6

0

Probability Rules

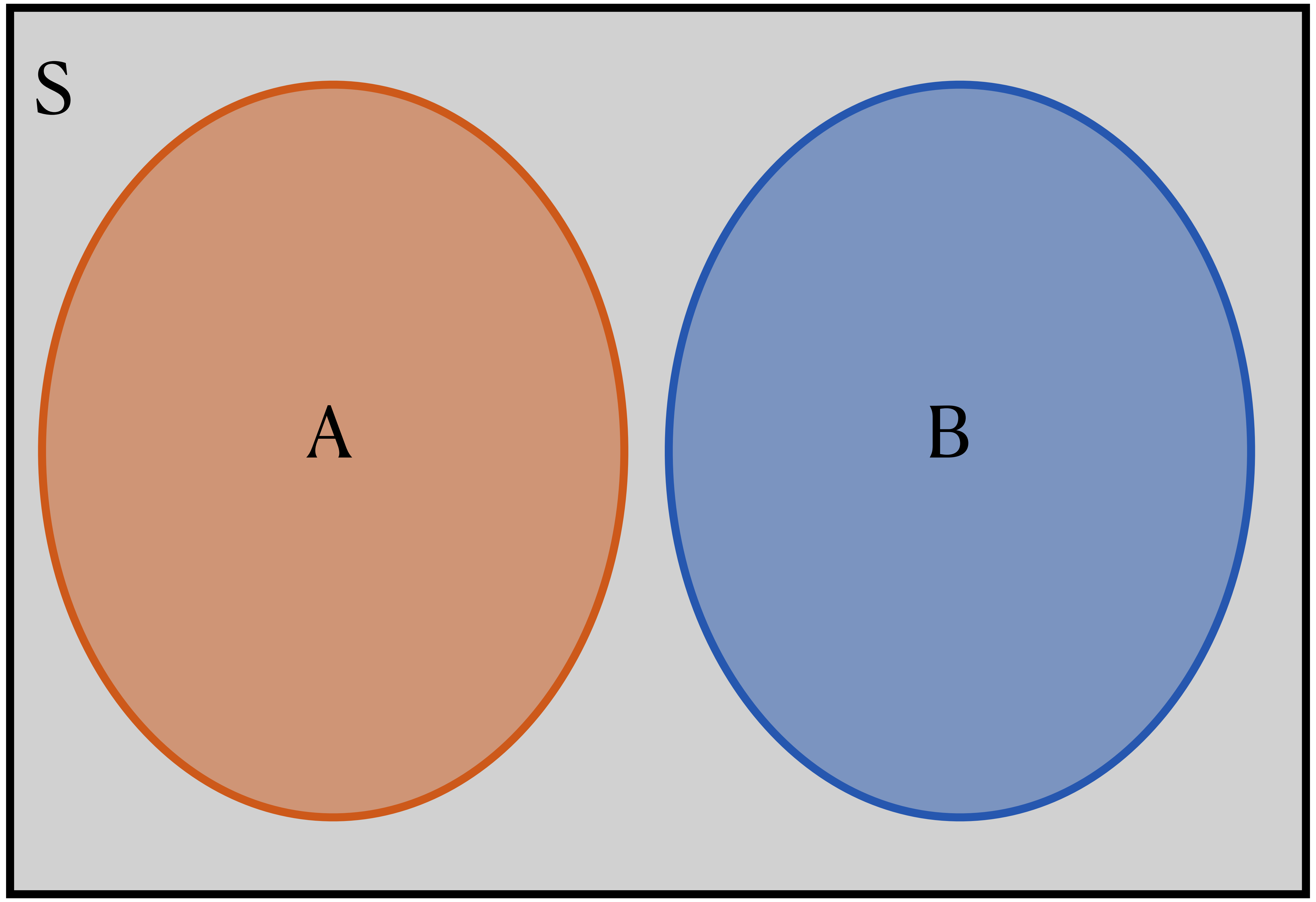
Terminology review

- A **random** circumstance is a circumstance with an uncertain outcome 
- The **sample space** is the set of all possible outcomes of the random circumstance 
- **Probability** is how likely a particular event is (relative size of the event) 
- The **complement** of an event is everything in the sample space except the event 
- The **intersection** of events is the event of *both* of them happening 
- The **union** of events is the event of *at least one* of them happening 
- The **addition rule** gives $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and B})$ 

Probability Concepts

Mutually exclusive events

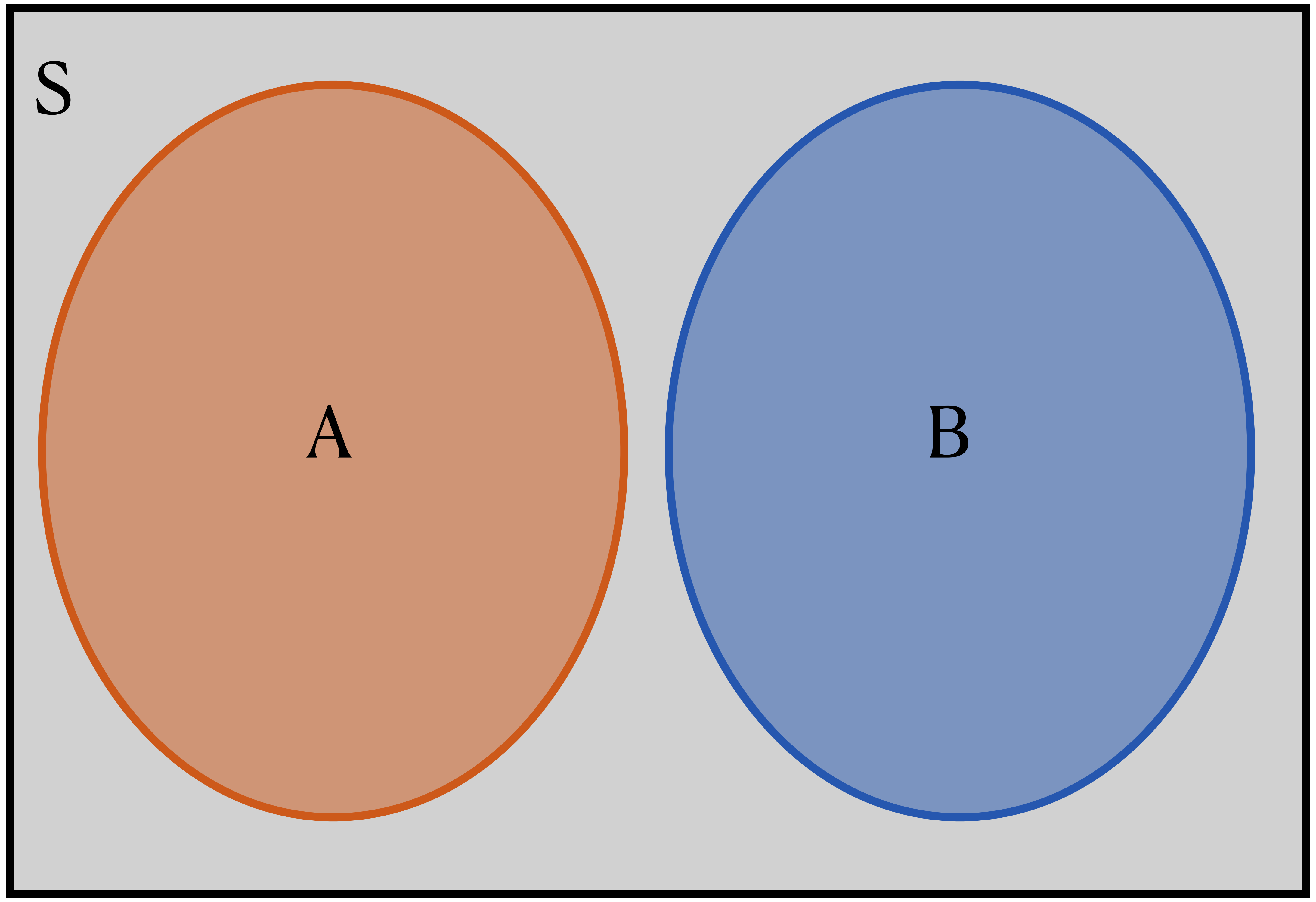
- What is P(A and B) here?



Probability Concepts

Mutually exclusive events

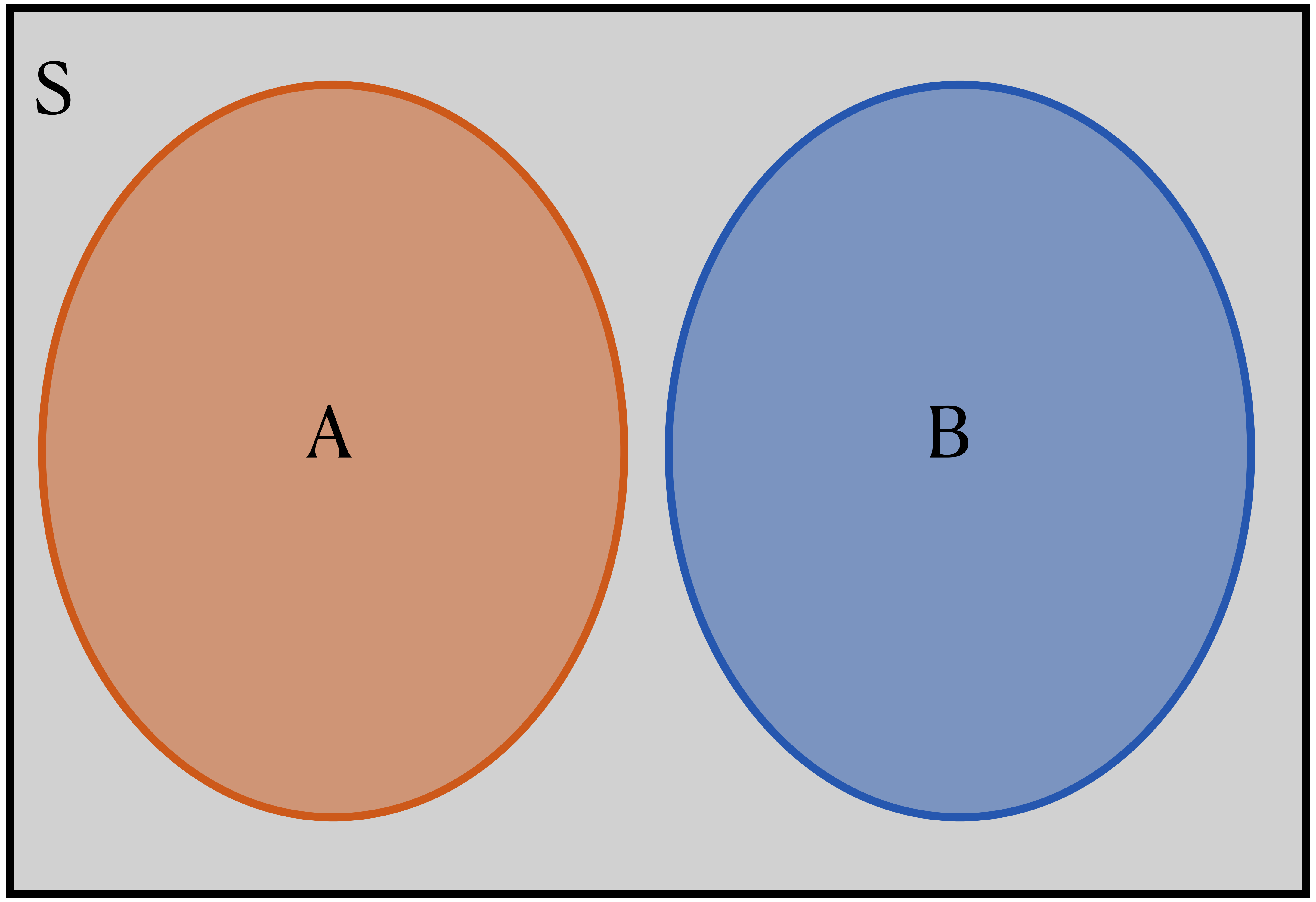
- What is P(A and B) here?
  - P(A and B) = 0
  - The area of the overlap is 0
  - Special name for this: A and B are **mutually exclusive** events



Probability Concepts

Mutually exclusive events

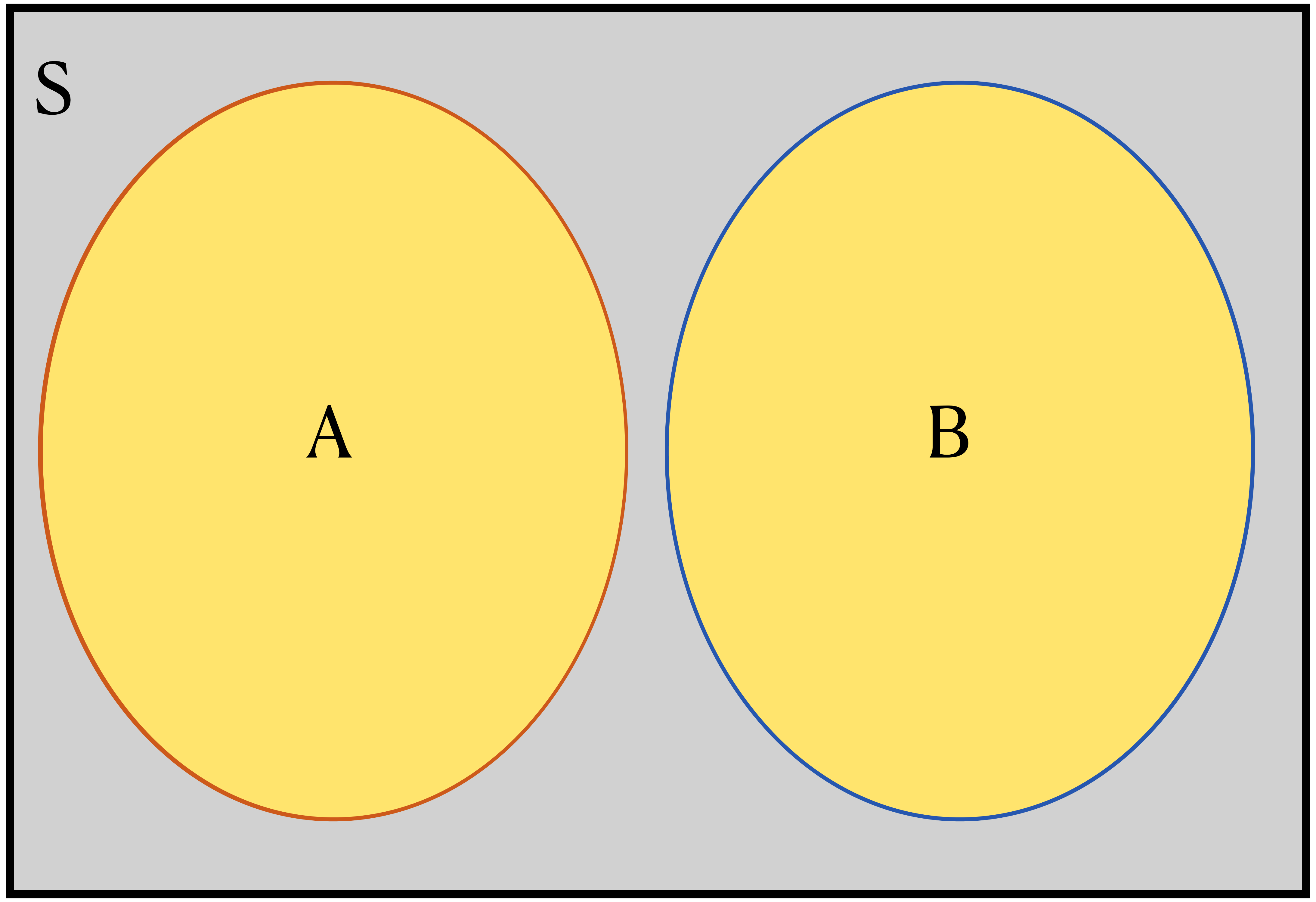
- What is P(A and B) here?
  - P(A and B) = 0
  - The area of the overlap is 0
  - Special name for this: A and B are **mutually exclusive** events
- A = *the event that a rolled die yields a 1*
- B = *the event that a rolled die yields a number divisible by 3*



Probability Concepts

Mutually exclusive events

- What is P(A and B) here?
  - P(A and B) = 0
  - The area of the overlap is 0
  - Special name for this: A and B are **mutually exclusive** events
- What is P(A or B) for mutually exclusive events A and B?
  - P(A or B) = P(A) + P(B) - P(A and B)

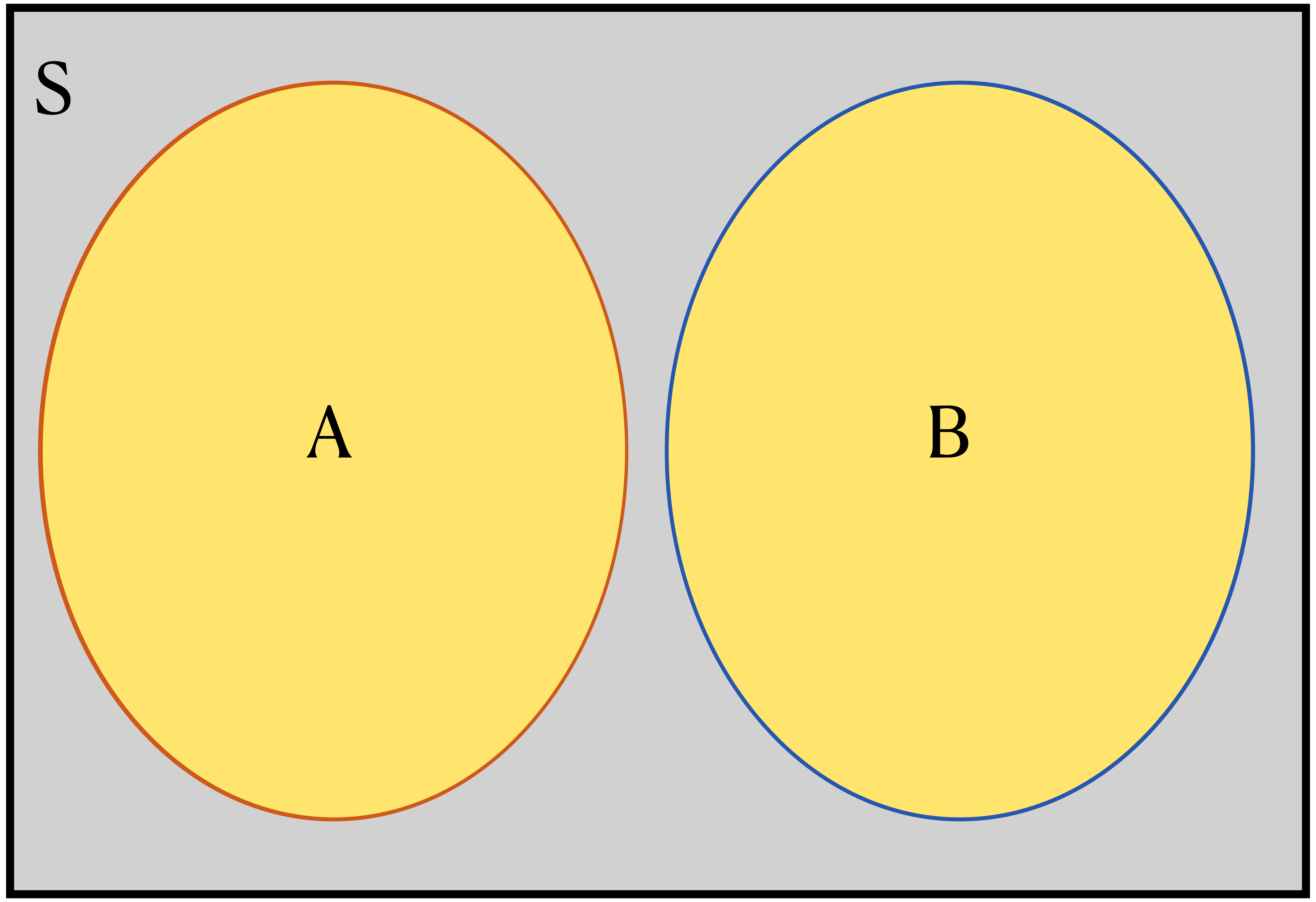


Probability Concepts

Mutually exclusive events

- What is P(A and B) here?
  - P(A and B) = 0
  - The area of the overlap is 0
  - Special name for this: A and B are **mutually exclusive** events
- What is P(A or B) for mutually exclusive events A and B?
  - P(A or B) = P(A) + P(B) - **<u>P(A and B)</u>**

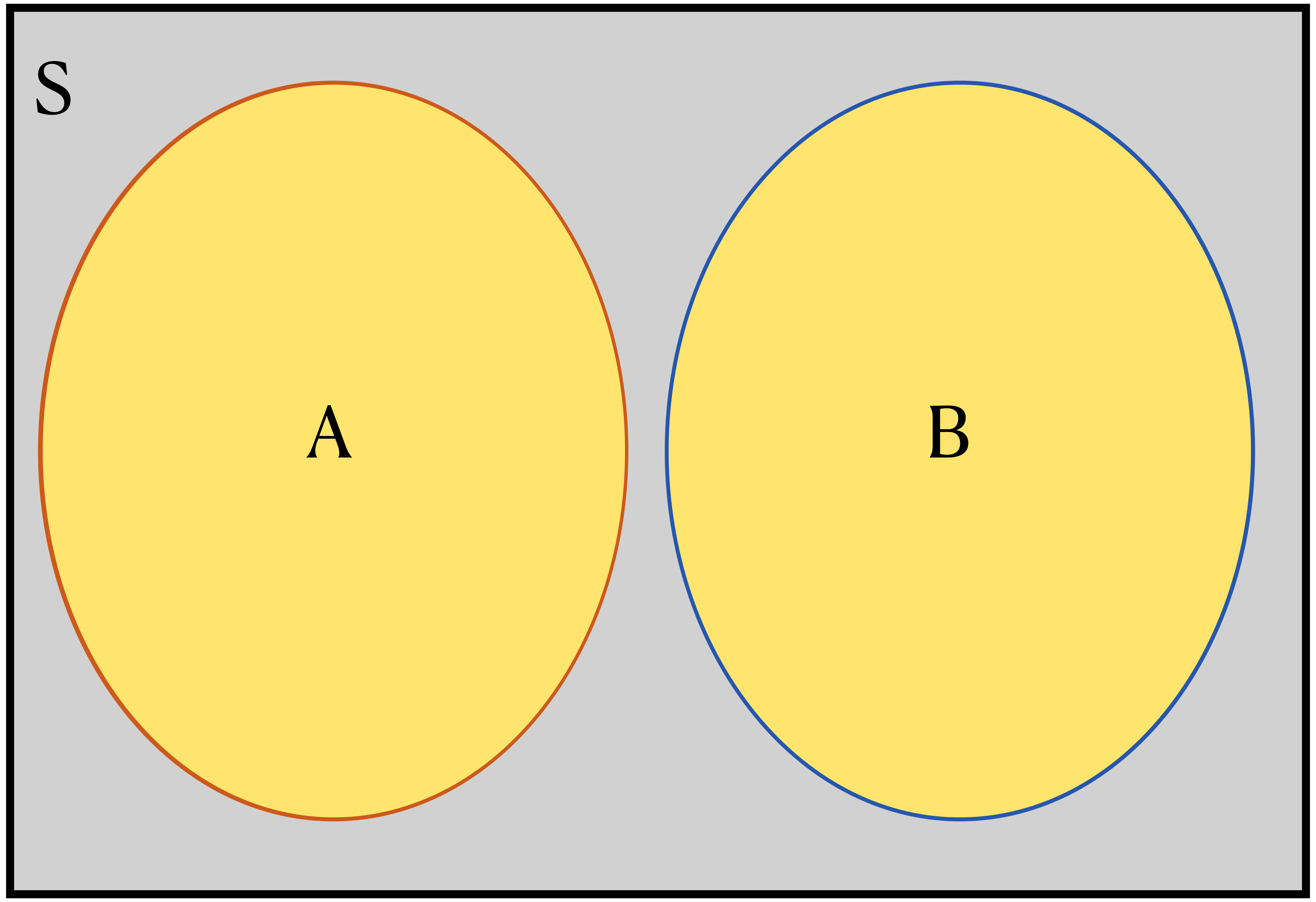
= 0 for m.e. events



Probability Concepts

Mutually exclusive events

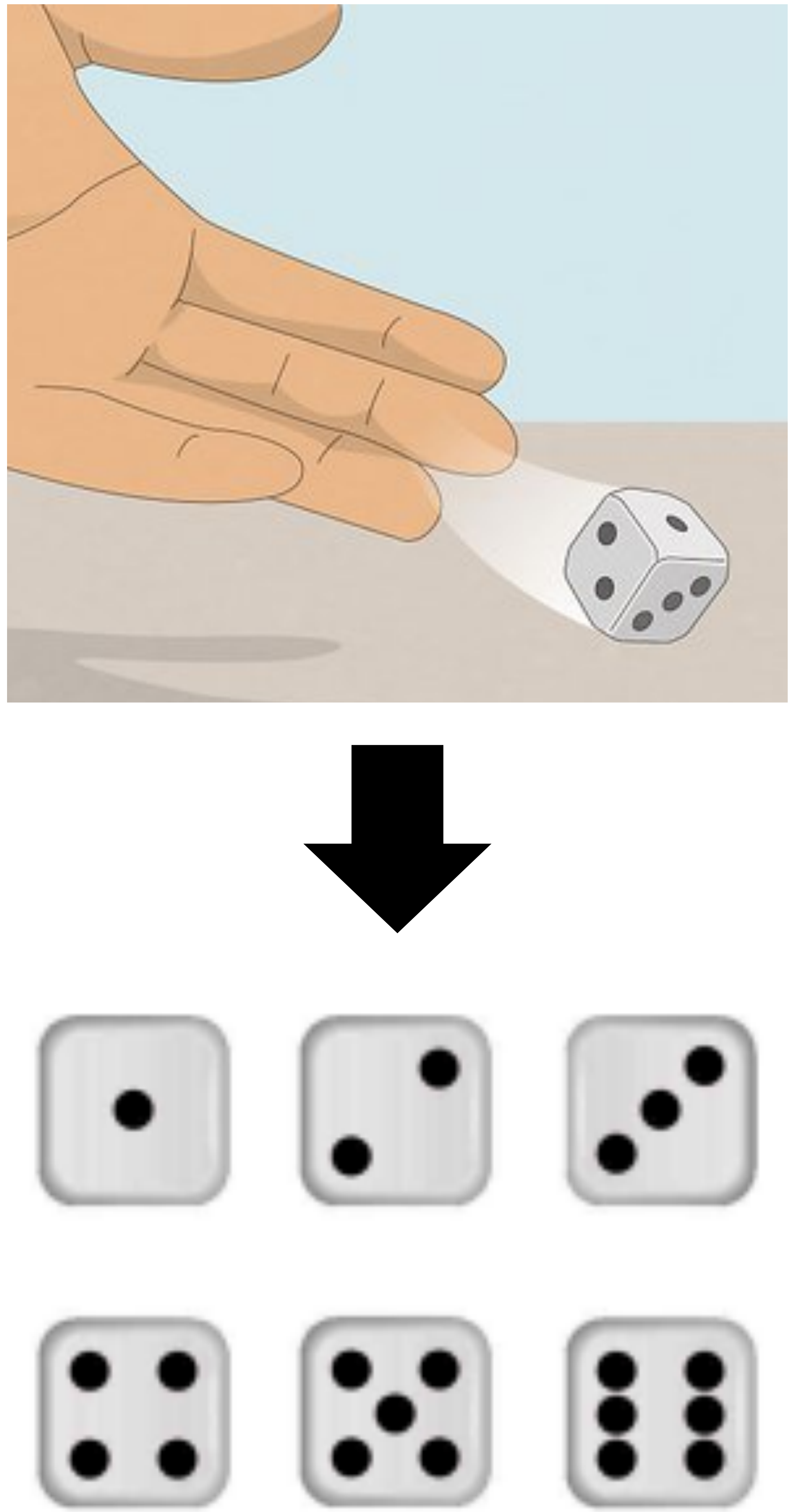
- What is P(A and B) here?
  - P(A and B) = 0
  - The area of the overlap is 0
  - Special name for this: A and B are **mutually exclusive** events
- What is P(A or B) for mutually exclusive events A and B?
  - P(A or B) = P(A) + P(B)



Probability Rules

Dice example

- One fair, six-sided die is rolled (so the probability of seeing any number 1-6 is 1/6)
- Event A = *the event that a rolled die yields a 1* = **1/6**
- Event B = *the event that a rolled die yields a number divisible by 3* = **2/6**
- Event C = *the event that a rolled die yields a 6* = **1/6**
- Which pairs of events are **mutually exclusive**?
  - A, B? (Just discussed this one)
  - A, C?
  - B, C?



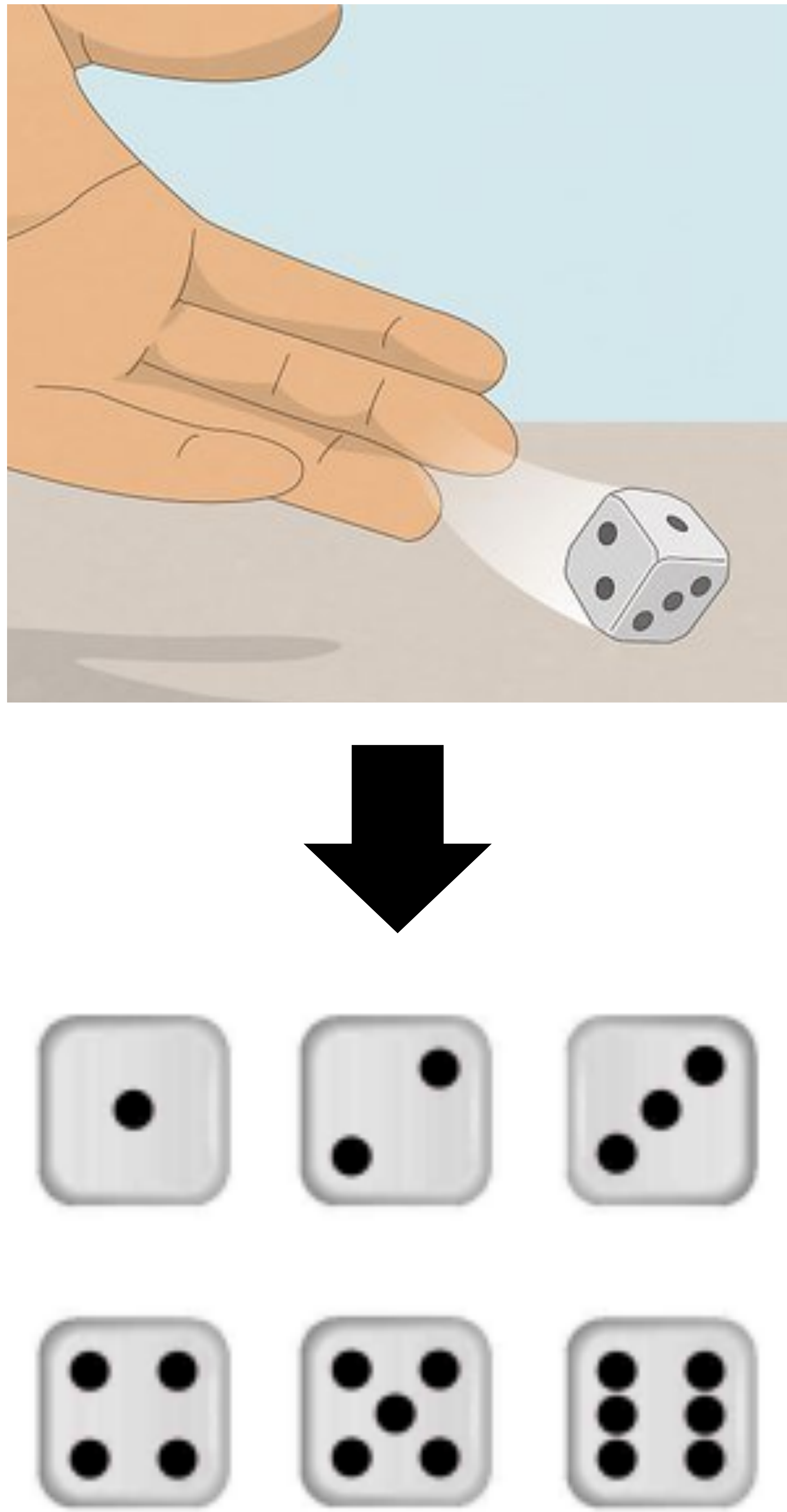
Probability Rules

Dice example

- One fair, six-sided die is rolled (so the probability of seeing any number 1-6 is 1/6)
- Event A = *the event that a rolled die yields a 1* = **1/6**
- Event B = *the event that a rolled die yields a number divisible by 3* = **2/6**
- Event C = *the event that a rolled die yields a 6* = **1/6**

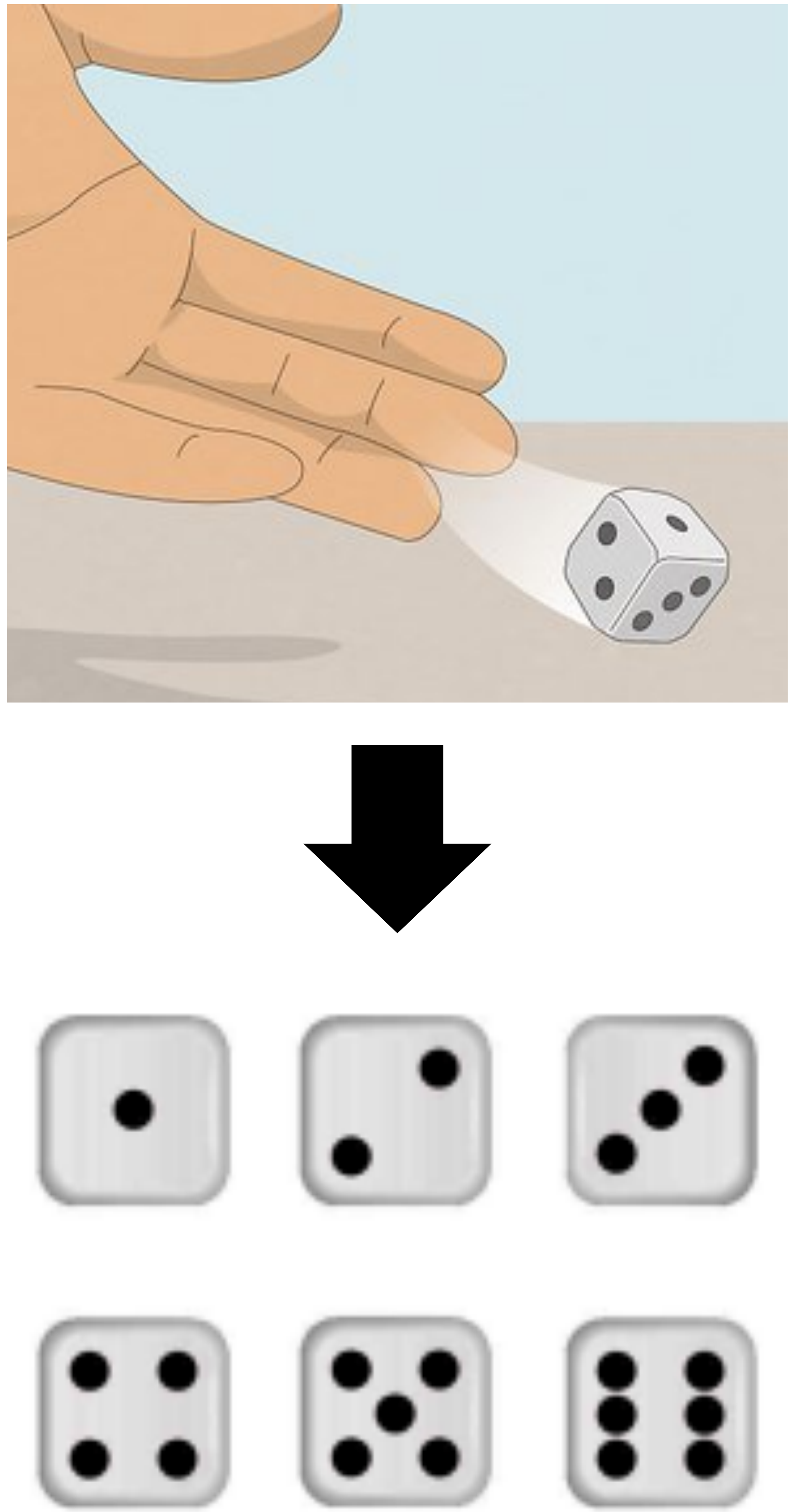
‣ Which pairs of events are **mutually exclusive** (cannot both happen)?

  - A, B? → P(1 *and* divisible by 3) = 0 ✓
  - A, C?
  - B, C?



Probability Rules

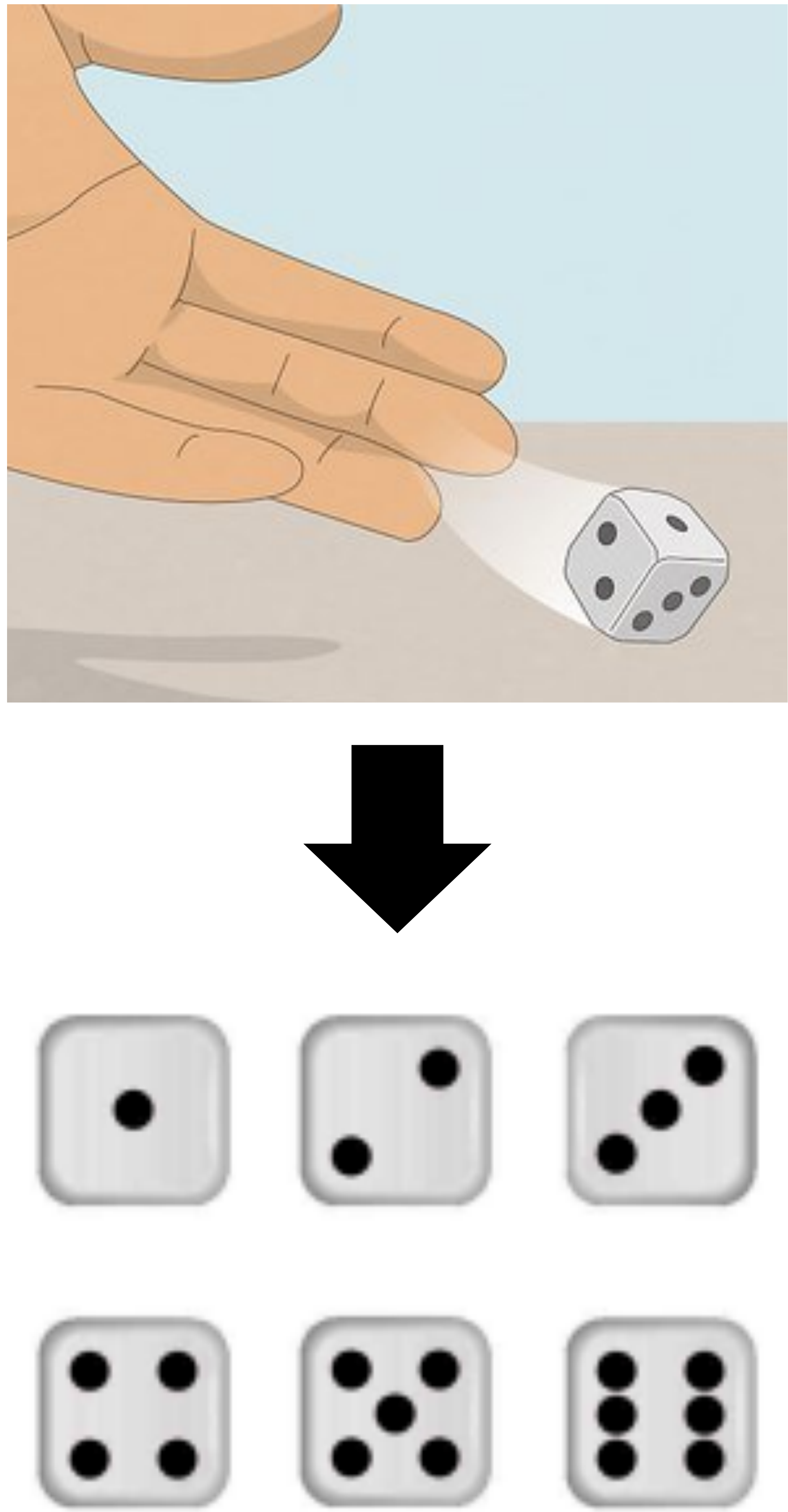
Dice example



- One fair, six-sided die is rolled (so the probability of seeing any number 1-6 is 1/6)
- Event A = *the event that a rolled die yields a 1* = **1/6**
- Event B = *the event that a rolled die yields a number divisible by 3* = **2/6**
- Event C = *the event that a rolled die yields a 6* = **1/6**
- Which pairs of events are **mutually exclusive** (cannot both happen)?
  - A, B? → P(1 *and* divisible by 3) = 0 ✓
  - A, C? → P(1 *and* 6) = 0 ✓
  - B, C?

Probability Rules

Dice example

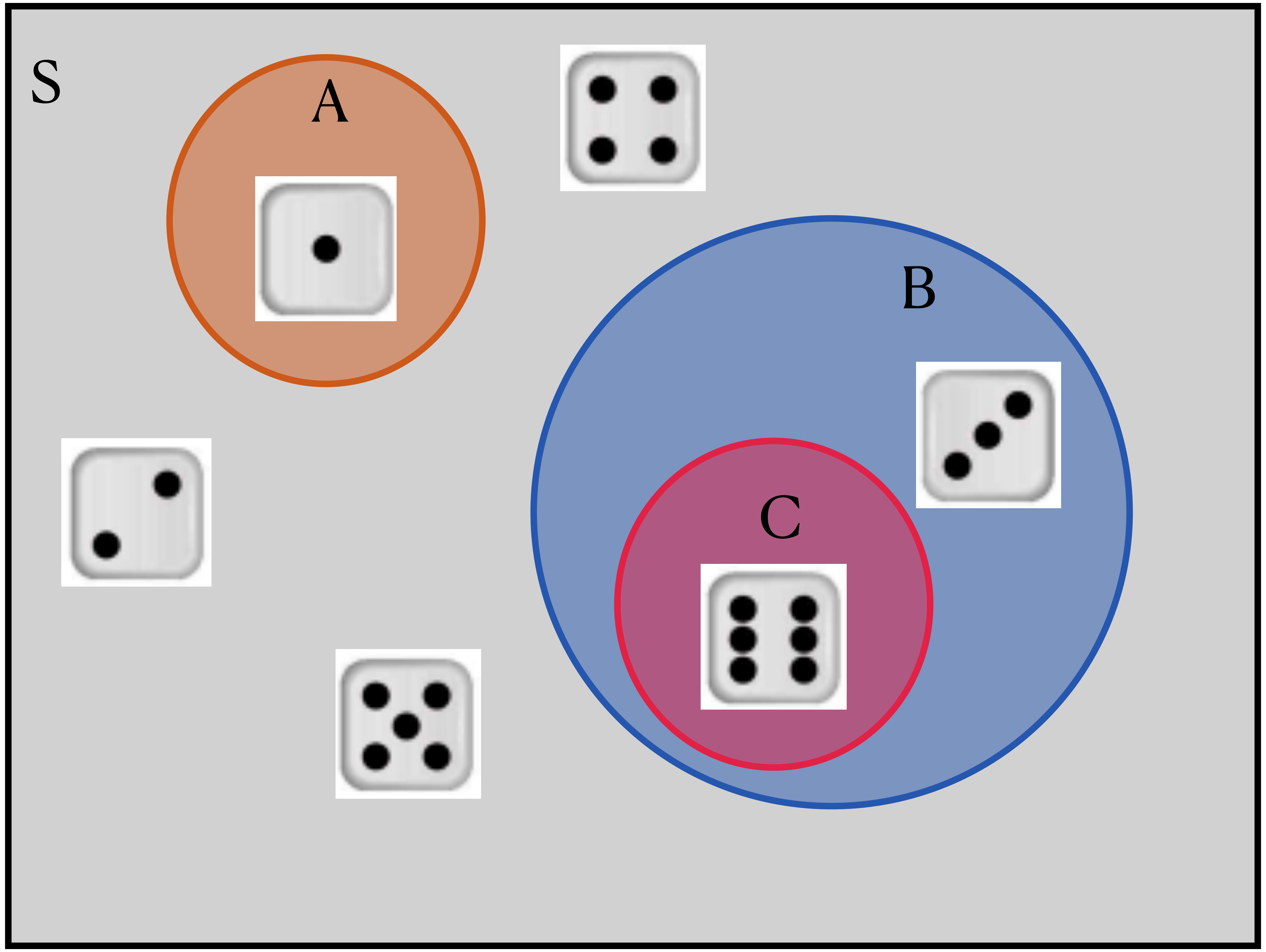


- One fair, six-sided die is rolled (so the probability of seeing any number 1-6 is 1/6)
- Event A = *the event that a rolled die yields a 1* = **1/6**
- Event B = *the event that a rolled die yields a number divisible by 3* = **2/6**
- Event C = *the event that a rolled die yields a 6* = **1/6**
- Which pairs of events are **mutually exclusive** (cannot both happen)?
  - A, B? → P(1 *and* divisible by 3) = 0 ✓
  - A, C? → P(1 *and* 6) = 0 ✓
  - B, C? → P(divisible by 3 *and* 6) ≠ 0 ✗

Probability Rules

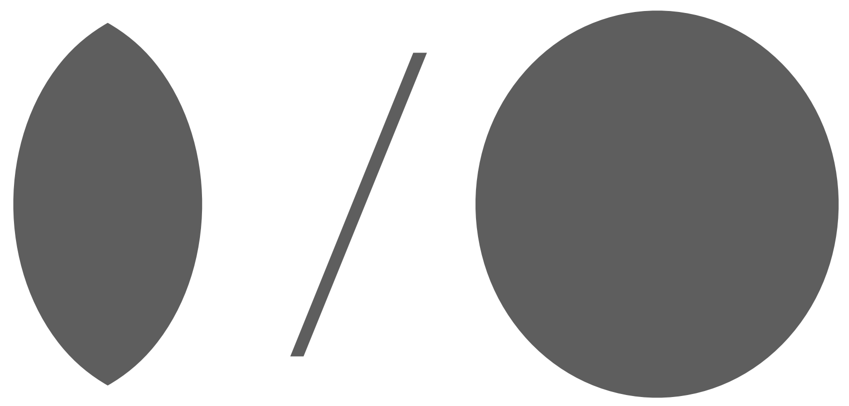
Dice example

- One fair, six-sided die is rolled (so the probability of seeing any number 1-6 is 1/6)
- Event A = *the event that a rolled die yields a 1* = **1/6**
- Event B = *the event that a rolled die yields a number divisible by 3* = **2/6**
- Event C = *the event that a rolled die yields a 6* = **1/6**
- A, B are mutually exclusive; A, C are mutually exclusive



Probability Concepts

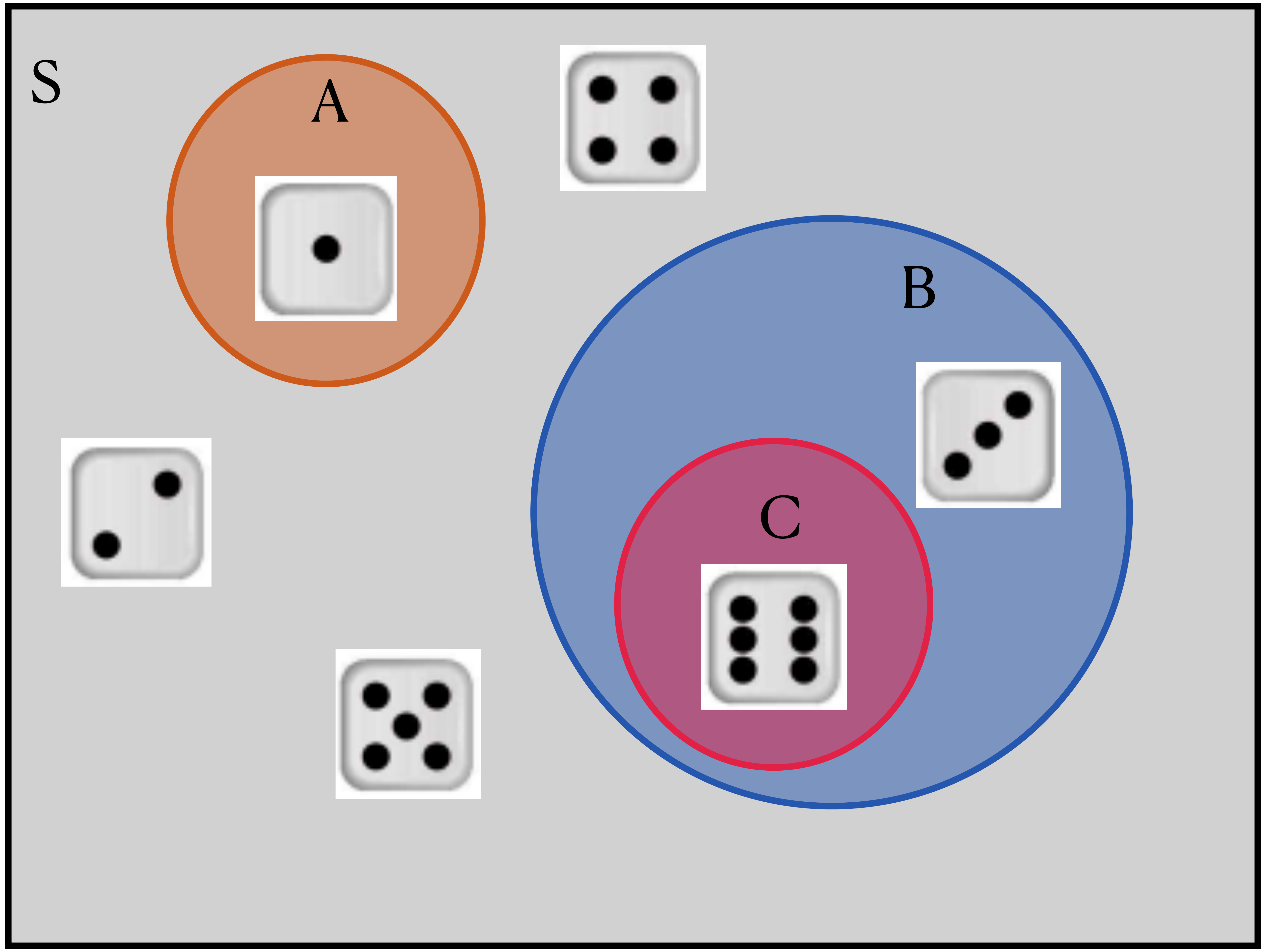
Conditional probability

- The **conditional probability** of event B *given* event A is the probability of B happening, given the extra information that A has happened/will happen
- (Same concept the conditional distributions we talked about in Topics 2 & 3!)
  - Notation: P(B | A) = "the probability of B given A" or "the probability of B conditional on A"
  - $P(B \mid A) = \dfrac{P(A \text{ and } B)}{P(A)} =$ 
  - Intuition: you can think of this as changing the **sample space** for the event B. Let's return to the dice…

Probability Concepts

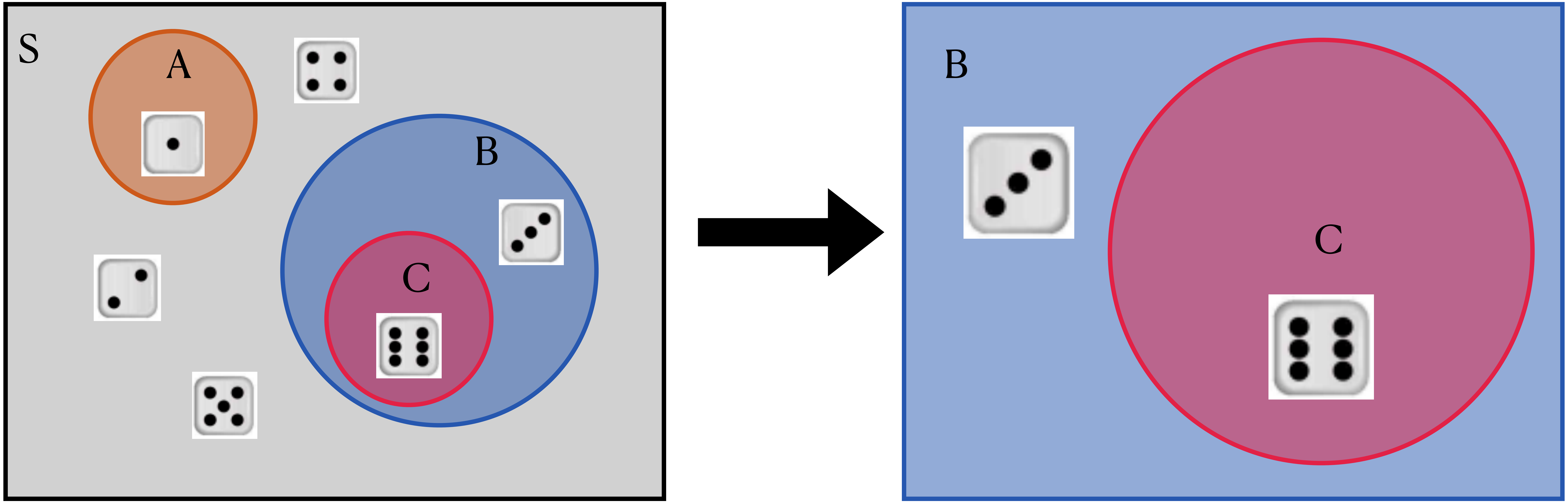
Conditional probability

- Suppose we want the **conditional probability** of C given B, or P(C | B)
  - B: roll a number divisible by 3
  - C: roll a 6
- "C given B" means that we are told that B will happen *for sure*, which means that the new **sample space** must only be what's in B
- Old sample space: {1, 2, 3, 4, 5, 6}
- New sample space conditional on B: {3, 6}



Probability Concepts

Conditional probability



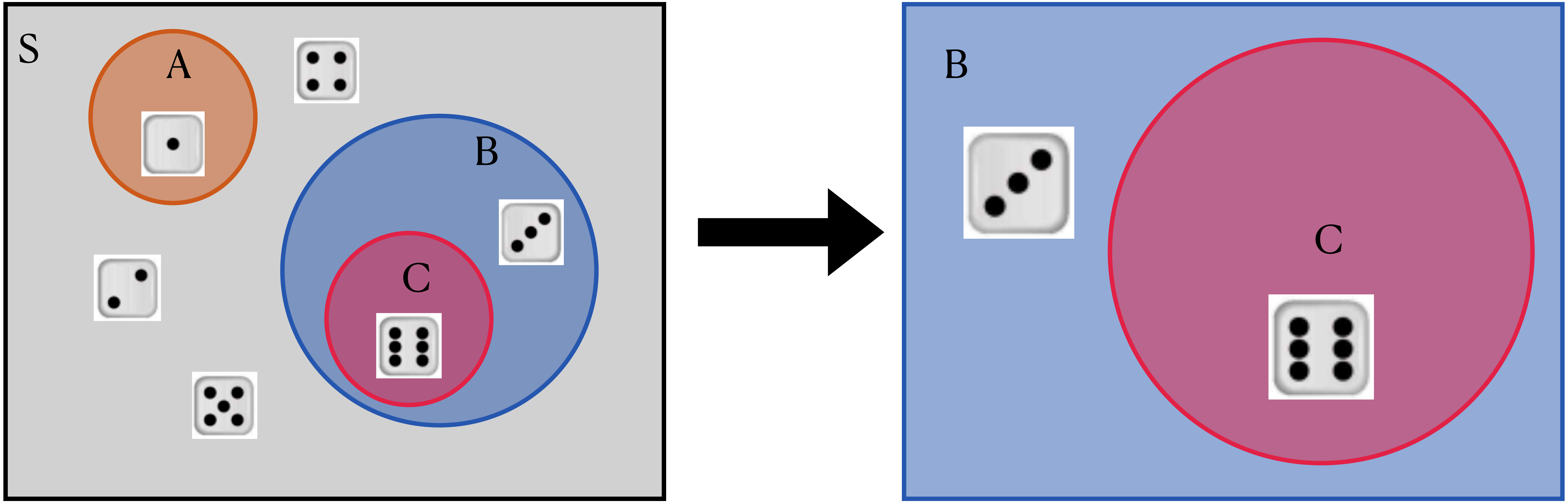
Area of S = $P(S) = 1$
Area of B = $P(B)$
Area of C = $P(C)$

Area of B = $P(B \mid B) = 1$

Area of C = $P(C \mid B)$

Probability Concepts

Conditional probability

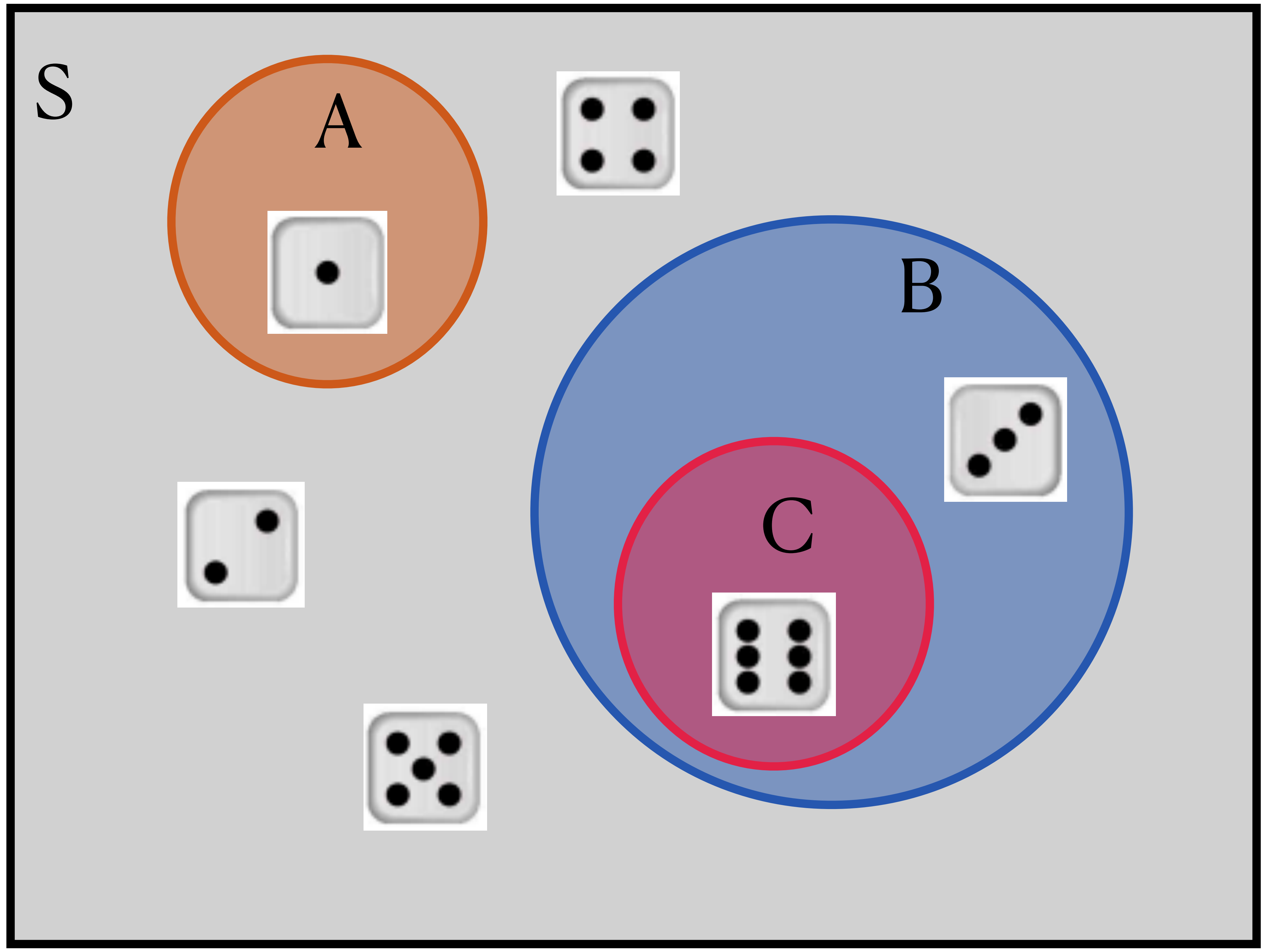


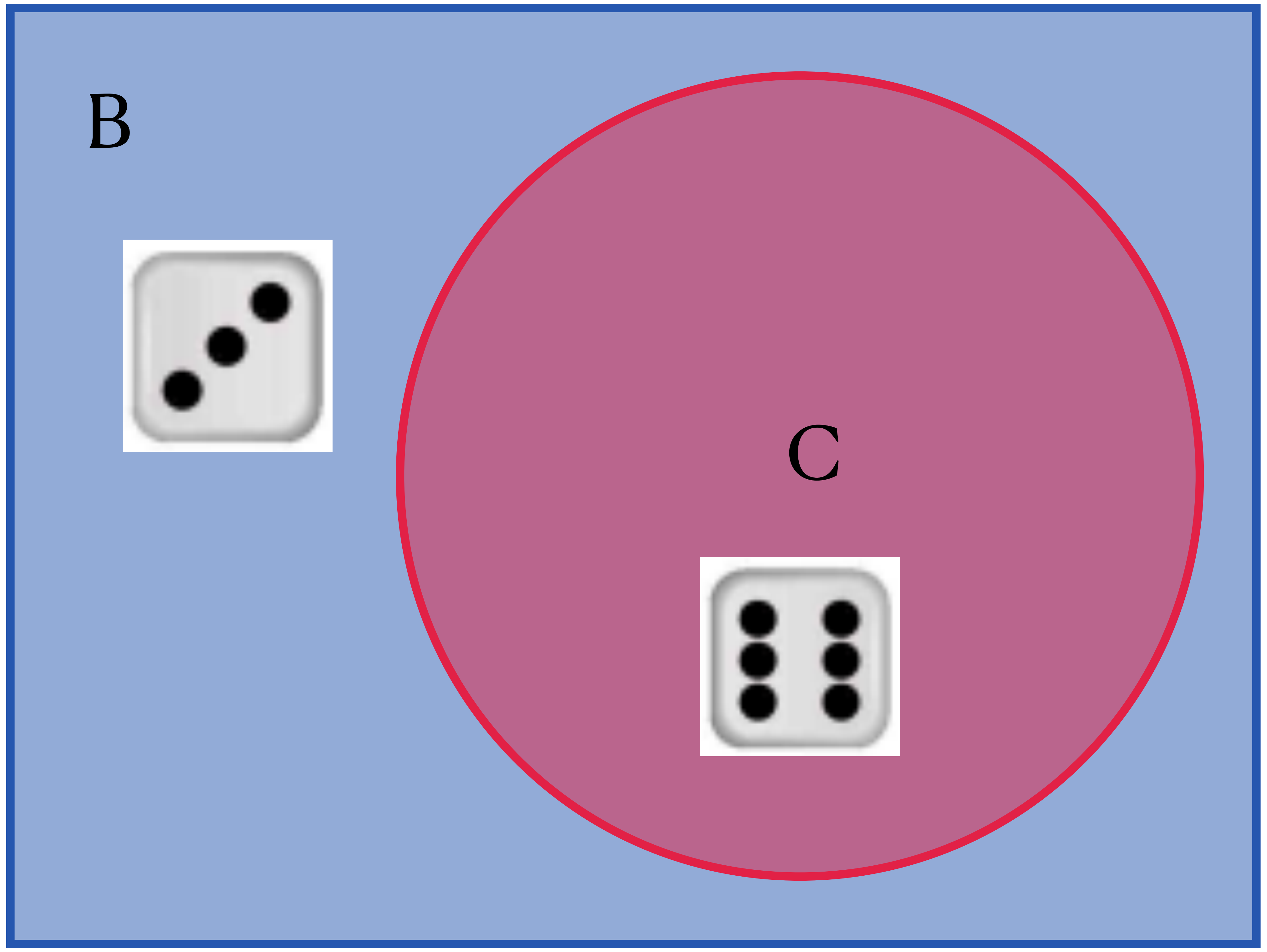
P(C) = relative size of event C compared to S

P(C | B) = relative size of event C compared to B

Probability Concepts

Conditional probability





Area of C = P(roll 6) = **1/6**

Area of C = P(roll 6 | roll divisible. by 3)
= (1/6) / (2/6) = **1/2**

→*Knowing B gave us more info about C*

Probability Concepts

Independence

- In the die example, knowing B happened gave us more information about whether C might happen,

$$P(C) \neq P(C \mid B)$$

- Are there events where knowing one does not change how likely the other is?
  - Ex: you roll *two* fair, six-sided dice; event D = first die rolls a 1, event E = second die rolls a 1
    - Suppose we know that D happened. Does this change how likely E is?
    - $P(E) = 1/6$; $P(E \mid D) = \ldots$

Probability Concepts

Independence

- In the die example, knowing B happened gave us more information about whether C might happen,

$$P(C) \neq P(C \mid B)$$

- Are there events where knowing one does not change how likely the other is?
  - Ex: you roll *two* fair, six-sided dice; event D = first die rolls a 1, event E = second die rolls a 1
    - Suppose we know that D happened. Does this change how likely E is?
    - $P(E) = 1/6$; $P(E \mid D) = 1/6$

Probability Concepts

Independence

- Are there events where knowing one does not change how likely the other is?
  - Ex: you roll *two* fair, six-sided dice; event D = first die rolls a 1, event E = second die rolls a 1
    - Suppose we know that D happened. Does this change how likely E is?
    - P(E) = 1/6; P(E | D) = 1/6
- **Independent events:** the knowledge that one event has happened does not affect the probability of the other event.
- Mathematically, two events A and B are independent if (and only if):
  - $P(A \mid B) = P(A)$ or
  - $P(A \text{ and } B) = P(A)P(B)$

Independence

PollEv.com/stats8

- Suppose I have a standard deck of 52 cards: 13 of each *suit* (clubs, spades, diamonds, hearts), and 4 of each *rank* (2-10, Jack, Queen, King, Ace).
  - I draw one card at random.
  - Without putting it back, I draw another card.
    - Suppose I want to know the probability that I draw a King.
    - Would it be helpful to know what the first card was?

Independence

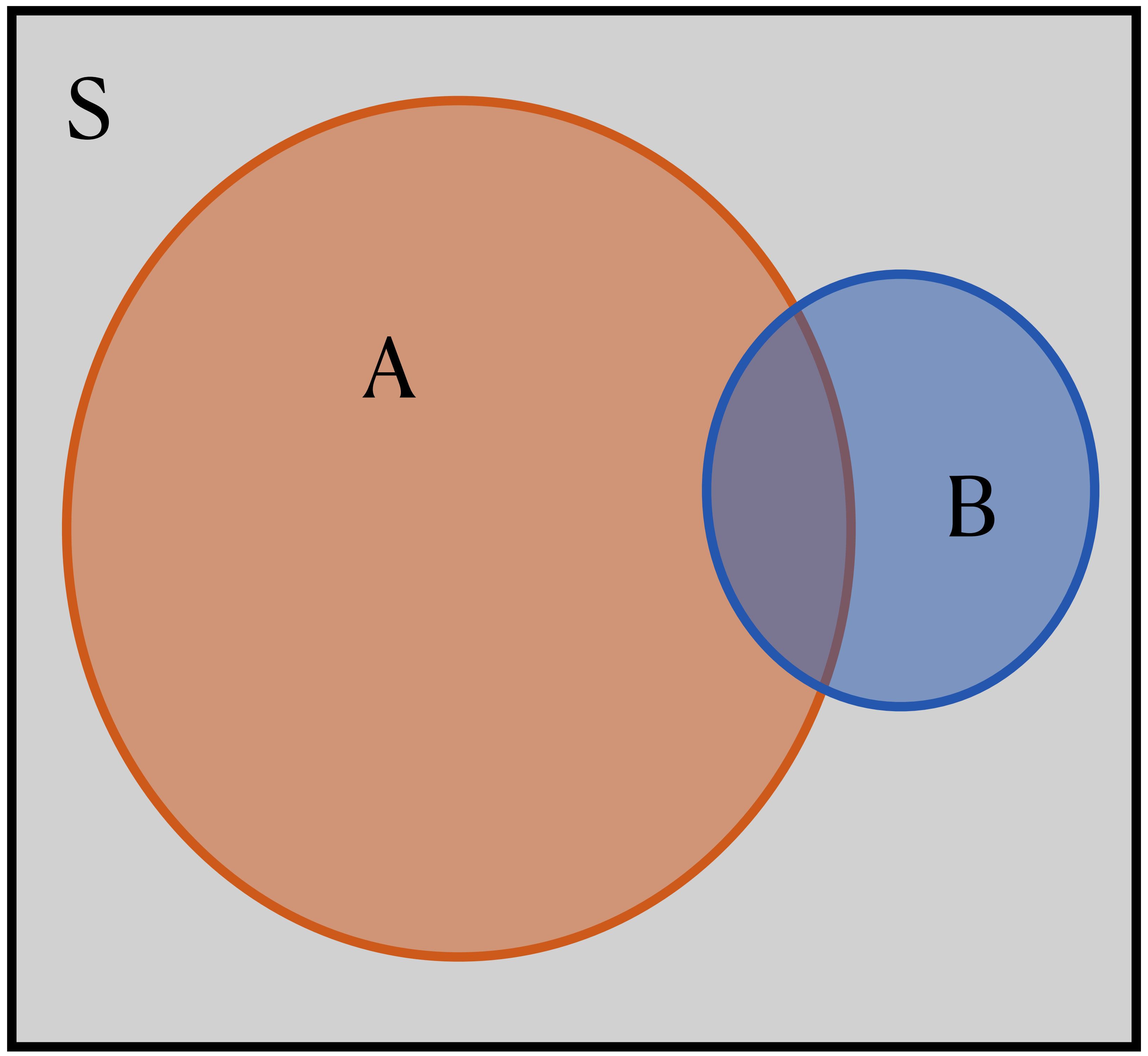
PollEv.com/stats8

- Suppose I have a standard deck of 52 cards: 13 of each *suit* (clubs, spades, diamonds, hearts), and 4 of each *rank* (2-10, Jack, Queen, King, Ace).
  - I draw one card at random.
  - **After putting it back**, I draw another card.
    - Suppose I want to know the probability that I draw a King.
    - Would it be helpful to know what the first card was?

Probability Concepts

Law of Total Probability

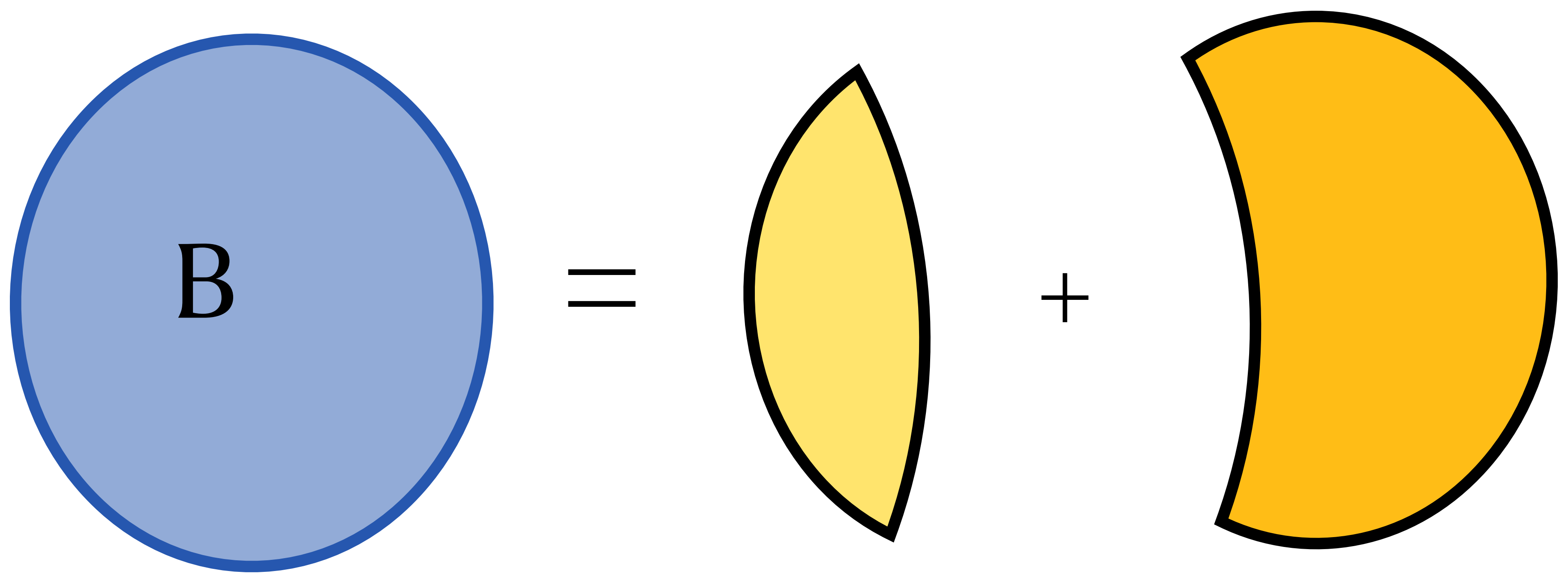
- The **Law of Total Probability** gives us a way to relate **marginal probabilities** to **joint probabilities**

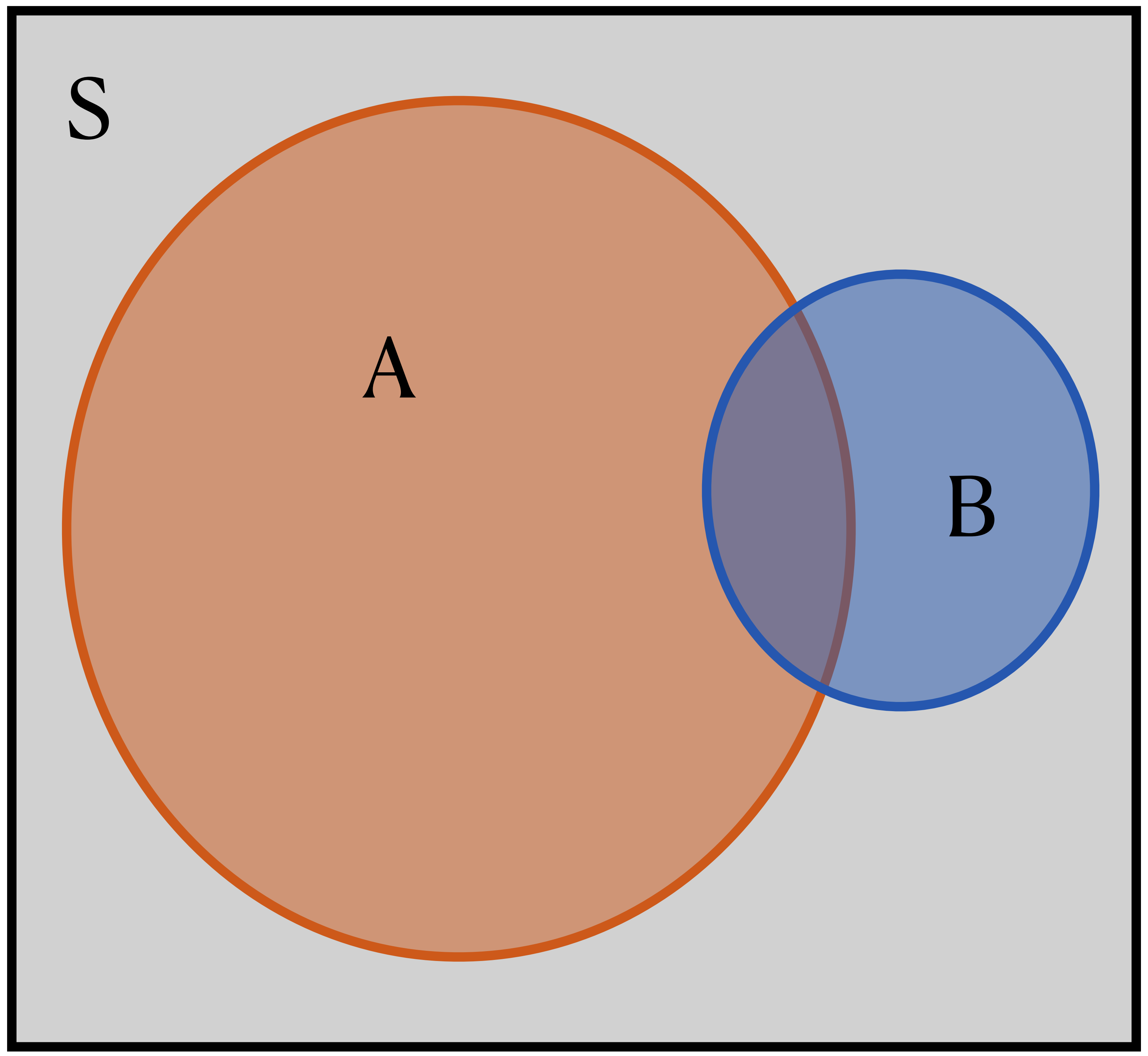


Probability Concepts

Law of Total Probability

- The **Law of Total Probability** gives us a way to relate **marginal probabilities** to **joint probabilities**
- Suppose we want to get $P(B)$ (*marginal*) using joint probabilities of A and B. Let's do it like a puzzle:

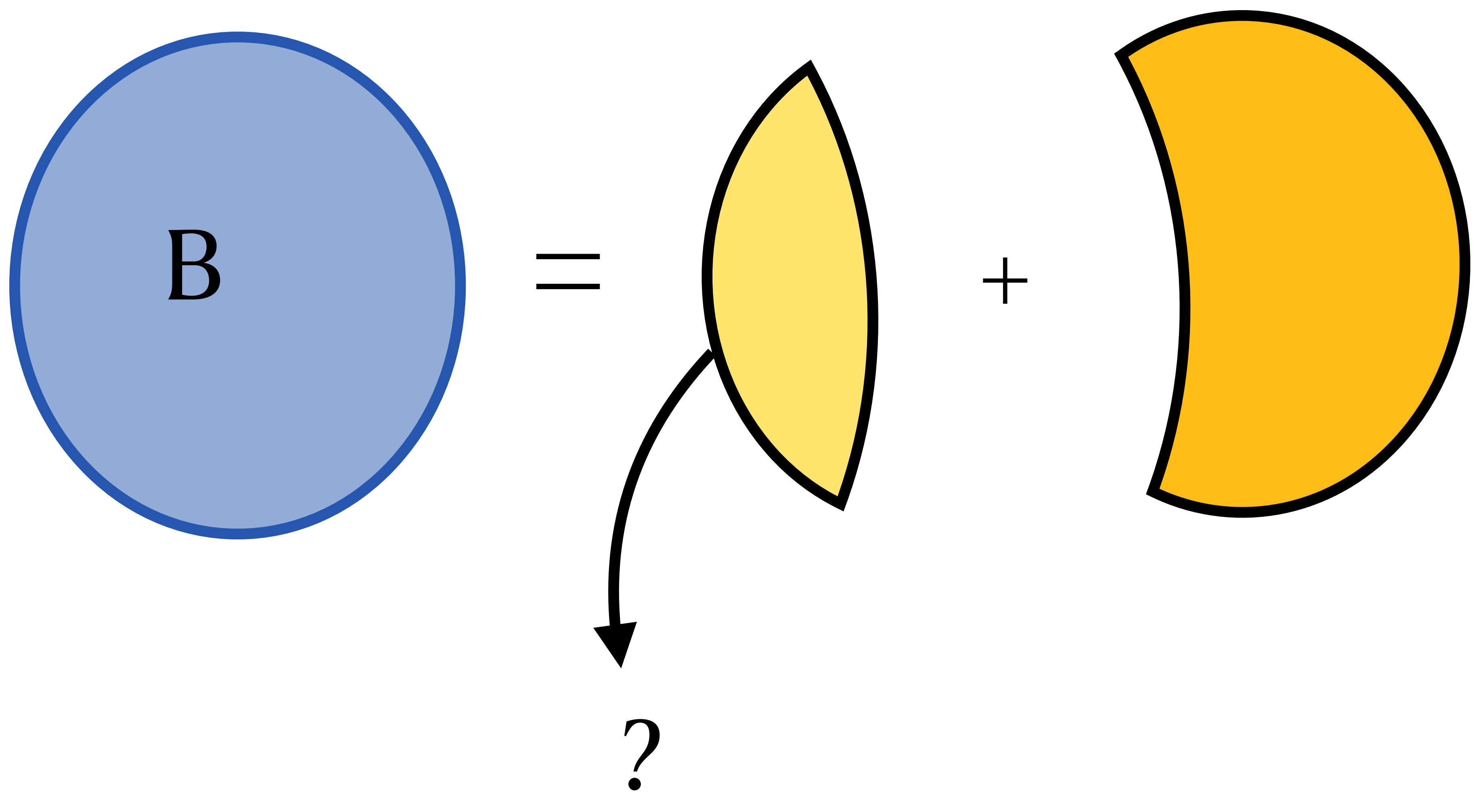


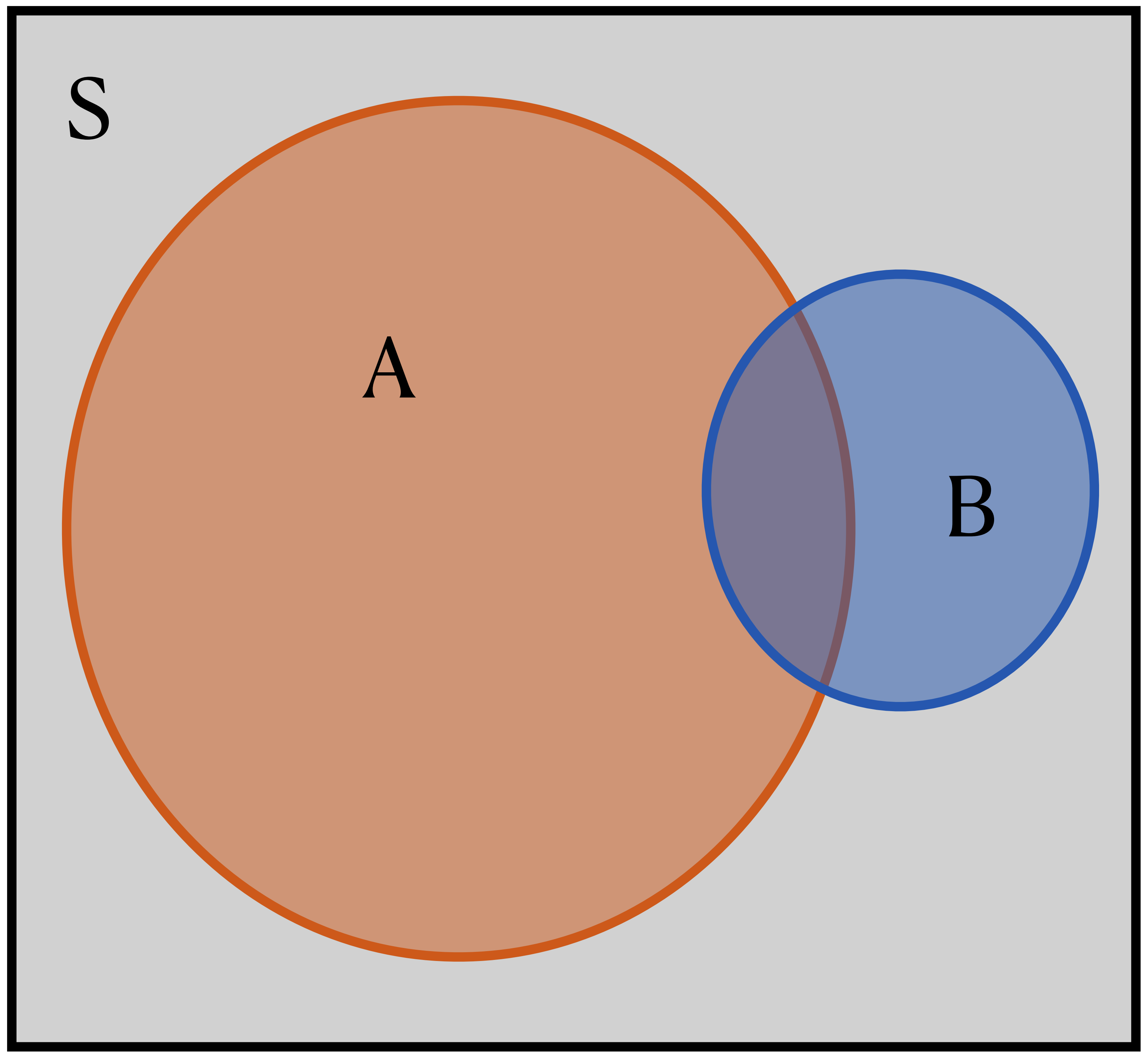


Probability Concepts

Law of Total Probability

- The **Law of Total Probability** gives us a way to relate **marginal probabilities** to **joint probabilities**
- Suppose we want to get $P(B)$ (*marginal*) using joint probabilities of A and B. Let's do it like a puzzle:



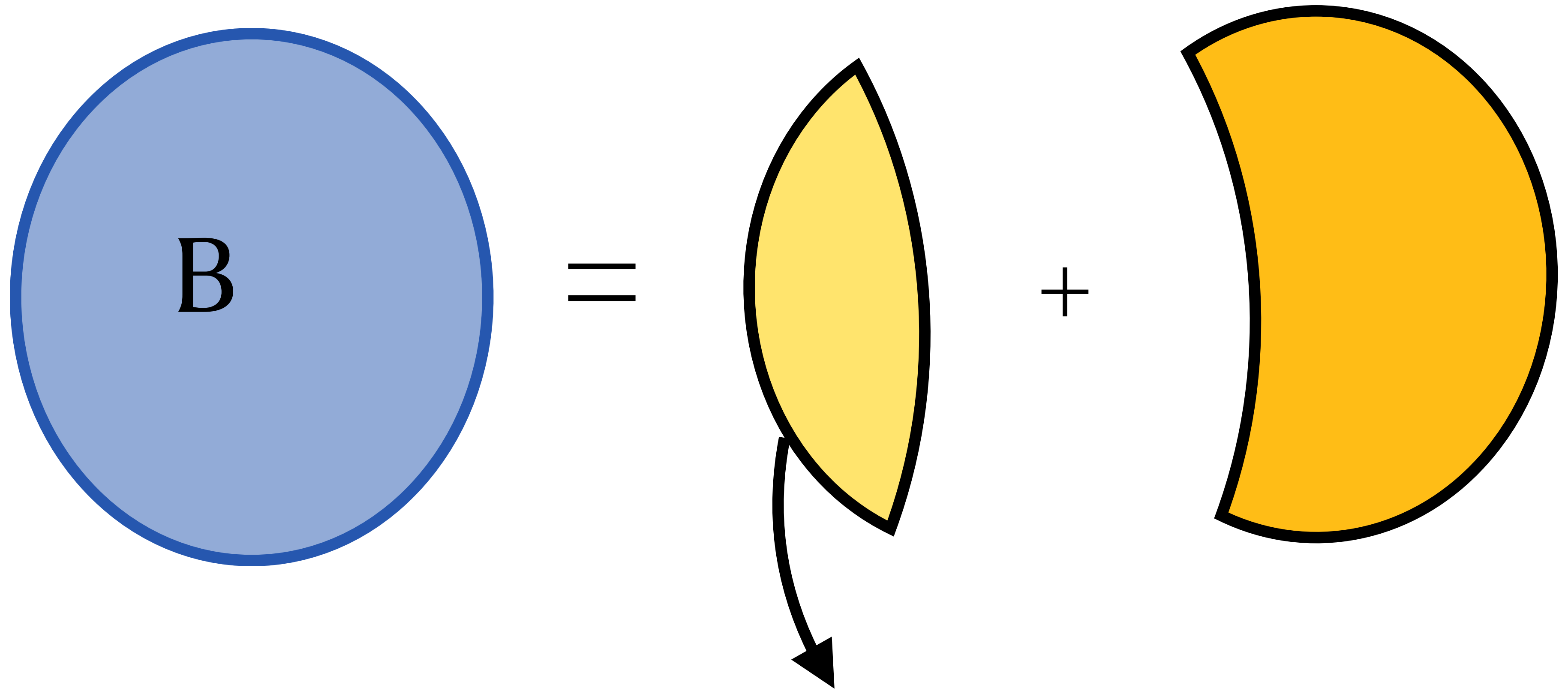


Probability Concepts

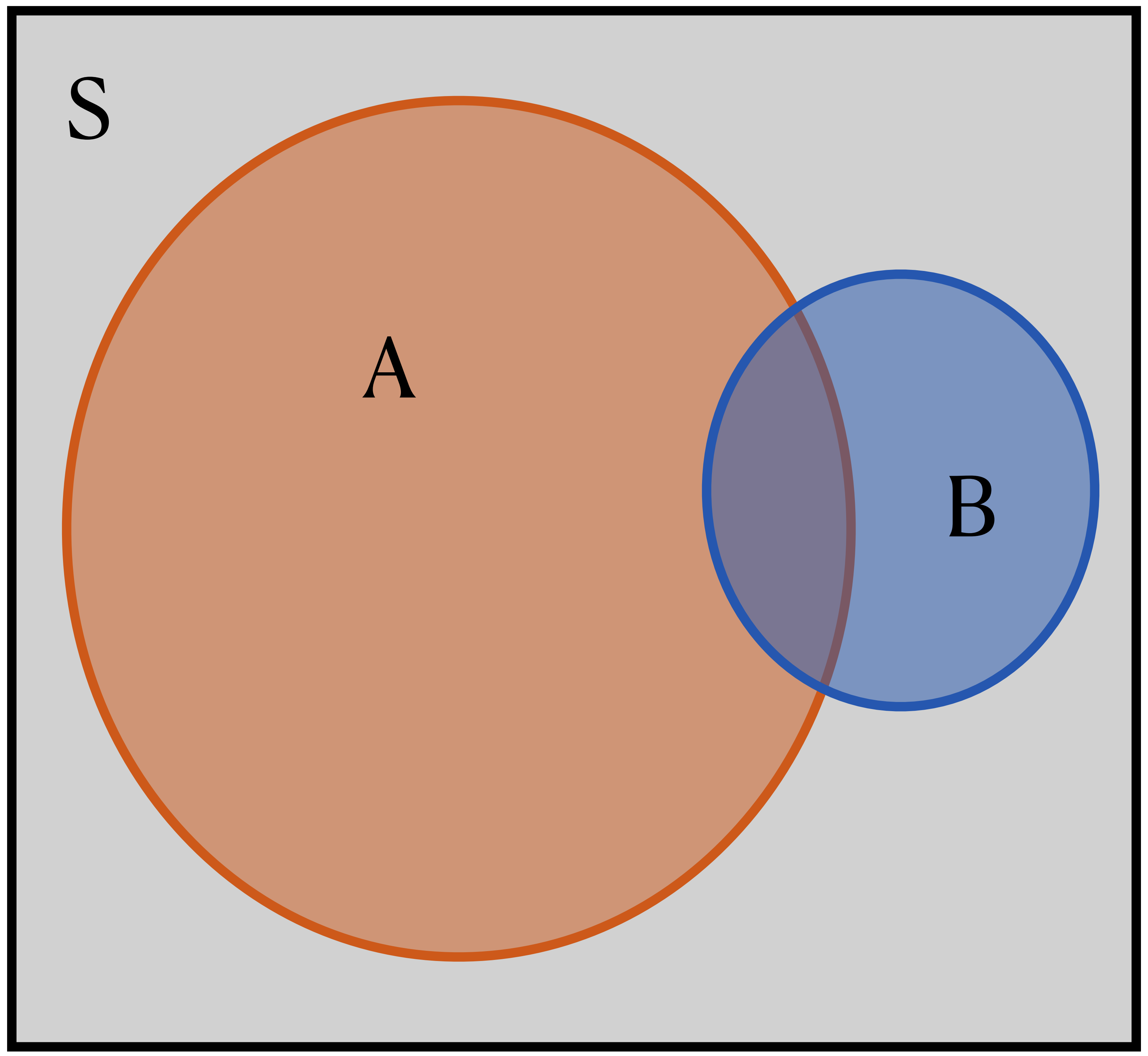
Law of Total Probability

- The **Law of Total Probability** gives us a way to relate **marginal probabilities** to **joint probabilities**

- Suppose we want to get $P(B)$ (*marginal*) using joint probabilities of A and B. Let's do it like a puzzle:



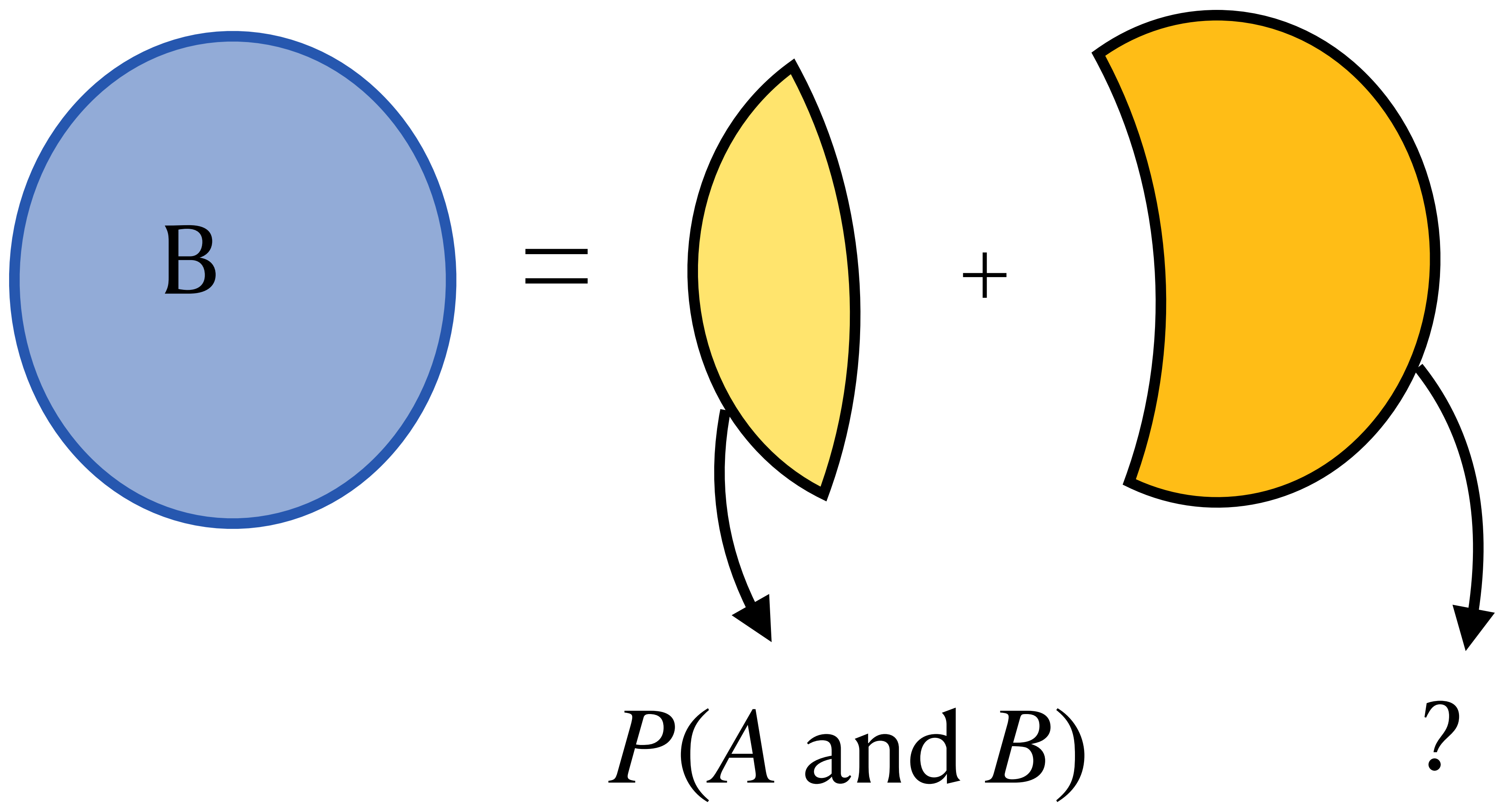
$P(A \text{ and } B)$

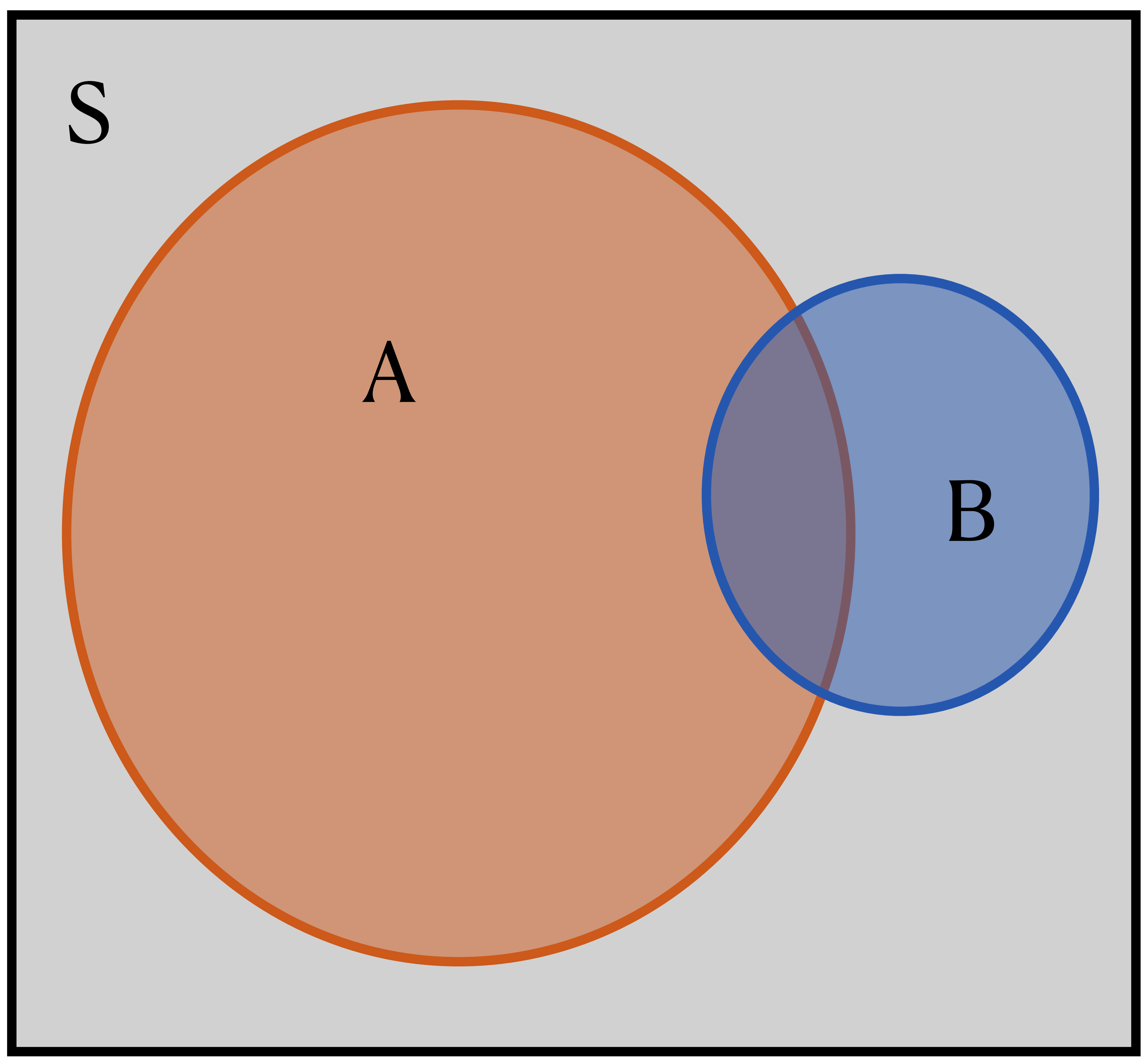


Probability Concepts

Law of Total Probability

- The **Law of Total Probability** gives us a way to relate **marginal probabilities** to **joint probabilities**
- Suppose we want to get $P(B)$ (*marginal*) using joint probabilities of A and B. Let's do it like a puzzle:

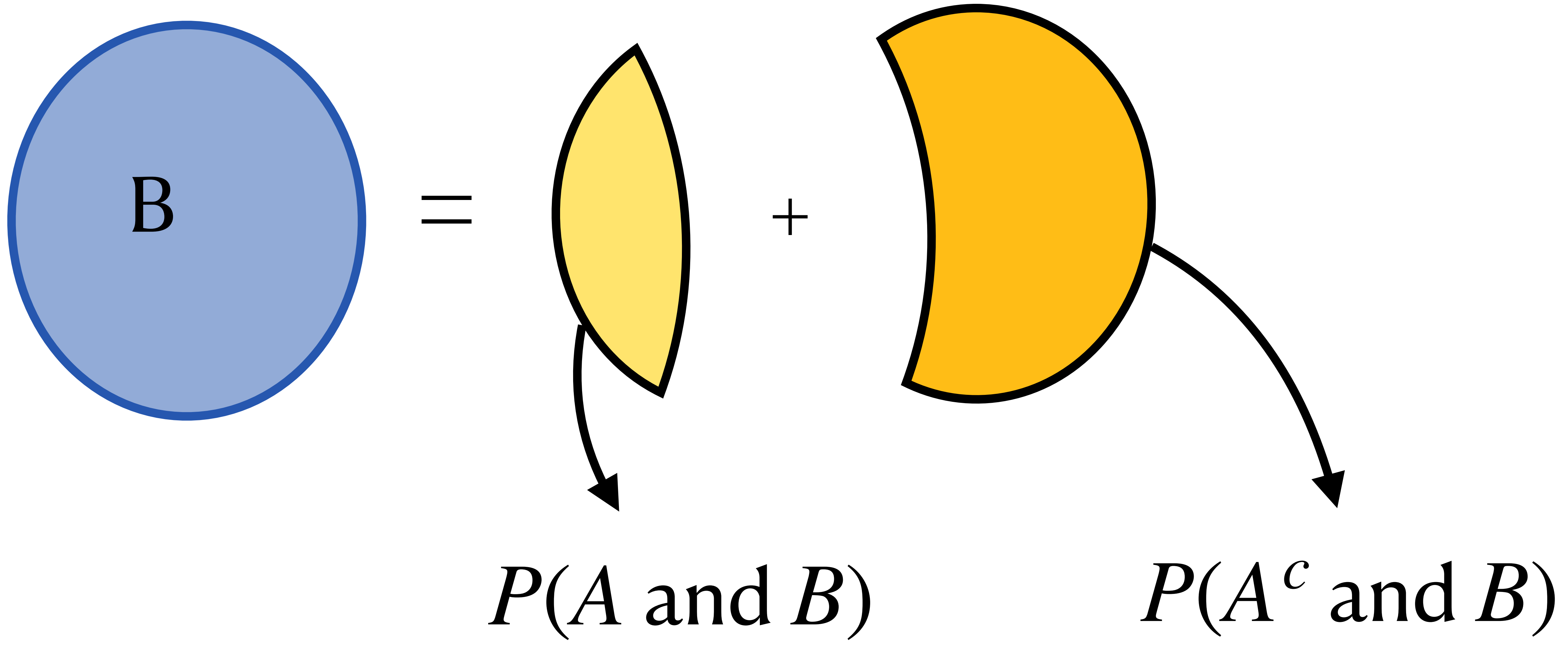


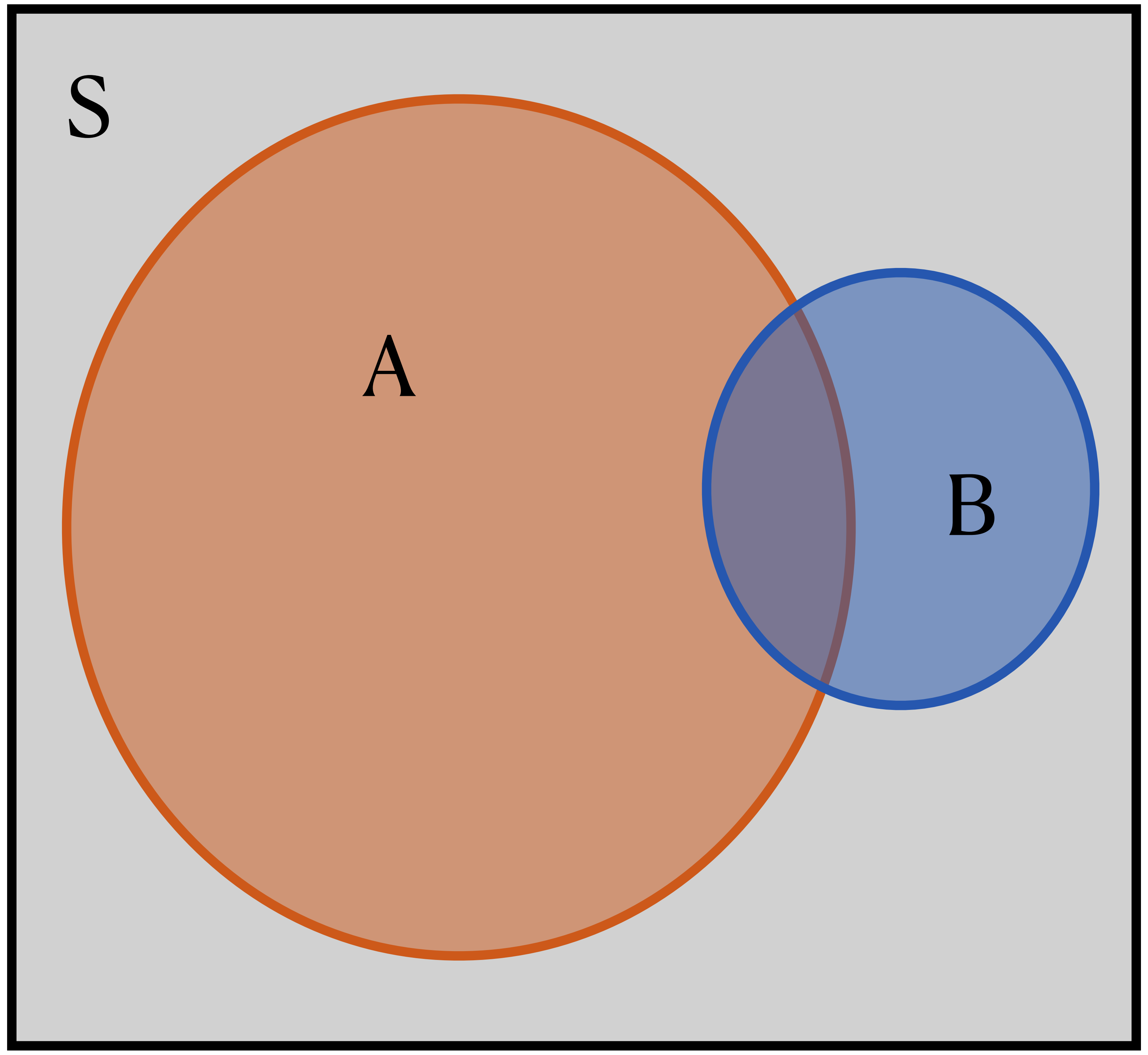


Probability Concepts

Law of Total Probability

- The **Law of Total Probability** gives us a way to relate **marginal probabilities** to **joint probabilities**
- Suppose we want to get $P(B)$ (*marginal*) using joint probabilities of A and B. Let's do it like a puzzle:

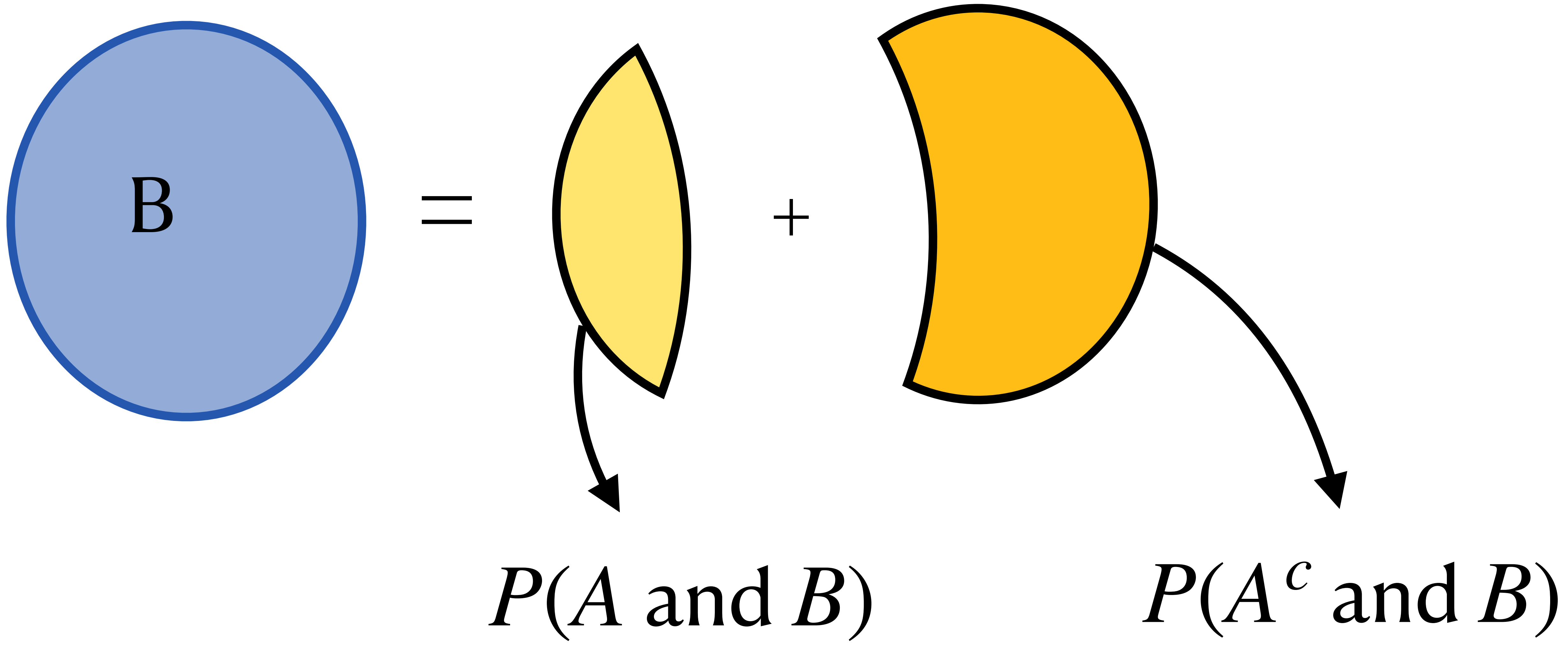




Probability Concepts

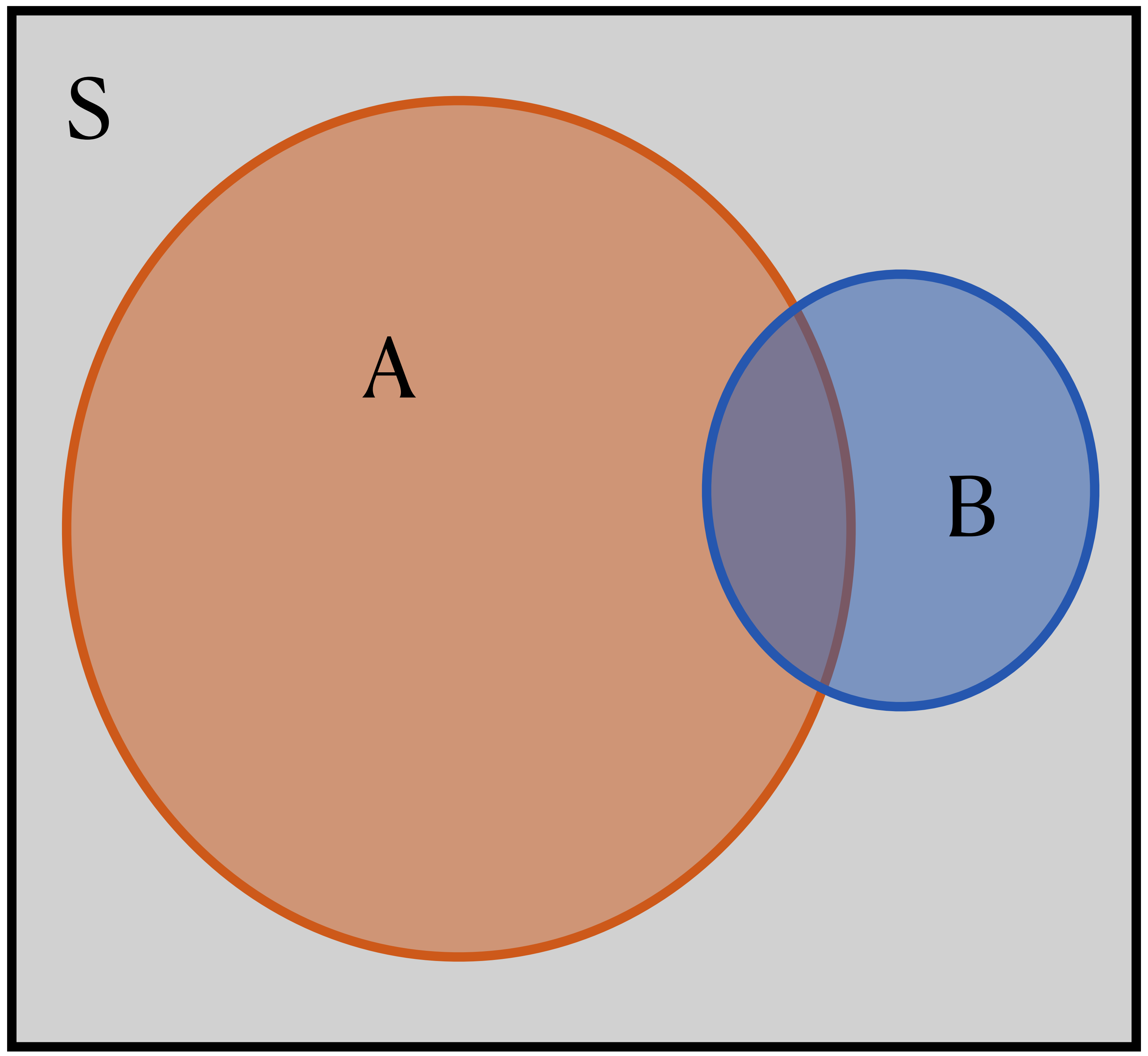
Law of Total Probability

- The **Law of Total Probability** gives us a way to relate **marginal probabilities** to **joint probabilities**



- The **Law of Total Probability** states:

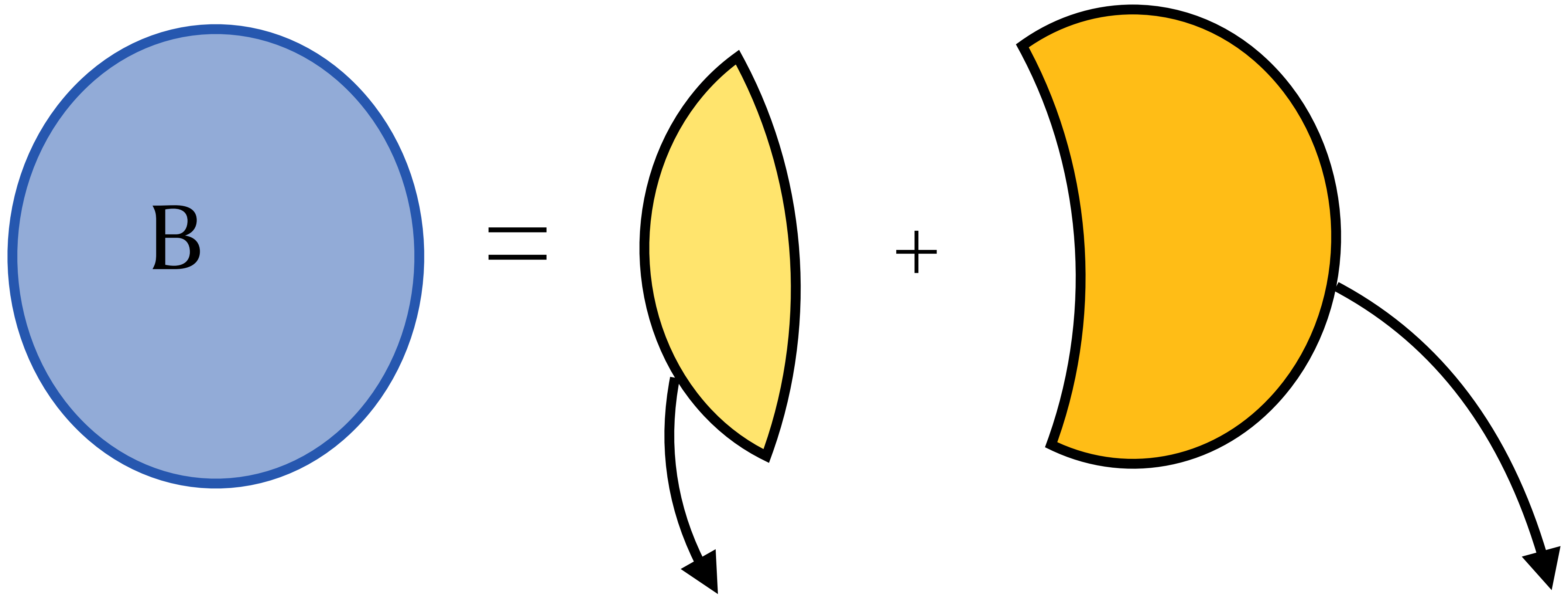
$$P(B) = P(A \text{ and } B) + P(A^c \text{ and } B)$$



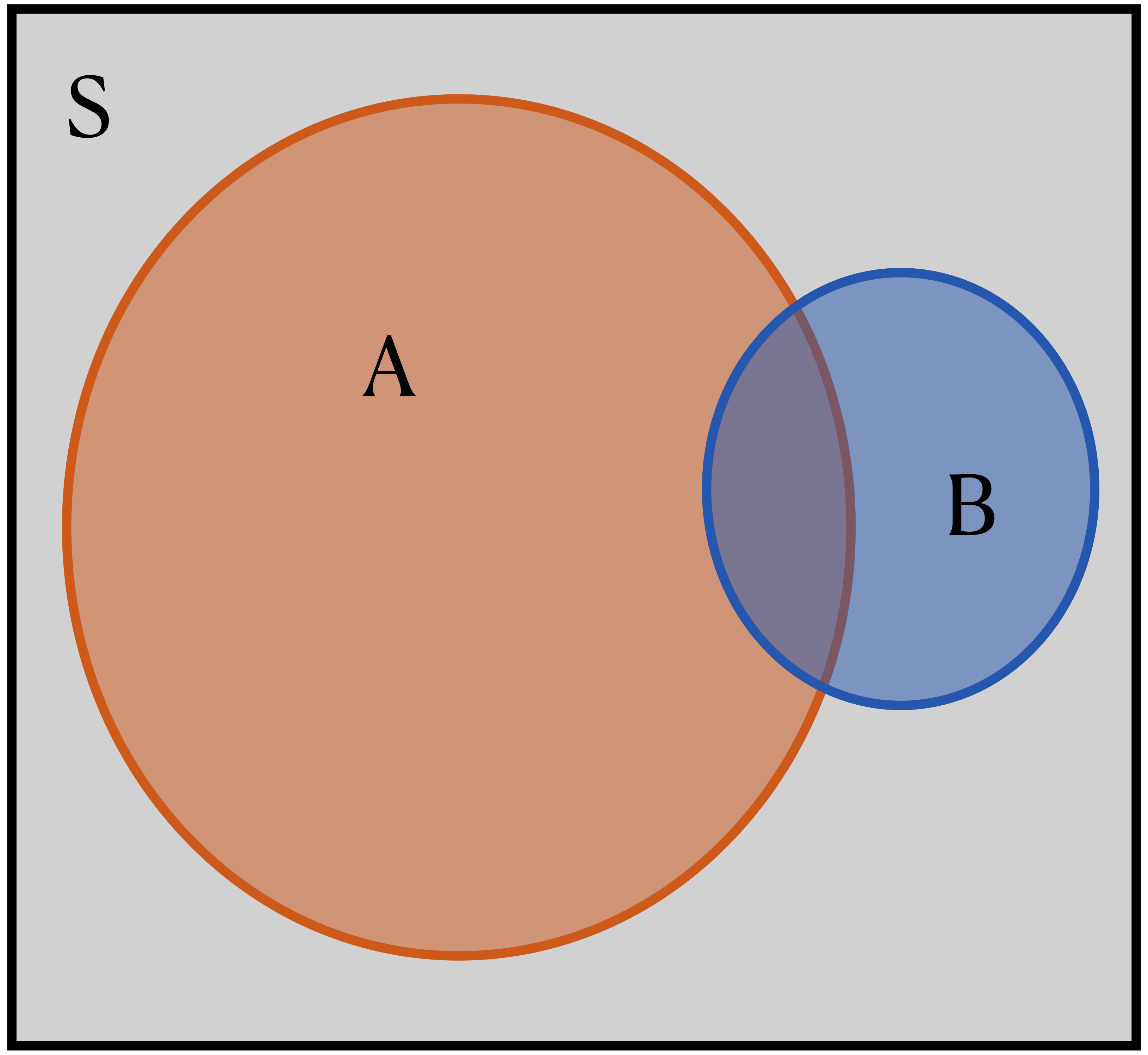
Probability Concepts

Law of Total Probability

- The **Law of Total Probability** gives us a way to relate **marginal probabilities** to **joint probabilities**



$P(A \text{ and } B)$ $P(A^c \text{ and } B)$



- The **Law of Total Probability** states:

$$P(B) = P(A \text{ and } B) + P(A^c \text{ and } B)$$

(This is why I love Venn diagrams — you barely have to memorize anything if you can just draw it to figure it out)

Common Confusions with Probability Concepts

Independent vs. Mutually Exclusive

- Events are **mutually exclusive** if both cannot happen, so

$$P(A \text{ and } B) = 0.$$

- Events are **independent** if knowledge of one does not affect the probability of the other, so

$$P(A \mid B) = P(A) \text{ or, equivalently, } P(A \text{ and } B) = P(A)P(B).$$

Common Confusions with Probability Concepts

Confusion of the Inverse

- We've talked about this one before! (Topic 3)
- Back to the COVID birth/ICU example →
- A news report says, "COVID isn't a big deal for people giving birth! Actually, it's better to have COVID while giving birth, since the proportion of people in the ICU after giving birth is overwhelmingly people who do not have COVID!"
  - P(COVID | ICU) = 11.0%
  - P(no COVID | ICU) = 89.0%

	Had COVID during birth	**Did not have COVID during birth**	**Total**
Admitted to ICU	977	7943	8920
Not admitted to ICU	17738	842241	859979
Total	18715	850184	868899

Common Confusions with Probability Concepts

Confusion of the Inverse

- We've talked about this one before! (Topic 3)
- Back to the COVID birth/ICU example →
- A news report says, "COVID isn't a big deal for people giving birth! Actually, it's better to have COVID while giving birth, since the proportion of people in the ICU after giving birth is overwhelmingly people who do not have COVID!"
  - Yes, but only because there are way more non-COVID births…

	Had COVID during birth	Did not have COVID during birth	Total
Admitted to ICU	977	7943	8920
Not admitted to ICU	17738	842241	859979
Total	18715	850184	868899

Common Confusions with Probability Concepts

Confusion of the Inverse

- We've talked about this one before! (Topic 3)
- Back to the COVID birth/ICU example →
- A news report says, "COVID isn't a big deal for people giving birth! Actually, it's better to have COVID while giving birth."
- Desired comparison:
  - P(ICU | COVID) = 5.2%
  - P(ICU | no COVID) = 0.9%
  - ...doesn't look so good now, does it?

	Had COVID during birth	**Did not have COVID during birth**	**Total**
Admitted to ICU	977	7943	8920
Not admitted to ICU	17738	842241	859979
Total	18715	850184	868899

Common Confusions with Probability Concepts

Counterintuitive Results

- Counterintuitive but interesting result regarding *screening tests*
- For a hypothetical rare disease (say, 1 in 1 million people have this),
  - Sensitivity = the probability of a positive test for a patient who has the disease, $P(T \mid D) = 95\%$
  - Specificity = the probability of a negative test for a patient who does not have the disease, $P(T^c \mid D^c) = 99\%$
- Suppose that a random person is chosen and given the test for this rare disease. They test positive. What's the probability they actually have it?

Common Confusions with Probability Concepts

Counterintuitive Results

- $P(T \mid D) = 95\%$, $P(T^c \mid D^c) = 99\%$, $P(D) = 0.000001$

- Suppose that a random person is chosen and given the test for this rare disease. They test positive. What's the probability they actually have it?
  - $P(D \mid T) \rightarrow$ probability they have disease, given positive test
    - How do we get this? We need to try rewriting this in terms of probabilities we actually have.
    - (Try this on your own. You have to use a lot of probability rules, and eventually you get to something you can plug into)

Common Confusions with Probability Concepts

Counterintuitive Results

- $P(T \mid D) = 95\%$, $P(T^c \mid D^c) = 99\%$, $P(D) = 0.000001$
- Suppose that a random person is chosen and given the test for this rare disease. They test positive. What's the probability they actually have it?
  - $P(D \mid T) \rightarrow$ probability they have disease, given positive test
    - How do we get this? We need to try rewriting this in terms of probabilities we actually have.
    - It ends up that $P(D \mid T)$ is only about 0.0095% because the disease is so rare… counterintuitive!

Probability Concepts

Overview

- Make sure you know the terms **sample space**, **complement**, **union**, **intersection**, **joint probability**, **marginal probability**, **addition rule**, **law of total probability**, **mutually exclusive**, **conditional probability**, **independent**
- Make sure you know the notation P(A), P(A^c), P(A **or** B), P(A **and** B), P(A | B)
- Watch out for common confusions like:
  - Mixing up independent and mutually exclusive
  - Confusion of the inverse with conditional probabilities